

1 Introduction

In this study, we numerically address an optimal control problem employing the Hamiltonian framework to derive the necessary conditions for optimality. The Hamiltonian formulation provides a systematic approach for deriving the governing equations.

Consider the following optimal control problem:

$$\begin{aligned} \min \quad & J(v, u) = \int_0^T \left(\frac{v^2}{2} + \frac{\alpha}{2} u^2 \right) dt, \\ & \dot{v} = g + \frac{u}{m} - \frac{k}{m} v, \\ & v(0) = v_0. \end{aligned} \tag{1}$$

The Hamiltonian, defined as

$$\mathcal{H}(v, u, \lambda) = \frac{v^2}{2} + \frac{\alpha}{2} u^2 + \lambda \left(g + \frac{u}{m} - \frac{k}{m} v \right),$$

is utilized to write the necessary optimality conditions for the above optimal control problem as follows:

$$\begin{aligned} \dot{v} &= \frac{\partial \mathcal{H}}{\partial \lambda}(v, u, \lambda) = g + \frac{u}{m} - \frac{k}{m} v, \\ \dot{\lambda} &= -\frac{\partial \mathcal{H}}{\partial v}(v, u, \lambda) = -v + \frac{k}{m} \lambda, \\ 0 &= \frac{\partial \mathcal{H}}{\partial u}(v, u, \lambda) = \alpha u + \frac{k}{m} \lambda, \\ v(0) &= v_0, \quad \lambda(T) = 0. \end{aligned} \tag{2}$$

To solve the resulting system of equations, we apply four numerical techniques, specifically the $SE1_1$, $SE1_2$, $SE2_1$ and $SE2_2$ methods. Each method begins with three initial points $\mathbf{z}_1 = (v_1(0), \lambda_1(0))$, $\mathbf{z}_2 = (v_2(0), \lambda_2(0))$ and $\mathbf{z}_3 = (v_3(0), \lambda_3(0))$, which form a triangle in the phase space. The area of this triangle is computed at the initial time $t = 0$, and the methods are applied, respectively, to determine the area of the triangle at the final time $t = T$. So, it allows us to identify which method effectively preserves the geometric properties of the triangle, particularly its area.

Preserving the triangle's area is important in this context, as it serves as a test for identifying symplectic algorithms. Symplectic algorithms are numerical methods designed to preserve the symplectic structure of Hamiltonian systems, which is directly related to the conservation of geometric properties like area in phase space.

By applying the $SE1_1$, $SE1_2$, $SE2_1$ and $SE2_2$ methods to (2), respectively, the resulting equations are as follows:

$$\begin{aligned} SE1_1 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_{n+1}, u_n, \lambda_n) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_{n+1}, u_n, \lambda_n) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \quad \lambda(0) = \lambda_0. \end{cases} \\ SE1_2 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_{n+1}, u_{n+1}, \lambda_n) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_{n+1}, u_{n+1}, \lambda_n) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \quad \lambda(0) = \lambda_0. \end{cases} \\ SE2_1 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_n, u_{n+1}, \lambda_{n+1}) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_n, u_{n+1}, \lambda_{n+1}) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \quad \lambda(0) = \lambda_0. \end{cases} \\ SE2_2 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_n, u_n, \lambda_{n+1}) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_n, u_n, \lambda_{n+1}) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \quad \lambda(0) = \lambda_0. \end{cases} \end{aligned} \tag{3}$$

Before presenting numerical results, we first demonstrate that the numerical methods $SE1_1$ and $SE2_1$ are symplectic integrators, which means that they exactly preserve the symplectic condition. In contrast, $SE1_2$ and $SE2_2$ are not symplectic integrators.

$SE1_1$:

$$v_{n+1} = v_n + \Delta t g + \frac{\Delta t}{m} u_n - \frac{k\Delta t}{m} v_{n+1} \Rightarrow \frac{m+k\Delta t}{m} v_{n+1} = v_n + \Delta t g + \frac{\Delta t}{m} u_n \Rightarrow$$

$$\boxed{v_{n+1} = \frac{m}{m+k\Delta t} v_n + \frac{m\Delta t}{m+k\Delta t} g + \frac{-k\Delta t}{m\alpha(m+k\Delta t)} \lambda_n.} \quad (4)$$

$$\lambda_{n+1} = \lambda_n - \Delta t v_{n+1} + \frac{k\Delta t}{m} \lambda_n \Rightarrow \boxed{\lambda_{n+1} = \frac{m+k\Delta t}{m} \lambda_n - \Delta t v_{n+1}.} \quad (5)$$

Taking differentials of both sides of equations (4) and (5) and using the linearity of the wedge product, we have

$$dv_{n+1} = \frac{m}{m+k\Delta t} dv_n + \frac{-k\Delta t}{m\alpha(m+k\Delta t)} d\lambda_n,$$

$$d\lambda_{n+1} = \frac{m+k\Delta t}{m} d\lambda_n - \Delta t dv_{n+1},$$

$$d\lambda_{n+1} \wedge dv_{n+1} = \frac{m+k\Delta t}{m} d\lambda_n \wedge dv_{n+1} = \frac{m+k\Delta t}{m} d\lambda_n \wedge \left(\frac{m}{m+k\Delta t} dv_n + \frac{-k\Delta t}{m\alpha(m+k\Delta t)} d\lambda_n \right) \Rightarrow$$

$$\boxed{d\lambda_{n+1} \wedge dv_{n+1} = \frac{m+k\Delta t}{m} d\lambda_n \wedge \frac{m}{m+k\Delta t} dv_n = d\lambda_n \wedge dv_n.}$$

$SE2_1$:

$$v_{n+1} = \frac{m-k\Delta t}{m} v_n + \Delta t g + \frac{\Delta t}{m} u_{n+1} \Rightarrow$$

$$\boxed{v_{n+1} = \frac{m-k\Delta t}{m} v_n + \Delta t g + \frac{-k\Delta t}{m^2\alpha} \lambda_{n+1}.} \quad (6)$$

$$\boxed{\lambda_{n+1} = \frac{m}{m-\Delta t k} (-\Delta t) v_n + \frac{m}{m-\Delta t k} \lambda_n.} \quad (7)$$

Similar to $SE1_1$, by taking differentials of both sides of equations (6) and (7) and using the wedge product, we have

$$dv_{n+1} = \frac{m-k\Delta t}{m} dv_n + \frac{-k\Delta t}{m^2\alpha} d\lambda_{n+1},$$

$$d\lambda_{n+1} = \frac{m}{m-\Delta t k} (-\Delta t) dv_n + \frac{m}{m-\Delta t k} d\lambda_n,$$

$$\boxed{dv_{n+1} \wedge d\lambda_{n+1} = \frac{m-k\Delta t}{m} dv_n \wedge d\lambda_{n+1} = dv_n \wedge d\lambda_n.}$$

$SE1_2$:

$$v_{n+1} = \frac{m}{m+k\Delta t} v_n + \frac{m}{m+k\Delta t} \Delta t g + \frac{\Delta t}{m+k\Delta t} u_{n+1} \Rightarrow$$

$$\boxed{v_{n+1} = \frac{m}{m+k\Delta t} v_n + \frac{m}{m+k\Delta t} \Delta t g + \frac{-k\Delta t}{m\alpha(m+k\Delta t)} \lambda_{n+1}.} \quad (8)$$

$$\boxed{\lambda_{n+1} = \frac{m+k\Delta t}{m} \lambda_n + (-\Delta t) v_{n+1}.} \quad (9)$$

Now by taking differentials of both sides of equations (8) and (9), we have

$$dv_{n+1} = \frac{m}{m+k\Delta t} dv_n + \frac{-k\Delta t}{m\alpha(m+k\Delta t)} d\lambda_{n+1}, \quad (10)$$

$$d\lambda_{n+1} = \frac{m+k\Delta t}{m} d\lambda_n + (-\Delta t) dv_{n+1}. \quad (11)$$

From equations (10) and (11), dv_{n+1} and $d\lambda_{n+1}$ can be expressed in terms of dv_n and $d\lambda_n$, so we have:

$$dv_{n+1} = \frac{m^2\alpha}{m\alpha(m+k\Delta t) - k(\Delta t)^2} dv_n + \frac{-k\Delta t(m+k\Delta t)}{m^2\alpha(m+k\Delta t) - km(\Delta t)^2} d\lambda_n \quad (12)$$

$$d\lambda_{n+1} = \frac{\alpha(m+k\Delta t)^2}{m\alpha(m+k\Delta t) - k(\Delta t)^2} d\lambda_n + \frac{-m^2\alpha\Delta t}{m\alpha(m+k\Delta t) - k(\Delta t)^2} dv_n \quad (13)$$

If we take the wedge product of (12) with (13), we get

$$\boxed{dv_{n+1} \wedge d\lambda_{n+1} = \left(1 + \frac{k(\Delta t)^2}{m\alpha(m+k\Delta t) - k(\Delta t)^2}\right) dv_n \wedge d\lambda_n.} \quad (14)$$

SE2₂ :

$$v_{n+1} = g\Delta t + \frac{m-k\Delta t}{m} v_n + \frac{-k\Delta t}{m^2\alpha} \lambda_n, \quad (15)$$

$$\lambda_{n+1} = \frac{-m\Delta t}{m-k\Delta t} v_n + \frac{m}{m-k\Delta t} \lambda_n. \quad (16)$$

Taking differentials of both sides of equations (15) and (16) and using the wedge product, we have

$$dv_{n+1} \wedge d\lambda_{n+1} = \left(1 - \frac{k(\Delta t)^2}{m\alpha(m-k\Delta t)}\right) dv_n \wedge d\lambda_n. \quad (17)$$

1.1 Linear optimal control problem

Let us consider the following linear optimal control problem:

$$\begin{aligned} J(\mathbf{v}, \mathbf{u}) &= \frac{1}{2} \int_a^b \mathbf{v}(t)^T \mathbf{Q} \mathbf{v}(t) + \mathbf{u}(t)^T \mathbf{R} \mathbf{u}(t) dt \\ \dot{\mathbf{v}}(t) &= \mathbf{A} \mathbf{v}(t) + \mathbf{B} \mathbf{u}(t), \\ \mathbf{v}(a) &= \mathbf{v}_0, \end{aligned} \quad (18)$$

where $\mathbf{Q}$ is a real symmetric positive semi-definite matrix and $\mathbf{R}$ is a real symmetric positive definite matrix.

The Hamiltonian is

$$\mathcal{H}(\mathbf{v}, \mathbf{u}, \boldsymbol{\lambda}) = \frac{1}{2} \mathbf{v}(t)^T \mathbf{Q} \mathbf{v}(t) + \frac{1}{2} \mathbf{u}(t)^T \mathbf{R} \mathbf{u}(t) + \boldsymbol{\lambda}^T \mathbf{A} \mathbf{v}(t) + \boldsymbol{\lambda}^T \mathbf{B} \mathbf{u}(t), \quad (19)$$

and necessary conditions for optimality are

$$\dot{\mathbf{v}}(t) = \frac{\partial \mathcal{H}(\mathbf{v}, \mathbf{u}, \boldsymbol{\lambda})}{\partial \boldsymbol{\lambda}} = \mathbf{A} \mathbf{v}(t) + \mathbf{B} \mathbf{u}(t), \quad (20)$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial \mathcal{H}(\mathbf{v}, \mathbf{u}, \boldsymbol{\lambda})}{\partial \mathbf{v}} = -\mathbf{Q} \mathbf{v}(t) - \mathbf{A}^T \boldsymbol{\lambda}(t), \quad (21)$$

$$\mathbf{0} = \frac{\partial \mathcal{H}(\mathbf{v}, \mathbf{u}, \boldsymbol{\lambda})}{\partial \mathbf{u}} = \mathbf{R} \mathbf{u}(t) + \mathbf{B}^T \boldsymbol{\lambda}(t). \quad (22)$$

Equation (22) can be solved for $\mathbf{u}$ to give

$$\mathbf{u}(t) = -\mathbf{R}^{-1} \mathbf{B}^T \boldsymbol{\lambda}(t), \quad (23)$$

the existence of $\mathbf{R}^{-1}$ is assured, since $\mathbf{R}$ is a positive definite matrix.

Let us apply the SE1₁ and SE2₁ schemes to equations (20)–(22), respectively. We want to show that both are symplectic integrators, that is, they preserve the symplectic condition, i.e. $d\mathbf{v}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} = d\mathbf{v}_n \wedge d\boldsymbol{\lambda}_n$ or $\det\left(\frac{\partial(\mathbf{v}_{n+1}, \boldsymbol{\lambda}_{n+1})}{\partial(\mathbf{v}_n, \boldsymbol{\lambda}_n)}\right) = 1$.

SE1₁ :

$$\begin{aligned} \frac{\mathbf{v}_{n+1} - \mathbf{v}_n}{\Delta t} &= \mathbf{A} \mathbf{v}_{n+1} + \mathbf{B} \mathbf{u}_n = \mathbf{A} \mathbf{v}_{n+1} - \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \boldsymbol{\lambda}_n \Rightarrow \\ \mathbf{v}_{n+1} &= (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{v}_n - \Delta t (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \boldsymbol{\lambda}_n, \end{aligned} \quad (24)$$

$$\begin{aligned} \frac{\boldsymbol{\lambda}_{n+1} - \boldsymbol{\lambda}_n}{\Delta t} &= -\mathbf{Q} \mathbf{v}_{n+1} - \mathbf{A}^T \boldsymbol{\lambda}_n \Rightarrow \\ \boldsymbol{\lambda}_{n+1} &= -\Delta t \mathbf{Q} \mathbf{v}_{n+1} + (\mathbf{I} - \Delta t \mathbf{A}^T) \boldsymbol{\lambda}_n. \end{aligned} \quad (25)$$

Taking differentials of (24) and (25), respectively, we obtain the following implicit system of equations:

$$d\mathbf{v}_{n+1} = (\mathbf{I} - \Delta t \mathbf{A})^{-1} d\mathbf{v}_n - \Delta t (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_n, \quad (26)$$

$$d\boldsymbol{\lambda}_{n+1} = -\Delta t \mathbf{Q} d\mathbf{v}_{n+1} + (\mathbf{I} - \Delta t \mathbf{A}^T) d\boldsymbol{\lambda}_n. \quad (27)$$

By taking the wedge product of (26) with (27) from the right, we have

$$\begin{aligned} d\mathbf{v}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} &= (\mathbf{I} - \Delta t \mathbf{A})^{-1} d\mathbf{v}_n \wedge (-\Delta t \mathbf{Q} d\mathbf{v}_{n+1}) + (\mathbf{I} - \Delta t \mathbf{A})^{-1} d\mathbf{v}_n \wedge (\mathbf{I} - \Delta t \mathbf{A}^T) d\boldsymbol{\lambda}_n + \\ &\quad (-\Delta t (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_n) \wedge (-\Delta t \mathbf{Q} d\mathbf{v}_{n+1}). \end{aligned} \quad (28)$$

Upon plugging (26) into (28) and using $d\mathbf{a} \wedge d(\mathbf{M} d\mathbf{b}) = (\mathbf{M}^T d\mathbf{a}) \wedge d\mathbf{b}$, we obtain the following equality

$$\begin{aligned} d\mathbf{v}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} &= (\mathbf{I} - \Delta t \mathbf{A})^{-1} d\mathbf{v}_n \wedge \Delta t^2 \mathbf{Q} (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_n + d\mathbf{v}_n \wedge d\boldsymbol{\lambda}_n + \\ &\quad - \Delta t (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_n \wedge -\Delta t \mathbf{Q} (\mathbf{I} - \Delta t \mathbf{A})^{-1} d\mathbf{v}_n \\ &= d\mathbf{v}_n \wedge d\boldsymbol{\lambda}_n - \Delta t^2 (\mathbf{I} - \Delta t \mathbf{A})^{-T} \mathbf{Q} (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_n \wedge d\mathbf{v}_n + \\ &\quad \Delta t^2 (\mathbf{I} - \Delta t \mathbf{A})^{-T} \mathbf{Q}^T (\mathbf{I} - \Delta t \mathbf{A})^{-1} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_n \wedge d\mathbf{v}_n \\ &= d\mathbf{v}_n \wedge d\boldsymbol{\lambda}_n. \end{aligned} \quad (29)$$

SE2₁ :

$$\mathbf{v}_{n+1} = (\mathbf{I} + \Delta t \mathbf{A}) \mathbf{v}_n - \Delta t \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T \boldsymbol{\lambda}_{n+1}, \quad (30)$$

$$\boldsymbol{\lambda}_{n+1} = -\Delta t (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} \mathbf{Q} \mathbf{v}_n + (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} \boldsymbol{\lambda}_n, \quad (31)$$

$$d\mathbf{v}_{n+1} = (\mathbf{I} + \Delta t \mathbf{A}) d\mathbf{v}_n - \Delta t \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_{n+1}, \quad (32)$$

$$d\boldsymbol{\lambda}_{n+1} = -\Delta t (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} \mathbf{Q} d\mathbf{v}_n + (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} d\boldsymbol{\lambda}_n. \quad (33)$$

By taking the wedge product of (32) with (33) we obtain

$$\begin{aligned} d\boldsymbol{\lambda}_{n+1} \wedge d\mathbf{v}_{n+1} &= (-\Delta t (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} \mathbf{Q} d\mathbf{v}_n) \wedge (-\Delta t \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_{n+1}) + \\ &\quad (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge (\mathbf{I} + \Delta t \mathbf{A}) d\mathbf{v}_n + \\ &\quad (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge (-\Delta t \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T d\boldsymbol{\lambda}_{n+1}). \end{aligned} \quad (34)$$

Upon plugging (33) into (34), we obtain the desired equality (35).

$$\begin{aligned} d\boldsymbol{\lambda}_{n+1} \wedge d\mathbf{v}_{n+1} &= d\boldsymbol{\lambda}_n \wedge d\mathbf{v}_n + \\ &\quad - \Delta t^2 ((\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} \mathbf{Q})^T \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge d\mathbf{v}_n + \\ &\quad \Delta t^2 (\mathbf{B} \mathbf{R}^{-1} \mathbf{B}^T (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} \mathbf{Q})^T (\mathbf{I} + \Delta t \mathbf{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge d\mathbf{v}_n = \\ &= d\boldsymbol{\lambda}_n \wedge d\mathbf{v}_n. \end{aligned} \quad (35)$$

2 Numerical Results (linear case)

In this section, we present numerical results (summarized in Table 1) that compare the initial and final areas computed by each method. Figures 1–4 depict the area of the triangles calculated in discrete time steps ($t = i\Delta t$, $\Delta t = 0.1$) with different α over the interval $[0, 10]$. This stepwise analysis highlights how the area varies in the time range for each approach.

3 Optimal control problem

Consider the following basic optimal control problem,

$$\begin{aligned} \min \int_a^b j(\mathbf{y}, \mathbf{u}) dt \\ \dot{\mathbf{y}} &= \mathbf{f}(\mathbf{y}, \mathbf{u}) \\ \mathbf{y}(0) &= \mathbf{y}_0. \end{aligned} \quad (36)$$

Method	α	A_0	A_T
$SE1_1$	10	0.1000	0.1000
	7	0.1000	0.1000
	5	0.1000	0.1000
	2	0.1000	0.1000
	1	0.1000	0.1000
	0.5	0.1000	0.0991
$SE1_2$	10	0.1000	0.1095
	7	0.1000	0.1139
	5	0.1000	0.1200
	2	0.1000	0.1577
	1	0.1000	0.2493
	0.5	0.1000	0.4609
$SE2_1$	10	0.1000	0.1000
	7	0.1000	0.1000
	5	0.1000	0.1000
	2	0.1000	0.1000
	1	0.1000	0.1001
	0.5	0.1000	0.2500
$SE2_2$	10	0.1000	0.0895
	7	0.1000	0.0853
	5	0.1000	0.0801
	2	0.1000	0.0573
	1	0.1000	0.0327
	0.5	0.1000	0.0117

Table 1: Application of methods $SE1_1$, $SE1_2$, $SE2_1$ and $SE2_2$ to system (2) on the interval $[0, 10]$ with three initial points $\mathbf{z}_1 = (10, 0.1)^T$, $\mathbf{z}_2 = (11, 0.1)^T$ and $\mathbf{z}_3 = (10, -0.1)^T$; in addition, $k = 1$, $m = 1$, $\Delta t = 0.1$, $T = 10$. A_0 and A_T denote the areas of the triangle at time $t = 0$ and $t = T = 10$, respectively.

The necessary conditions are derived from the Hamiltonian H , defined as

$$H(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) = j(\mathbf{y}, \mathbf{u}) + \boldsymbol{\lambda}^T \mathbf{f}(\mathbf{y}, \mathbf{u}). \quad (37)$$

The necessary optimality conditions can be written in terms of the Hamiltonian:

$$\dot{\mathbf{y}} = H_{\boldsymbol{\lambda}} = \frac{\partial H}{\partial \boldsymbol{\lambda}}(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) \quad (38)$$

$$\dot{\boldsymbol{\lambda}} = -H_{\mathbf{y}} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}) \quad (39)$$

$$\mathbf{0} = H_{\mathbf{u}} = \frac{\partial H}{\partial \mathbf{u}}(\mathbf{y}, \mathbf{u}, \boldsymbol{\lambda}). \quad (40)$$

We want to show that $SE1_1$ and $SE2_1$ schemes are symplectic integrators, that is, $d\mathbf{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} = d\mathbf{y}_n \wedge d\boldsymbol{\lambda}_n$ if $H_{\mathbf{y}\mathbf{u}} = \frac{\partial^2 H}{\partial \mathbf{u} \partial \mathbf{y}} = \mathbf{0}$.

SE1₁ : Let $\mathbf{u}$ be a function of $\mathbf{y}$ and $\boldsymbol{\lambda}$, i.e. $\mathbf{u} = \mathbf{g}(\mathbf{y}, \boldsymbol{\lambda})$, so from (40) we have

$$H_{\mathbf{u}\mathbf{y}} + H_{\mathbf{u}\mathbf{u}}\mathbf{g}_{\mathbf{y}} = \mathbf{0} \Rightarrow \mathbf{g}_{\mathbf{y}} = -H_{\mathbf{u}\mathbf{u}}^{-1}H_{\mathbf{u}\mathbf{y}} \quad (41)$$

$$H_{\mathbf{u}\boldsymbol{\lambda}} + H_{\mathbf{u}\mathbf{u}}\mathbf{g}_{\boldsymbol{\lambda}} = \mathbf{0} \Rightarrow \mathbf{g}_{\boldsymbol{\lambda}} = -H_{\mathbf{u}\mathbf{u}}^{-1}H_{\mathbf{u}\boldsymbol{\lambda}}, \quad (42)$$

$$d\mathbf{u} = -H_{\mathbf{u}\mathbf{u}}^{-1}H_{\mathbf{u}\mathbf{y}}d\mathbf{y} - H_{\mathbf{u}\mathbf{u}}^{-1}H_{\mathbf{u}\boldsymbol{\lambda}}d\boldsymbol{\lambda} \quad (43)$$

where $\mathbf{g}_{\mathbf{y}} = \frac{\partial \mathbf{g}}{\partial \mathbf{y}}$ and $\mathbf{g}_{\boldsymbol{\lambda}} = \frac{\partial \mathbf{g}}{\partial \boldsymbol{\lambda}}$, respectively.

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \Delta t H_{\boldsymbol{\lambda}}(\mathbf{y}_{n+1}, \mathbf{u}_n, \boldsymbol{\lambda}_n). \quad (44)$$

Taking the differential of (44) and plugging (41)–(43) into (45) we obtain the equation:

$$d\mathbf{y}_{n+1} = d\mathbf{y}_n + \Delta t (H_{\boldsymbol{\lambda}\mathbf{y}}d\mathbf{y}_{n+1} + H_{\boldsymbol{\lambda}\mathbf{u}}d\mathbf{u}_n + H_{\boldsymbol{\lambda}\boldsymbol{\lambda}}d\boldsymbol{\lambda}_n), \quad (45)$$

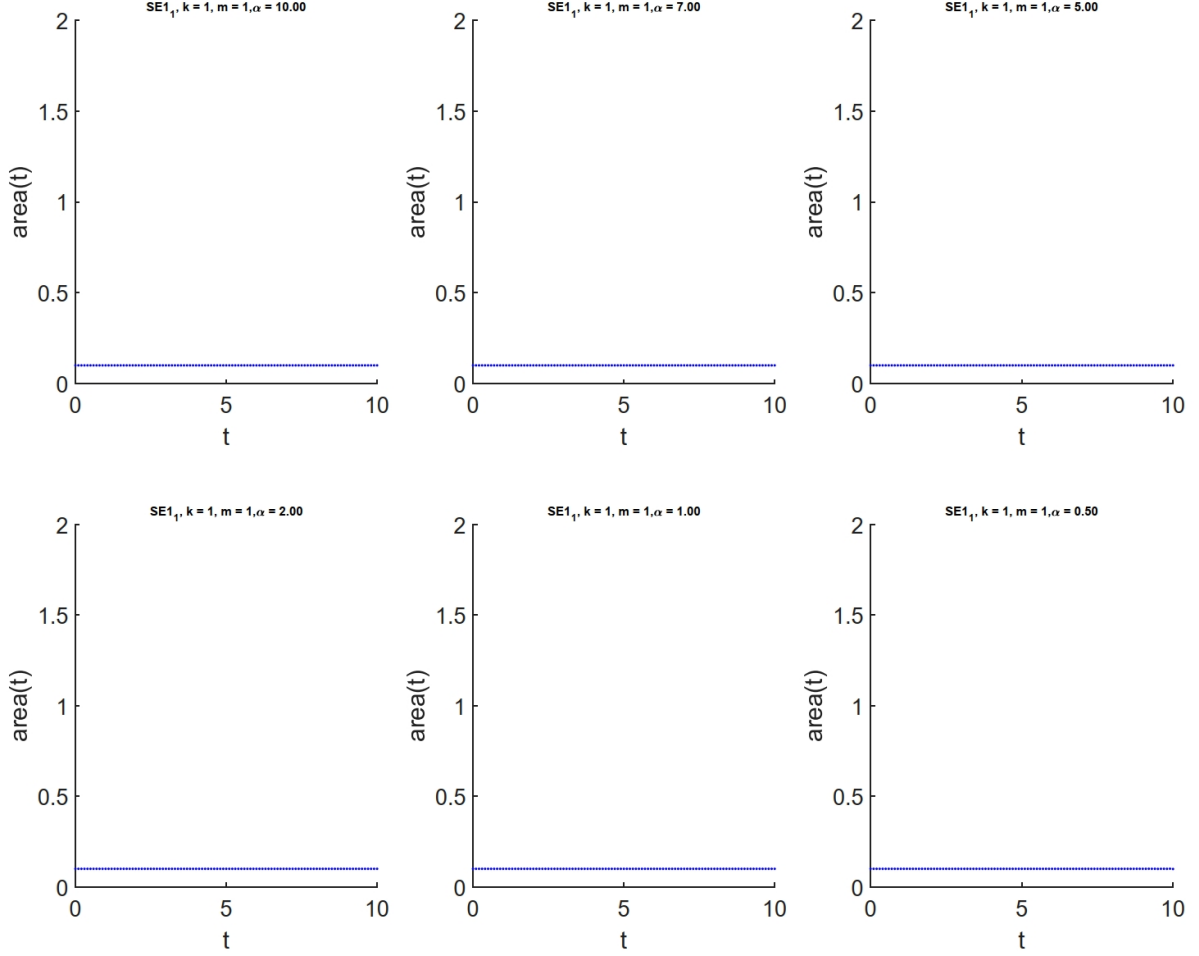


Figure 1: $SE1_1$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

$$d\mathbf{y}_{n+1} = d\mathbf{y}_n + \Delta t H_{\lambda y} d\mathbf{y}_{n+1} - \Delta t H_{\lambda u} H_{uu}^{-1} H_{uy} d\mathbf{y}_n - \Delta t H_{\lambda u} H_{uu}^{-1} H_{u\lambda} d\lambda_n + \Delta t H_{\lambda\lambda} d\lambda_n \quad (46)$$

$$\boxed{d\mathbf{y}_{n+1} = (\mathbf{I} - \Delta t H_{\lambda y})^{-1} (\mathbf{I} - \Delta t \mathbf{A}) d\mathbf{y}_n + \Delta t (\mathbf{I} - \Delta t H_{\lambda y})^{-1} (-\mathbf{B} + H_{\lambda\lambda}) d\lambda_n}, \quad (47)$$

where $\mathbf{A} = H_{\lambda u} H_{uu}^{-1} H_{uy}$ and $\mathbf{B} = H_{\lambda u} H_{uu}^{-1} H_{u\lambda}$.

$$\lambda_{n+1} = \lambda_n - \Delta t H_y(\mathbf{y}_{n+1}, \mathbf{u}_n, \lambda_n) \quad (48)$$

$$d\lambda_{n+1} = d\lambda_n - \Delta t (H_{yy} d\mathbf{y}_{n+1} + H_{yu} d\mathbf{u}_n + H_{y\lambda} d\lambda_n) \quad (49)$$

Upon plugging (41)–(43) into (49), we have

$$d\lambda_{n+1} = d\lambda_n - \Delta t H_{yy} d\mathbf{y}_{n+1} + \Delta t H_{yu} H_{uu}^{-1} H_{uy} d\mathbf{y}_n + \Delta t H_{yu} H_{uu}^{-1} H_{u\lambda} d\lambda_n - \Delta t H_{y\lambda} d\lambda_n \quad (50)$$

$$d\lambda_{n+1} = (\mathbf{I} - \Delta t H_{y\lambda} + \Delta t \mathbf{A}^T) d\lambda_n + \Delta t \mathbf{D} d\mathbf{y}_n - \Delta t H_{yy} d\mathbf{y}_{n+1} \quad (51)$$

where $\mathbf{D} = H_{yu} H_{uu}^{-1} H_{uy}$.

By taking the wedge product of (51) with $d\mathbf{y}_{n+1}$ we obtain

$$\begin{aligned} d\lambda_{n+1} \wedge d\mathbf{y}_{n+1} = & (\mathbf{I} - \Delta t H_{y\lambda} + \Delta t \mathbf{A}^T) d\lambda_n \wedge (\mathbf{I} - \Delta t H_{\lambda y})^{-1} (\mathbf{I} - \Delta t \mathbf{A}) d\mathbf{y}_n + \\ & \Delta t (\mathbf{I} - \Delta t H_{\lambda y})^{-1} (\mathbf{B} - H_{\lambda\lambda}) d\lambda_n \wedge \Delta t \mathbf{D} d\mathbf{y}_n. \end{aligned}$$

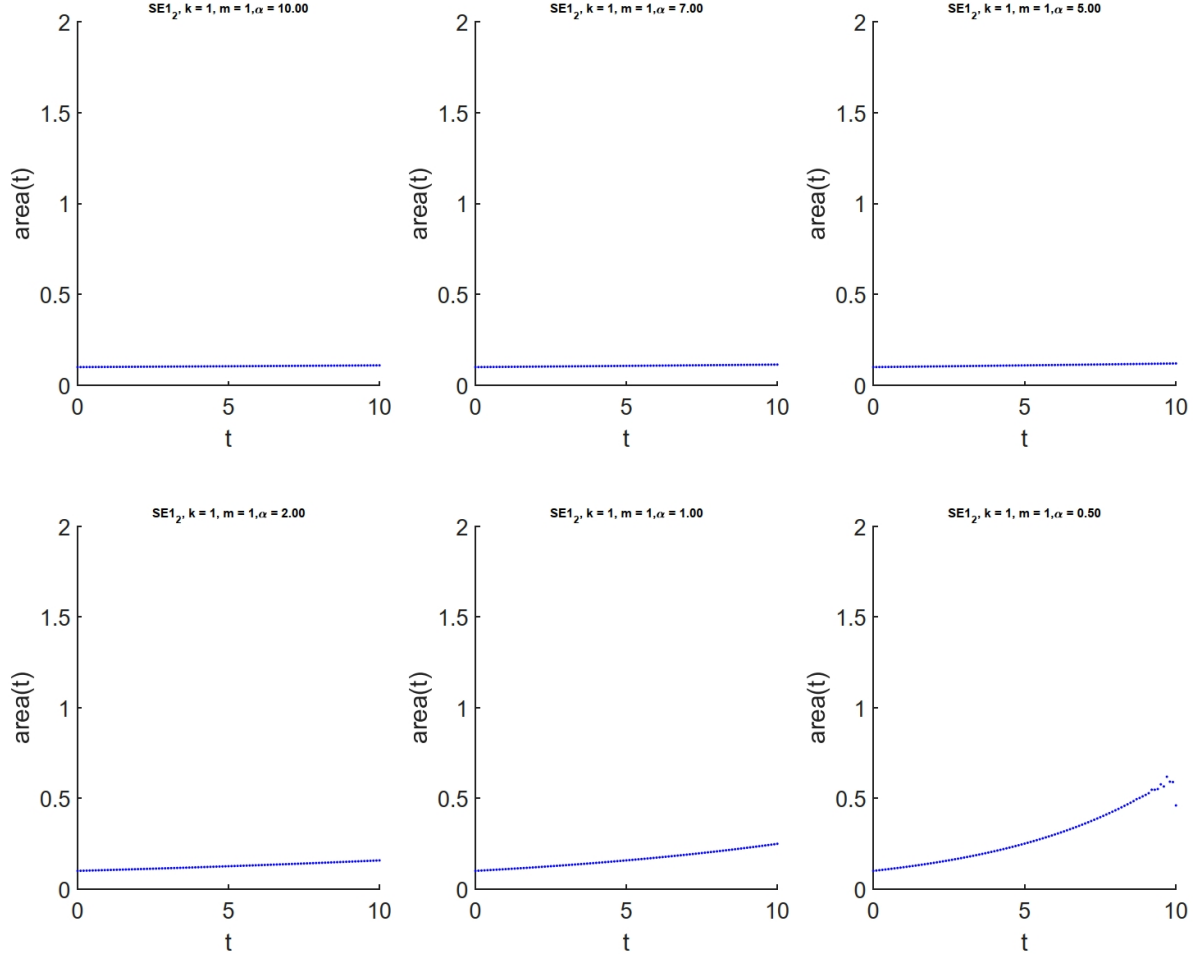


Figure 2: $SE1_2$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

If $H_{yu} = \mathbf{0}$, then $\mathbf{A} = \mathbf{0}$, $\mathbf{A}^T = \mathbf{0}$, $\mathbf{D} = \mathbf{0}$ so we have

$$d\lambda_{n+1} \wedge dy_{n+1} = (\mathbf{I} - \Delta t H_{y\lambda}) d\lambda_n \wedge (\mathbf{I} - \Delta t H_{\lambda y})^{-1} dy_n = \quad (52)$$

$$(\mathbf{I} - \Delta t H_{\lambda y})^{-T} (\mathbf{I} - \Delta t H_{y\lambda}) d\lambda_n \wedge dy_n = \quad (53)$$

$$(\underbrace{\mathbf{I} - \Delta t H_{\lambda y}^T}_{H_{y\lambda} = H_{\lambda y}^T})^{-1} (\mathbf{I} - \Delta t H_{y\lambda}) d\lambda_n \wedge dy_n = \quad (54)$$

$$(\mathbf{I} - \Delta t H_{y\lambda})^{-1} (\mathbf{I} - \Delta t H_{y\lambda}) d\lambda_n \wedge dy_n \quad (55)$$

$$= d\lambda_n \wedge dy_n. \quad (56)$$

$SE2_1$:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \Delta t H_{\lambda}(\mathbf{y}_n, \mathbf{u}_{n+1}, \lambda_{n+1}) \quad (57)$$

$$\begin{aligned} d\mathbf{y}_{n+1} &= d\mathbf{y}_n + \Delta t H_{\lambda y} d\mathbf{y}_n - \Delta t H_{\lambda u} H_{uu}^{-1} H_{uy} d\mathbf{y}_{n+1} - \\ &\quad \Delta t H_{\lambda u} H_{uu}^{-1} H_{u\lambda} d\lambda_{n+1} + \Delta t H_{\lambda\lambda} d\lambda_{n+1} \\ &= (\mathbf{I} + \Delta t H_{\lambda u} H_{uu}^{-1} H_{uy})^{-1} (\mathbf{I} + \Delta t H_{\lambda y}) d\mathbf{y}_n + \\ &\quad (\mathbf{I} + \Delta t H_{\lambda u} H_{uu}^{-1} H_{uy})^{-1} (-\Delta t H_{\lambda u} H_{uu} H_{u\lambda} + \Delta t H_{\lambda\lambda}) d\lambda_{n+1}, \end{aligned} \quad (58)$$

$$\lambda_{n+1} = \lambda_n - \Delta t H_y(\mathbf{y}_n, \mathbf{u}_{n+1}, \lambda_{n+1}) \quad (59)$$

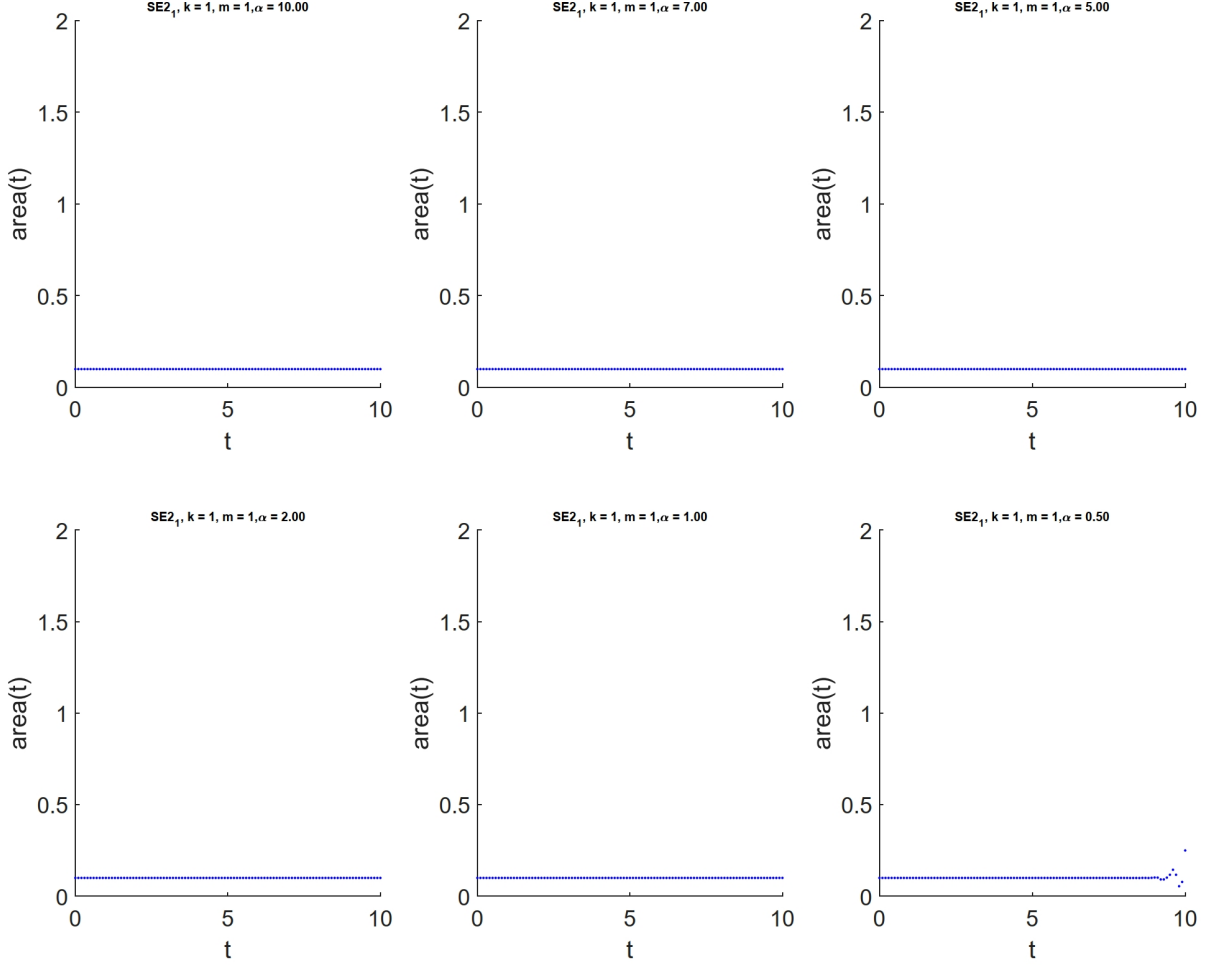


Figure 3: $SE2_1$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

$$\begin{aligned}
d\lambda_{n+1} &= d\lambda_n - \Delta t H_{yy} dy_n + \Delta t H_{yu} H_{uu}^{-1} H_{uy} dy_{n+1} + \\
&\quad \Delta t H_{yu} H_{uu}^{-1} H_{u\lambda} d\lambda_{n+1} - \Delta t H_{y\lambda} d\lambda_{n+1} \\
&= (\mathbf{I} - \Delta t H_{yu} H_{uu}^{-1} H_{u\lambda} + \Delta t H_{y\lambda})^{-1} d\lambda_n - \\
&\quad \Delta t (\mathbf{I} - \Delta t H_{yu} H_{uu}^{-1} H_{u\lambda} + \Delta t H_{y\lambda})^{-1} H_{yy} dy_n + \\
&\quad \Delta t (\mathbf{I} - \Delta t H_{yu} H_{uu}^{-1} H_{u\lambda} + \Delta t H_{y\lambda})^{-1} H_{yu} H_{uu}^{-1} H_{uy} dy_{n+1}.
\end{aligned} \tag{60}$$

If $H_{uy} = \mathbf{0}$, the equations (58) and (60) simplify to

$$dy_{n+1} = (\mathbf{I} + \Delta t H_{\lambda y}) dy_n + (-\Delta t H_{\lambda u} H_{uu}^{-1} H_{u\lambda} + \Delta t H_{\lambda\lambda}) d\lambda_{n+1} \tag{61}$$

$$d\lambda_{n+1} = (\mathbf{I} + \Delta t H_{y\lambda})^{-1} dy_n - \Delta t (\mathbf{I} + \Delta t H_{y\lambda})^{-1} H_{yy} dy_n \tag{62}$$

Taking the wedge product of (61) with $d\lambda_{n+1}$ and substituting equation (62) into the resulting expression, we obtain:

$$\begin{aligned}
dy_{n+1} \wedge d\lambda_{n+1} &= (\mathbf{I} + \Delta t H_{\lambda y}) dy_n \wedge (\mathbf{I} + \Delta t H_{y\lambda})^{-1} d\lambda_n \\
&= \underbrace{(\mathbf{I} + \Delta t H_{y\lambda}^T)^{-1}}_{H_{y\lambda}^T = H_{\lambda y}} (\mathbf{I} + \Delta t H_{\lambda y}) dy_n \wedge d\lambda_n \\
&= dy_n \wedge d\lambda_n.
\end{aligned} \tag{63}$$

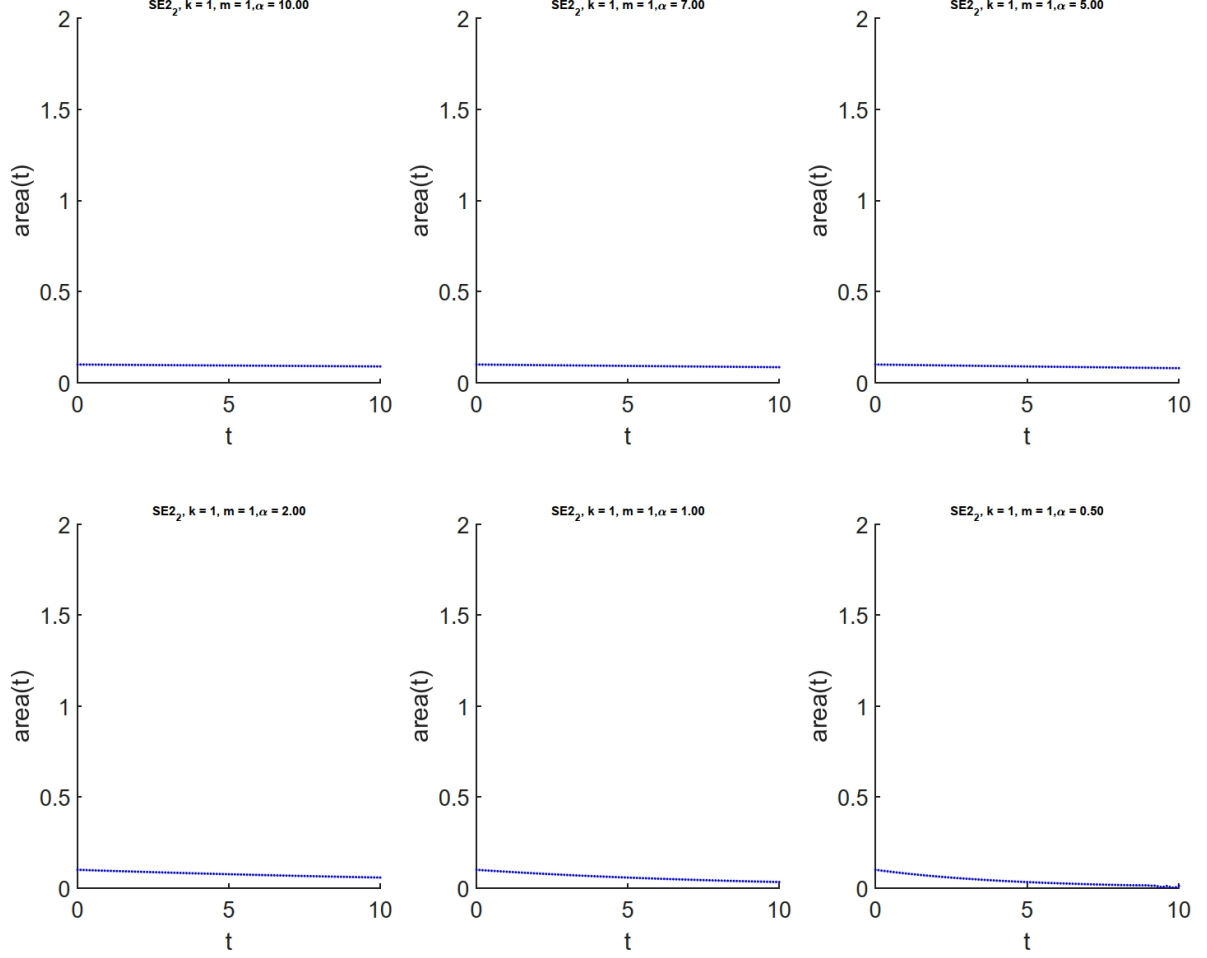


Figure 4: $SE2_2$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

Modified MidPoint: (M.M.P)

$$\begin{aligned}
 \mathbf{y}_{n+1} &= \mathbf{y}_n + \Delta t H_{\lambda}(\mathbf{y}_{n+\frac{1}{2}}, \mathbf{u}(\mathbf{y}_{n+\frac{1}{2}}, \boldsymbol{\lambda}_{n+\frac{1}{2}}), \boldsymbol{\lambda}_{n+\frac{1}{2}}), \\
 d\mathbf{y}_{n+1} &= d\mathbf{y}_n + \frac{\Delta t}{2} H_{\lambda \mathbf{y}}(d\mathbf{y}_{n+1} + d\mathbf{y}_n) + \frac{\Delta t}{2} H_{\lambda \mathbf{u}} d\mathbf{u} + \underbrace{\frac{\Delta t}{2} H_{\lambda \lambda}(d\boldsymbol{\lambda}_{n+1} + d\boldsymbol{\lambda}_n)}_0, \\
 d\mathbf{y}_{n+1} &= d\mathbf{y}_n + \frac{\Delta t}{2} H_{\lambda \mathbf{y}}(d\mathbf{y}_{n+1} + d\mathbf{y}_n) + \frac{\Delta t}{2} H_{\lambda \mathbf{u}}(-H_{\mathbf{u}\mathbf{u}}^{-1} H_{\mathbf{u}\mathbf{y}}(d\mathbf{y}_{n+1} + d\mathbf{y}_n) - H_{\mathbf{u}\mathbf{u}}^{-1} H_{\mathbf{u}\lambda}(d\boldsymbol{\lambda}_{n+1} + d\boldsymbol{\lambda}_n)), \\
 \boxed{d\mathbf{y}_{n+1} &= (\mathbf{I} - \frac{\Delta t}{2} \mathbf{A})^{-1} (\mathbf{I} + \frac{\Delta t}{2} \mathbf{A}) d\mathbf{y}_n + (-\frac{\Delta t}{2}) (\mathbf{I} - \frac{\Delta t}{2} \mathbf{A})^{-1} \mathbf{B} d\boldsymbol{\lambda}_n + (-\frac{\Delta t}{2}) (\mathbf{I} - \frac{\Delta t}{2} \mathbf{A})^{-1} \mathbf{B} d\boldsymbol{\lambda}_{n+1},}
 \end{aligned} \tag{64}$$

where $\mathbf{A} = H_{\lambda \mathbf{y}} - H_{\lambda \mathbf{u}} H_{\mathbf{u}\mathbf{u}}^{-1} H_{\mathbf{u}\mathbf{y}}$, $\mathbf{B} = H_{\lambda \mathbf{u}} H_{\mathbf{u}\mathbf{u}}^{-1} H_{\mathbf{u}\lambda}$.

Applying a similar procedure, we obtain:

$$\boxed{d\boldsymbol{\lambda}_{n+1} = (\mathbf{I} + \frac{\Delta t}{2} \mathbf{A}^T)^{-1} (\mathbf{I} - \frac{\Delta t}{2} \mathbf{A}^T) d\boldsymbol{\lambda}_n + (-\frac{\Delta t}{2}) (\mathbf{I} + \frac{\Delta t}{2} \mathbf{A}^T)^{-1} \mathbf{C} d\mathbf{y}_{n+1} + (-\frac{\Delta t}{2}) (\mathbf{I} + \frac{\Delta t}{2} \mathbf{A}^T)^{-1} \mathbf{C} d\mathbf{y}_n,} \tag{65}$$

where $\mathbf{C} = H_{\mathbf{y}\mathbf{y}} - H_{\mathbf{y}\mathbf{u}} H_{\mathbf{u}\mathbf{u}}^{-1} H_{\mathbf{u}\mathbf{y}}$ and

$$\boldsymbol{\lambda}_{n+1} = \boldsymbol{\lambda}_n - \Delta t H_{\mathbf{y}}(\mathbf{y}_{n+\frac{1}{2}}, \mathbf{u}(\mathbf{y}_{n+\frac{1}{2}}, \boldsymbol{\lambda}_{n+\frac{1}{2}}), \boldsymbol{\lambda}_{n+\frac{1}{2}}).$$

By taking the wedge product of (64) with $d\lambda_{n+1}$ we have:

$$\begin{aligned}
d\mathbf{y}_{n+1} \wedge d\lambda_{n+1} &= (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)d\mathbf{y}_n \wedge d\lambda_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge d\lambda_{n+1} = \\
&\quad (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)d\mathbf{y}_n \wedge (I + \frac{\Delta t}{2}A^T)^{-1}(I - \frac{\Delta t}{2}A^T)d\lambda_n + \\
&\quad (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)d\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_n. \tag{66}
\end{aligned}$$

Using the facts $d\mathbf{a} \wedge d\mathbf{c} = -d\mathbf{c} \wedge d\mathbf{a}$ and $d\mathbf{a} \wedge M d\mathbf{c} = M^T d\mathbf{a} \wedge d\mathbf{c}$, equation (66) can be written as:

$$\begin{aligned}
d\mathbf{y}_{n+1} \wedge d\lambda_{n+1} &= (I - \frac{\Delta t}{2}A)(I + \frac{\Delta t}{2}A)^{-1}(I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)d\mathbf{y}_n \wedge d\lambda_n + \\
&\quad (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)^{-1}d\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n. \tag{67}
\end{aligned}$$

Since $(I - \frac{\Delta t}{2}A)(I + \frac{\Delta t}{2}A) = (I + \frac{\Delta t}{2}A)(I - \frac{\Delta t}{2}A)$, it follows that $(I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)^{-1} = (I + \frac{\Delta t}{2}A)^{-1}(I - \frac{\Delta t}{2}A)^{-1}$, so equation (67) can be rewritten as:

$$\begin{aligned}
d\mathbf{y}_{n+1} \wedge d\lambda_{n+1} &= d\mathbf{y}_n \wedge d\lambda_n + \\
&\quad (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)^{-1}d\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n. \tag{68}
\end{aligned}$$

$$\begin{aligned}
d\mathbf{y}_{n+1} \wedge d\lambda_{n+1} &= \left(I - \frac{\Delta t^2}{4}B^T(I - \frac{\Delta t}{2}A^T)^{-1}(I + \frac{\Delta t}{2}A^T)^{-1}C \right) d\mathbf{y}_n \wedge d\lambda_n + \\
&\quad (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)^{-1}d\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1}. \tag{69}
\end{aligned}$$

By taking the wedge product of $d\mathbf{y}_{n+1}$ with $(-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1}$ we have the following:

$$\begin{aligned}
\underbrace{d\mathbf{y}_{n+1} \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1}}_0 &= (I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)d\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
&\quad (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_{n+1} \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1}. \tag{70}
\end{aligned}$$

Rewriting equation (70) by isolating $(-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_{n+1} \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1}$, we obtain the following equivalent form:

$$\begin{aligned}
(-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} \wedge (-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_{n+1} &= \\
(I - \frac{\Delta t}{2}A)^{-1}(I + \frac{\Delta t}{2}A)d\mathbf{y}_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} + \\
(-\frac{\Delta t}{2})(I - \frac{\Delta t}{2}A)^{-1}Bd\lambda_n \wedge (-\frac{\Delta t}{2})(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1}. \tag{71}
\end{aligned}$$

Upon plugging (71) into (69), we get:

$$\begin{aligned}
d\mathbf{y}_{n+1} \wedge d\lambda_{n+1} &= \left(I - \frac{\Delta t^2}{4}B^T(I - \frac{\Delta t}{2}A^T)^{-1}(I + \frac{\Delta t}{2}A^T)^{-1}C \right) d\mathbf{y}_n \wedge d\lambda_n \\
&\quad + \frac{\Delta t^2}{4}B^T(I - \frac{\Delta t}{2}A^T)^{-1}(I + \frac{\Delta t}{2}A^T)^{-1}Cd\mathbf{y}_{n+1} \wedge d\lambda_{n+1}. \tag{72}
\end{aligned}$$

By solving (72) for $d\mathbf{y}_{n+1} \wedge d\lambda_{n+1}$, we have:

$$\begin{aligned}
\left(I - \frac{\Delta t^2}{4}B^T(I - \frac{\Delta t}{2}A^T)^{-1}(I + \frac{\Delta t}{2}A^T)^{-1}C \right) d\mathbf{y}_{n+1} \wedge d\lambda_{n+1} &= \\
\left(I - \frac{\Delta t^2}{4}B^T(I - \frac{\Delta t}{2}A^T)^{-1}(I + \frac{\Delta t}{2}A^T)^{-1}C \right) d\mathbf{y}_n \wedge d\lambda_n. \tag{73}
\end{aligned}$$

3.1 Numerical results (nonlinear case)

We consider the following nonlinear optimal control problems and apply $SE1_1$, $SE2_1$, and the implicit midpoint method. As analytically proven, the condition $H_{yu} = \mathbf{0}$ in Problem (75) ensures the symplecticity of $SE1_1$ and $SE2_1$, thus guaranteeing the preservation of the phase-space area. This property is numerically validated in Table 2, which compares the absolute relative errors in the preserved area at the initial and final times. In contrast, problem (74) does not satisfy the condition $H_{yu} = \mathbf{0}$, and as shown in Table 2, the $SE1_1$ and $SE2_1$ methods lose their symplectic property in this case, leading to nonpreservation of the phase-space area.

These simulations employ three distinct initial points to construct a triangular shape at the initial time; see Figure 5. To compute the area of this triangle, we split it into four smaller non-overlapping triangles, calculate their individual areas, and add them to determine the total area.

$$\begin{aligned} \min \int_0^2 \frac{1}{2}v^2 + \frac{\alpha}{2}u^2 dt \\ \dot{v} = g + \frac{u}{m} - \frac{k}{m}v^2 \\ v(0) = v_0 \end{aligned} \quad (74)$$

$$\begin{aligned} \min \int_0^2 \frac{1}{2}v^2 + \frac{\alpha}{2}u^2 dt \\ \dot{v} = g + \frac{u}{m} - \frac{k}{m}uv \\ v(0) = v_0 \end{aligned} \quad (75)$$

Necessary optimality conditions for (74) and (75), respectively,

$$\begin{aligned} H(v, u, \lambda) &= \frac{1}{2}v^2 + \frac{\alpha}{2}u^2 + \lambda(g + \frac{1}{m}u - \frac{k}{m}v^2) \\ \begin{cases} \dot{v} = H_\lambda(v, u, \lambda) = g + \frac{u}{m} - \frac{k}{m}v^2 \\ \dot{\lambda} = -H_v(v, u, \lambda) = -v + \frac{2k}{m}\lambda v \\ 0 = H_u(v, u, \lambda) = \alpha u + \frac{1}{m}\lambda \end{cases} \\ H_{vu} &= 0. \end{aligned} \quad (76)$$

$$\begin{aligned} H(v, u, \lambda) &= \frac{1}{2}v^2 + \frac{\alpha}{2}u^2 + \lambda(g + \frac{1}{m}u - \frac{k}{m}uv) \\ \begin{cases} \dot{v} = H_\lambda(v, u, \lambda) = g + \frac{u}{m} - \frac{k}{m}uv \\ \dot{\lambda} = -H_v(v, u, \lambda) = -v + \frac{k}{m}\lambda u \\ 0 = H_u(v, u, \lambda) = \alpha u + \frac{1}{m}\lambda - \frac{k}{m}\lambda v \end{cases} \\ H_{vu} &\neq 0. \end{aligned} \quad (77)$$

$\alpha = 0.5, N.T = 1553$	Δt	$ A_0 - A_T , H_{vu} = 0$	$\frac{ A_0 - A_T }{A_0}, H_{vu} = 0$	$ A_0 - A_T , H_{vu} \neq 0$	$\frac{ A_0 - A_T }{A_0}, H_{vu} \neq 0$
$SE1_1$	0.02	3.0978×10^{-5}	2.478×10^{-2}	1.1293×10^{-3}	0.9032
	0.01	2.2191×10^{-5}	1.7753×10^{-2}	1.1883×10^{-3}	0.9506
$SE2_1$	0.02	1.1769×10^{-5}	9.415×10^{-3}	1.2610×10^{-4}	0.1008
	0.01	1.3989×10^{-5}	1.1191×10^{-2}	1.5809×10^{-4}	0.1264
MidPoint	0.02	1.7307×10^{-5}	1.3845×10^{-2}	2.0385×10^{-4}	0.1630
	0.01	1.7222×10^{-5}	1.3777×10^{-2}	1.9555×10^{-4}	0.1564

Table 2: Application of $SE1_1$, $SE2_1$ and implicit midpoint methods to systems (76) and (77) on the interval $[0, 2]$ with three initial points $\mathbf{z}_1 = (0, 1)^T$, $\mathbf{z}_2 = (0.05, 1)^T$ and $\mathbf{z}_3 = (0, 1.05)^T$, where $k = 1$, $m = 20$, $T = 2$. $A_0 = 0.0013$ and A_T denote the areas of the triangle at time $t = 0$ and $t = T = 2$, respectively.

Method, $\alpha = 0.5$	Δt	$\ A_0^T J A_0 - J\ , H_{uv} \neq \mathbf{0}$
$SE1_1$	0.5	0.0174
	0.2	0.0028
	0.02	2.7721×10^{-5}
$SE2_1$	0.5	0.0166
	0.2	0.0028
	0.02	2.7712×10^{-5}
MidPoint	0.5	0.0079
	0.2	2.7735×10^{-4}
	0.02	1.5254×10^{-7}
Modified $SE1_1$	0.5	2.5931×10^{-8}
	0.2	3.6732×10^{-9}
	0.02	4.9304×10^{-10}
Modified $SE2_1$	0.5	1.0876×10^{-8}
	0.2	3.6732×10^{-9}
	0.02	4.9304×10^{-10}
Modified MidPoint	0.5	2.4509×10^{-8}
	0.2	5.0484×10^{-9}
	0.02	5.1423×10^{-10}

Table 3: Numerical evaluation of symplecticity for $SE1_1$, $SE2_1$, $M.P$, $M.SE1_1$, $M.SE2_1$, and $M.M.P$ methods (applied to system (77)) with different time step sizes (Δt). $k = 1$, $m = 20$. $A_0 = \frac{\partial z_{n+1}}{\partial z_n}(v_n, \lambda_n)$.

4 Nonlinear optimal control problems (Inverted pendulum problem)

Consider the optimal control problem of the inverted pendulum:

$$\begin{aligned} \min_u \quad & \frac{1}{2} \int_0^T \|\mathbf{x} - \mathbf{x}_d\|^2 dt + \frac{\alpha}{2} \int_0^T u^2 dt \\ & m\ddot{\mathbf{x}} + \frac{\partial U}{\partial \mathbf{x}} = \mathbf{0}, \\ & \mathbf{x}(0) = \mathbf{x}_0, \quad \dot{\mathbf{x}}(0) = \mathbf{v}_0, \end{aligned}$$

where

$$\begin{aligned} U &= \frac{1}{2}k(L - l_0)^2 - \mathbf{a}^T \mathbf{x}, \quad L = \|\mathbf{x} - (u, 0)^T\|, \\ \frac{\partial U}{\partial \mathbf{x}} &= k\left(\frac{L-l_0}{L}\right)(\mathbf{x} - (u, 0)^T) - \mathbf{a}. \end{aligned}$$

By introducing an auxiliary variable $\mathbf{v}$ and setting it to $\dot{\mathbf{x}}$, the second-order differential equation is converted into a first-order system of differential equations, so the control problem is formulated as follows:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{v}, u} \quad & J(\mathbf{x}, u) \\ & \dot{\mathbf{x}} - \mathbf{v} = \mathbf{0}, \\ & \dot{\mathbf{v}} + \frac{1}{m} \frac{\partial U}{\partial \mathbf{x}} = \mathbf{0}, \\ & \mathbf{x}(0) = \mathbf{x}_0, \quad \mathbf{v}(0) = \mathbf{v}_0. \end{aligned} \tag{78}$$

To derive the necessary conditions for optimality, we introduce the Hamiltonian function H :

$$H(\mathbf{v}, \mathbf{x}, u, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f(\mathbf{x}, u) + \boldsymbol{\lambda}^T \left(\frac{-1}{m}\right) \frac{\partial U}{\partial \mathbf{x}} + \boldsymbol{\mu}^T \mathbf{v},$$

where $f(\mathbf{x}, u) = r(\mathbf{x}) + q(u)$. By applying Pontryagin's maximum principle to the Hamiltonian function,

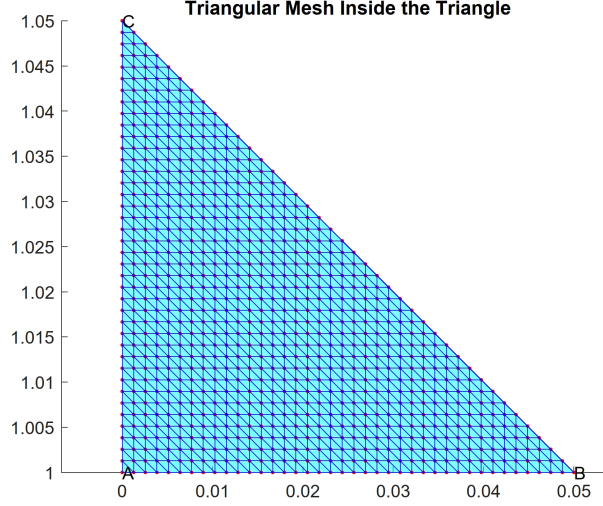


Figure 5: Splitting the triangle with vertices A, B and C into 1553 smaller non-overlapping triangles.

the necessary conditions for optimality are given as follows:

$$\begin{aligned} \text{Adjoint equations: } & \begin{cases} \dot{\lambda} = -\frac{\partial H}{\partial v}, & \lambda(T) = \mathbf{0} \\ \dot{\mu} = -\frac{\partial H}{\partial x}, & \mu(T) = \mathbf{0} \end{cases} \\ \text{Control equation: } & \left\{ \frac{\partial H}{\partial u} = 0 \right. \\ \text{State equations: } & \begin{cases} \dot{v} = \frac{\partial H}{\partial \lambda}, & v(0) = v_0 \\ \dot{x} = \frac{\partial H}{\partial \mu}, & x(0) = x_0 \end{cases} \end{aligned}$$

This system of equations forms a two-point boundary value problem (TPBVP), with boundary conditions specified at both the initial and terminal times.

$$\begin{aligned} & \begin{cases} \dot{\lambda} + \mu = \mathbf{0}, & \lambda(T) = \mathbf{0}, \\ \dot{\mu} + (x - x_d) - \left(\frac{kl_0}{mL^3} (x - (u, 0)^T)(x - (u, 0)^T)^T + \frac{k}{m} \left(\frac{L-l_0}{L} \right) \mathbf{I} \right) \lambda = \mathbf{0}, & \mu(T) = \mathbf{0}, \\ \alpha u - \lambda^T \left(\left[\frac{-kl_0}{mL^3} \mathbf{e}_1^T (x - (u, 0)^T) \right] (x - (u, 0)^T) + \frac{k}{m} \left(\frac{L-l_0}{L} \right) \mathbf{e}_1 \right) = 0, \\ \dot{v} + \frac{k}{m} \left(\frac{L-l_0}{L} \right) (x - (u, 0)^T) - \frac{1}{m} \mathbf{a} = \mathbf{0}, & v(0) = v_0, \\ \dot{x} - v = \mathbf{0}, & x(0) = x_0 \end{cases} \end{aligned} \quad (79)$$

To solve (79), six numerical schemes are employed as follows:

$$\text{Set } \mathbf{y} := (v, x)^T, \quad \mathbf{p} := (\lambda, \mu)^T.$$

$$\begin{aligned} SE1 : & \begin{cases} \frac{\Delta \mathbf{y}}{\Delta t} = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{y}_{n+1}, u_n, \mathbf{p}_n) \\ \frac{\Delta \mathbf{p}}{\Delta t} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}_{n+1}, u_n, \mathbf{p}_n) \\ H_u(\mathbf{y}_n, u_n, \mathbf{p}_n) = 0 \end{cases} & M.SE1 : & \begin{cases} \frac{\Delta \mathbf{y}}{\Delta t} = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{y}_{n+1}, u_n, \mathbf{p}_n) \\ \frac{\Delta \mathbf{p}}{\Delta t} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}_{n+1}, u_n, \mathbf{p}_n) \\ H_u(\mathbf{y}_{n+1}, u_n, \mathbf{p}_n) = 0 \end{cases} \\ SE2 : & \begin{cases} \frac{\Delta \mathbf{y}}{\Delta t} = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{y}_n, u_{n+1}, \mathbf{p}_{n+1}) \\ \frac{\Delta \mathbf{p}}{\Delta t} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}_n, u_{n+1}, \mathbf{p}_{n+1}) \\ H_u(\mathbf{y}_n, u_n, \mathbf{p}_n) = 0 \end{cases} & M.SE2 : & \begin{cases} \frac{\Delta \mathbf{y}}{\Delta t} = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{y}_n, u_n, \mathbf{p}_{n+1}) \\ \frac{\Delta \mathbf{p}}{\Delta t} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}_n, u_n, \mathbf{p}_{n+1}) \\ H_u(\mathbf{y}_n, u_n, \mathbf{p}_{n+1}) = 0 \end{cases} \\ M.P : & \begin{cases} \frac{\Delta \mathbf{y}}{\Delta t} = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{y}_{n+0.5}, u_{n+0.5}, \mathbf{p}_{n+0.5}) \\ \frac{\Delta \mathbf{p}}{\Delta t} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}_{n+0.5}, u_{n+0.5}, \mathbf{p}_{n+0.5}) \\ H_u(\mathbf{y}_n, u_n, \mathbf{p}_n) = 0 \end{cases} & M.M.P : & \begin{cases} \frac{\Delta \mathbf{y}}{\Delta t} = \frac{\partial H}{\partial \mathbf{p}}(\mathbf{y}_{n+0.5}, u_n, \mathbf{p}_{n+0.5}) \\ \frac{\Delta \mathbf{p}}{\Delta t} = -\frac{\partial H}{\partial \mathbf{y}}(\mathbf{y}_{n+0.5}, u_n, \mathbf{p}_{n+0.5}) \\ H_u(\mathbf{y}_{n+0.5}, u_n, \mathbf{p}_{n+0.5}) = 0 \end{cases} \end{aligned} \quad (80)$$

Table 4 presents the numerical results obtained by applying the six numerical schemes for different values of α .

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 0.1$	$\Delta t = 0.025$
SE1	6	6	6	6	8	14	-	-
M. SE1	6	6	6	6	7	15	55	104
SE2	6	6	6	6	77	38	84	113
M. SE2	6	6	6	6	26	28	29	41
MidPoint	6	6	6	6	25	24	67	51
M. MidPoint	6	6	6	6	19	19	29	61

Table 4: Number of iterations (N.I.) for different methods and values of α , with $\Delta t \in \{0.1, 0.025\}$, $k = 1$, $m = 1$, $\mathbf{x}_0 = (1, 1)^T$, $\mathbf{x}_d = (0, 2)^T$, $\mathbf{v}_0 = (0, 0)^T$, $l_0 = \sqrt{2}$, $T \in [0, 10]$.

Table 5: Numerical evaluation of symplecticity for SE1, SE2, M.P, M.SE1, M.SE2, and M.M.P methods with different time step sizes Δt . $\|\mathbf{A}_0^T \mathbf{J} \mathbf{A}_0 - \mathbf{J}\|$, where $\mathbf{A}_0 = \frac{\partial(\mathbf{y}_{n+1}, \mathbf{p}_{n+1})}{\partial(\mathbf{y}_n, \mathbf{p}_n)}$.

Method	$\alpha = 10.0$			$\alpha = 5.0$			$\alpha = 1.0$		
	$\Delta t = 1$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 1$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 1$	$\Delta t = 0.1$	$\Delta t = 0.025$
SE1	0.0614853	0.0030918	1.4210×10^{-4}	0.1148578	0.0031728	1.0397×10^{-4}	0.3654304	0.0045715	1.7756×10^{-4}
M. SE1	3.7994×10^{-10}	4.7898×10^{-10}	6.8444×10^{-10}	3.8148×10^{-10}	3.9087×10^{-10}	6.7330×10^{-10}	3.9091×10^{-10}	4.9181×10^{-10}	4.0767×10^{-10}
SE2	81.5165	5.5104×10^{-4}	8.2952×10^{-5}	29.9389	0.0021965	3.2923×10^{-5}	25.2454	0.1528280	5.3776×10^{-5}
M. SE2	4.1198×10^{-8}	1.2966×10^{-9}	5.8667×10^{-10}	4.3593×10^{-8}	6.6569×10^{-10}	4.4809×10^{-10}	1.9479×10^{-8}	5.4068×10^{-10}	7.0251×10^{-10}
MidPoint	0.1192	0.0020301	1.3219×10^{-4}	0.1403	0.0026914	1.7456×10^{-4}	0.4556	0.0121969	7.6003×10^{-4}
M. MidPoint	1.3738×10^{-9}	4.6400×10^{-10}	7.4220×10^{-10}	1.0867×10^{-9}	5.4220×10^{-10}	5.9989×10^{-10}	6.6610×10^{-10}	6.0072×10^{-10}	7.4891×10^{-10}

5 Algorithm Comparison

In this section, we compare the baseline algorithms (SE1, SE2, MidPoint) with their modified counterparts. The comparison is illustrated using three types of plots:

- **Iteration plots:** visualize convergence behavior over a grid of initial states $\mathbf{x}_0$. For each tested $\mathbf{x}_0$, the color encodes the number of iterations required to converge (up to a fixed maximum). If the method does not converge within the maximum number of iterations (30 in our experiments), the corresponding point is colored black. Additionally, an arrow is drawn from each $\mathbf{x}_0$ to the final state $\mathbf{x}_N$, indicating the achieved terminal state.
- **Control plots:** show the computed control sequence $u(t_i)$ (or u_i) over the discrete time index t_i , where $i \in \{0, \dots, N\}$.
- **Trajectory plots:** show the path of the system in state space as it evolves from $\mathbf{x}_0$ to $\mathbf{x}_N$. Unlike the iteration plot arrows, these plots display the full trajectory, not only the start and end points.

Before presenting the full set of results, we first explain these plots using a representative example.

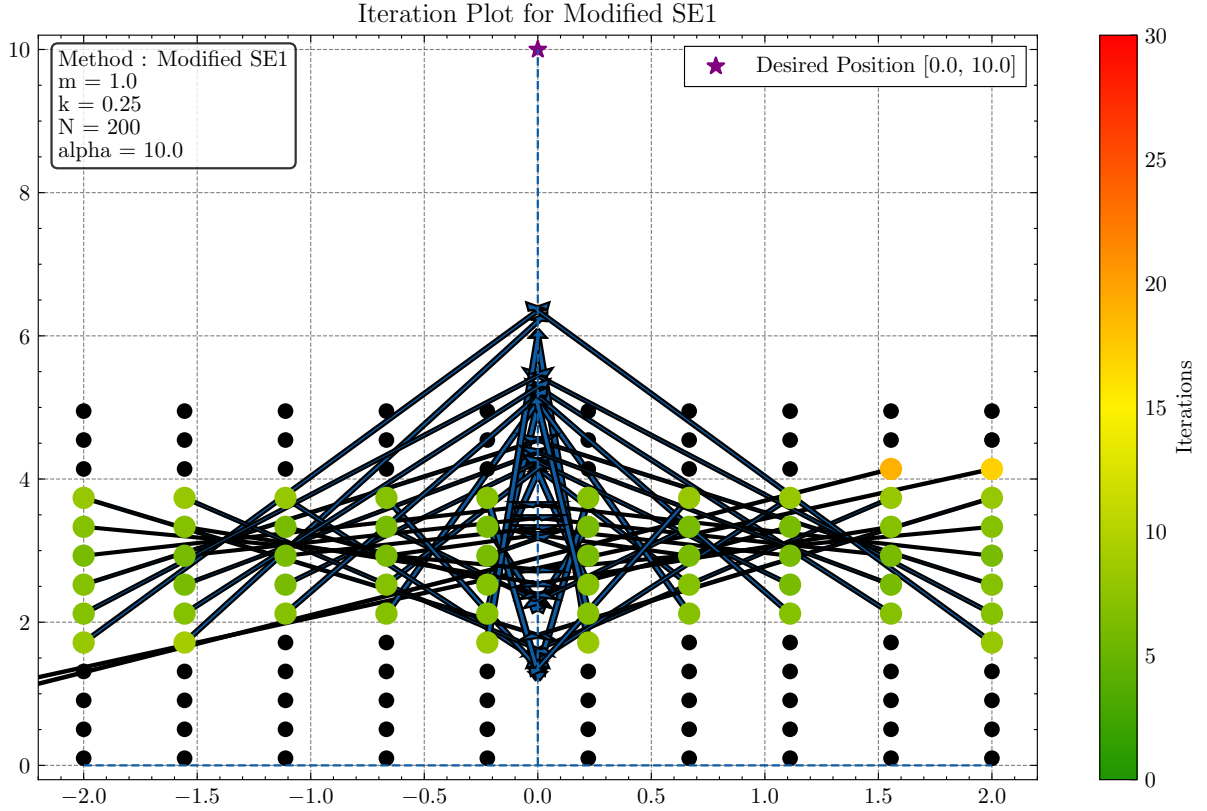


Figure 6: Iteration plot of Modified SE1 for $k = 0.25$, $h = 10.0$, $N = 200$, and $m = 1.0$.

Figure 6 shows an iteration plot for the *Modified SE1* method with $k = 0.25$, $h = 10.0$, $N = 200$, $m = 1.0$, and desired terminal state $\mathbf{x}_d = \begin{bmatrix} 0.00 \\ 10.00 \end{bmatrix}$. Each dot corresponds to a different initial condition $\mathbf{x}_0$ in the sampled grid. The color indicates the number of iterations needed to converge; points that do not converge within the allowed maximum are colored black. Finally, each arrow maps $\mathbf{x}_0$ to the achieved terminal state $\mathbf{x}_N$. Note that these arrows do *not* represent the full trajectory through the state space—only the start and end points. The full state evolution is shown in the trajectory plot.

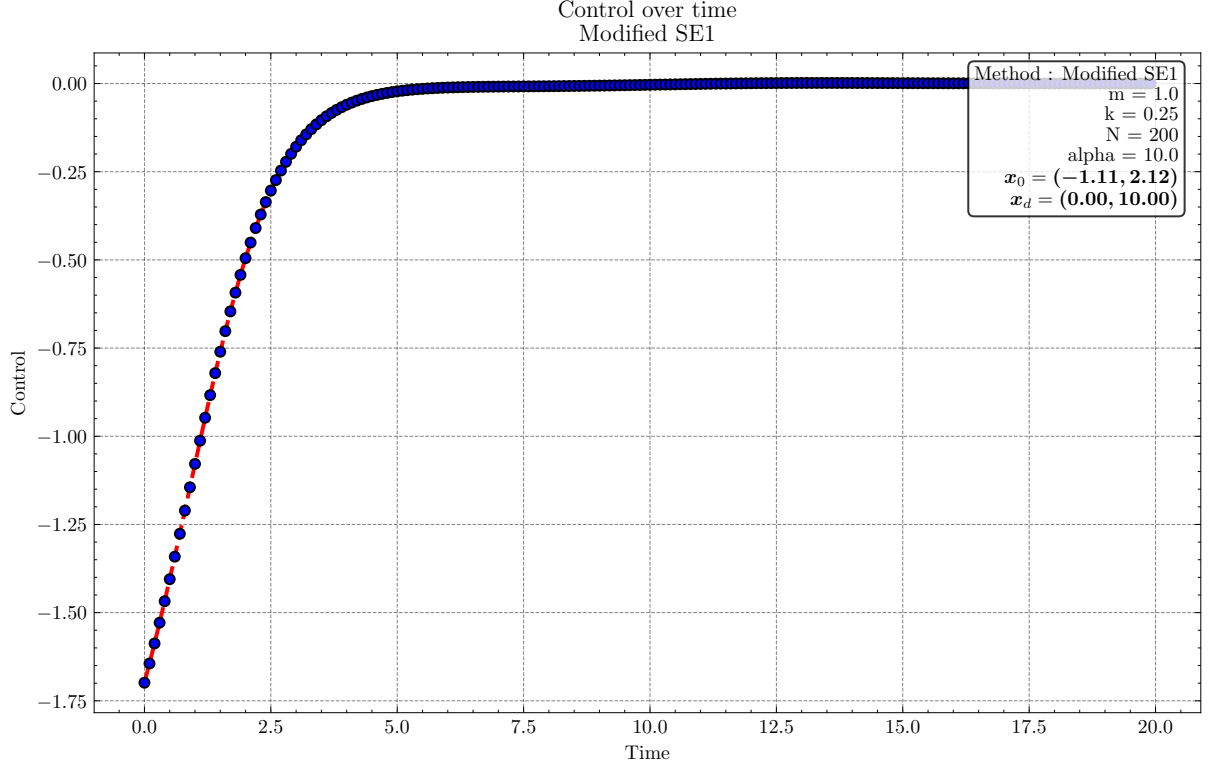


Figure 7: Control plot of Modified SE1 for $k = 0.25$, $h = 10.0$, $N = 200$, $m = 1.0$, and $\mathbf{x}_0 = \begin{bmatrix} -1.11 \\ 2.12 \end{bmatrix}$.

Figure 7 shows the computed control sequence for Modified SE1 with $k = 0.25$, $h = 10.0$, $N = 200$, $m = 1.0$, desired state $\mathbf{x}_d = \begin{bmatrix} 0.00 \\ 10.00 \end{bmatrix}$, and initial condition $\mathbf{x}_0 = \begin{bmatrix} -1.11 \\ 2.12 \end{bmatrix}$. For modified methods, the horizontal axis corresponds to discrete time samples t_i , where the exact interpretation depends on the modification. In practice, the sampling approaches the uniform grid as $N \rightarrow \infty$, i.e.,

$$t_i \rightarrow t_0 + \frac{T - t_0}{N} i.$$

For the unmodified methods, the grid is explicitly uniform:

$$t_i = t_0 + \frac{T - t_0}{N} i, \quad i \in \{0, \dots, N\}.$$

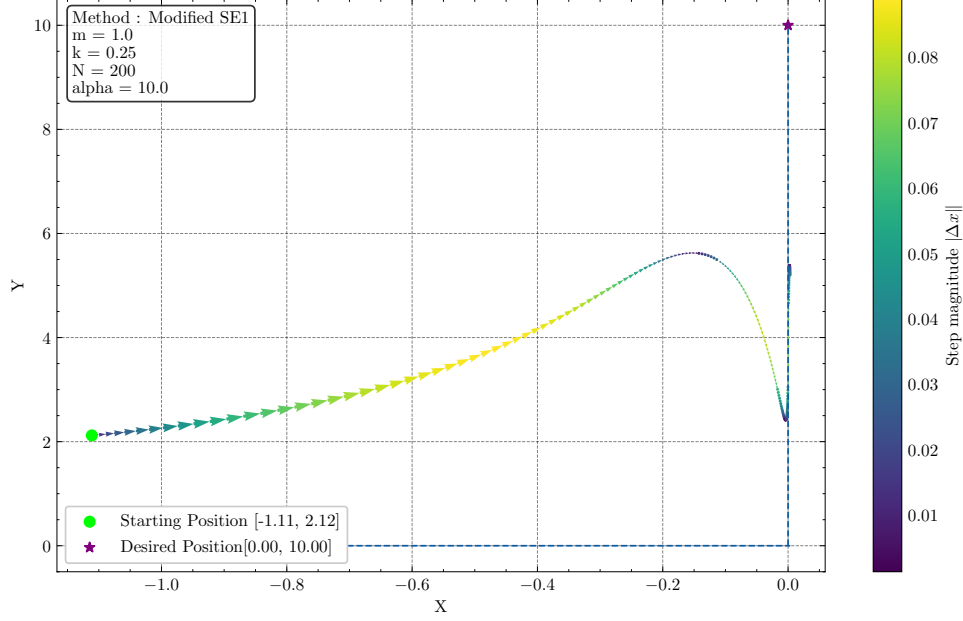


Figure 8: Trajectory plot of Modified SE1 for $k = 0.25$, $h = 10.0$, $N = 200$, $m = 1.0$, and $\mathbf{x}_0 = \begin{bmatrix} -1.11 \\ 2.12 \end{bmatrix}$.

Figure 8 shows the resulting trajectory in state space for the same configuration as in Figure 7. In particular, the plot visualizes the evolution of the state from $\mathbf{x}_0 = \begin{bmatrix} -1.11 \\ 2.12 \end{bmatrix}$ to the final state $\mathbf{x}_N$, which corresponds to the endpoint indicated in the iteration plot (Figure 6).

Finally, note that because the number of generated examples is large, we only include representative cases in which the difference between baseline and modified trajectories is pronounced (i.e., where the trajectory discrepancy is high). Moreover, when the desired height component of $\mathbf{x}_d$ is small, the converged solutions often correspond to trajectories in which the pendulum falls. Empirically, such cases can exhibit large differences in the projection plots even when the terminal states $\mathbf{x}_N$ are close.

5.1 SE1

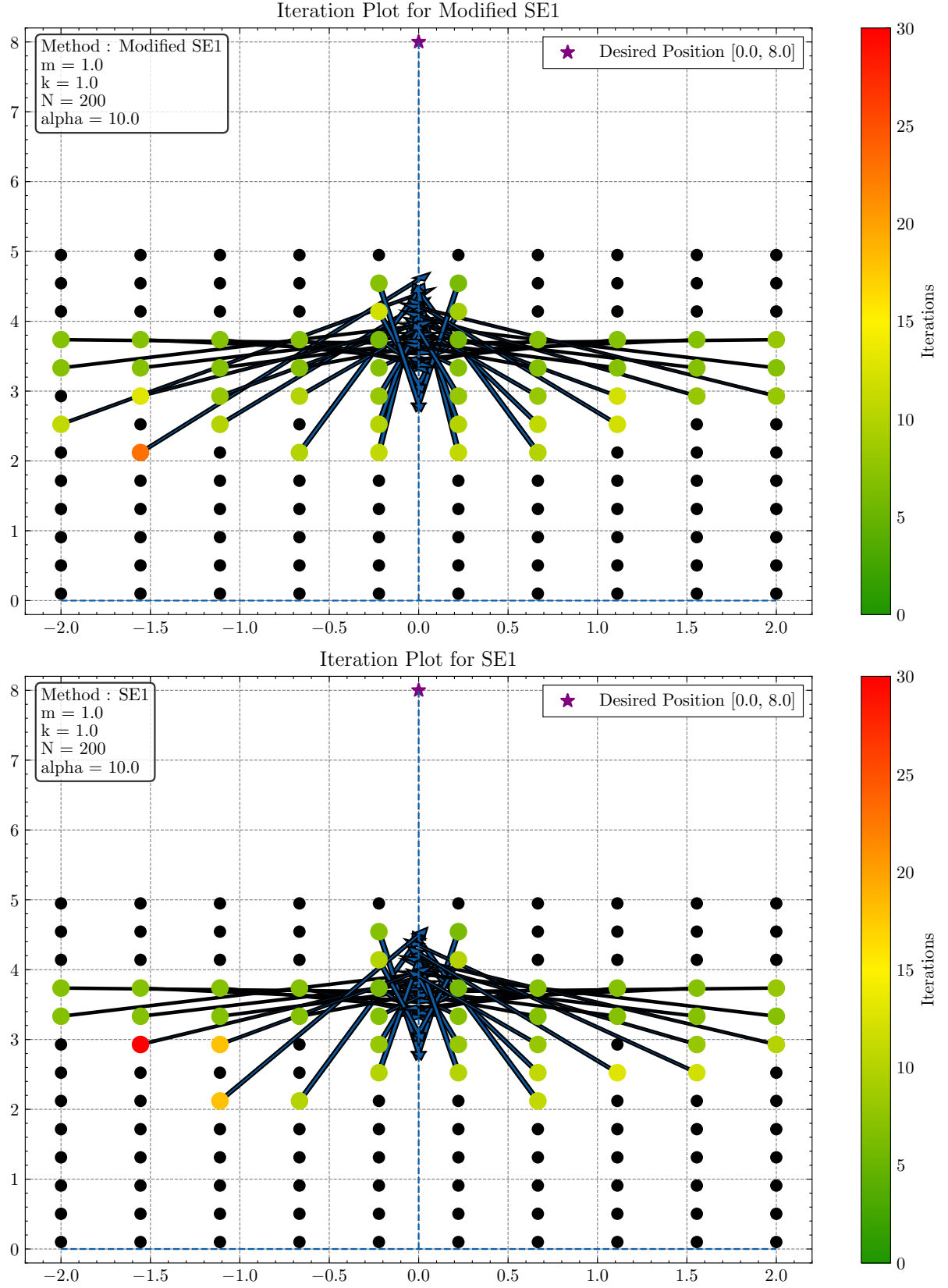
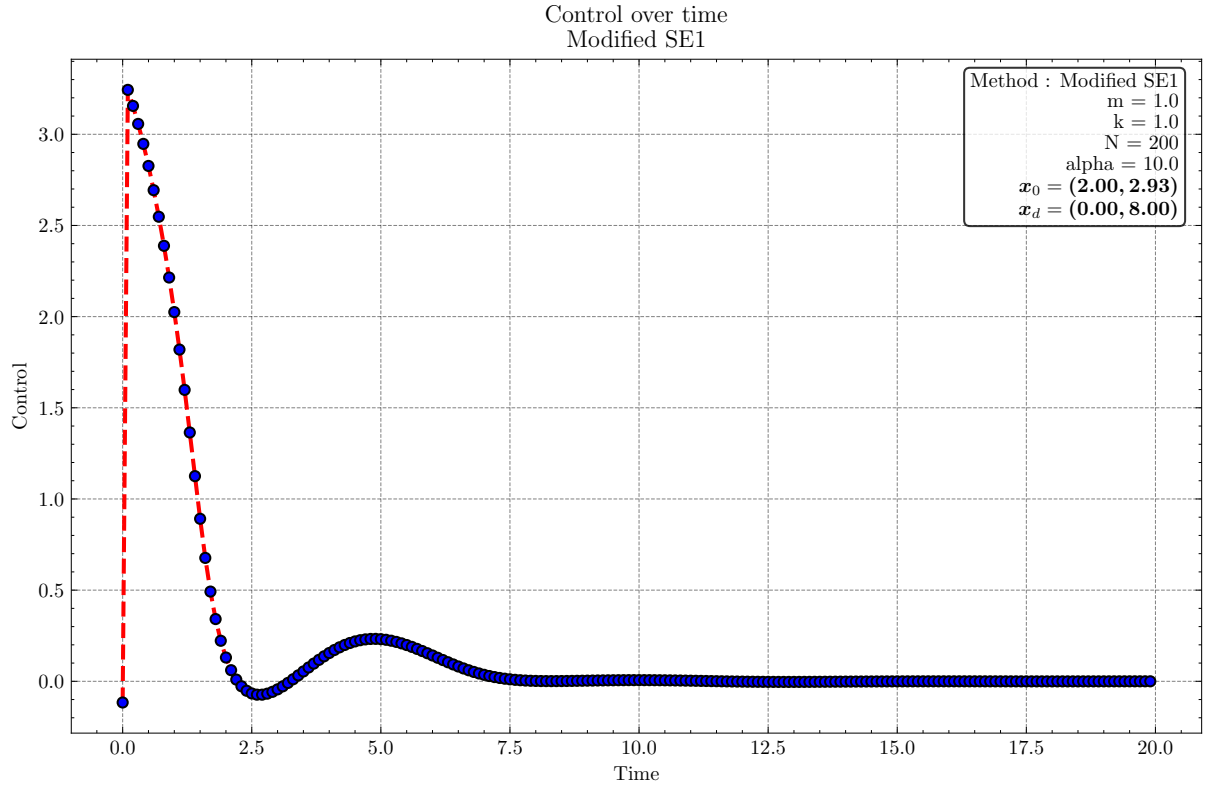
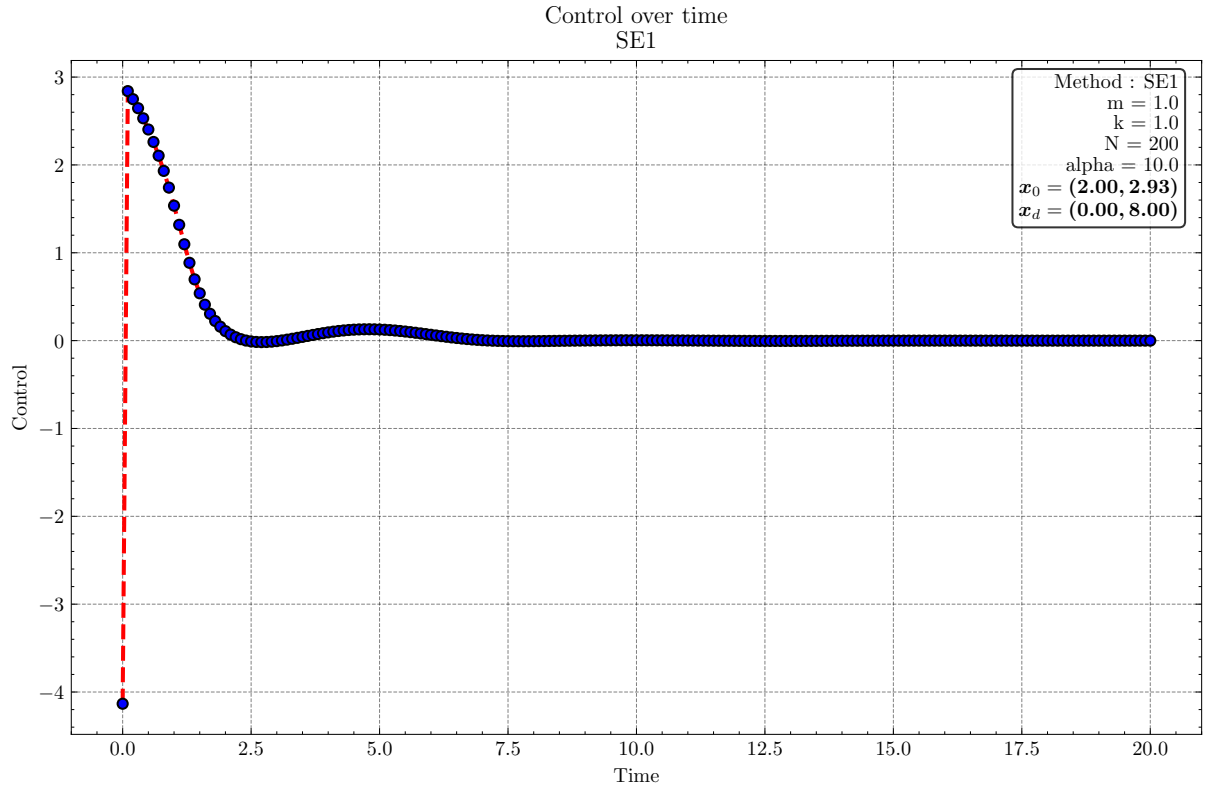


Figure 9: Iteration plots for SE1 and its modified variant with $k = 1.0$, $h = 8.0$, $N = 200$, $m = 1.0$.



(a) Modified SE1



(b) SE1

Figure 10: Control trajectories for $\mathbf{x}_0 = (2.00, 2.93)$ with $k = 1.0$, $h = 8.0$, $N = 200$, $m = 1.0$.

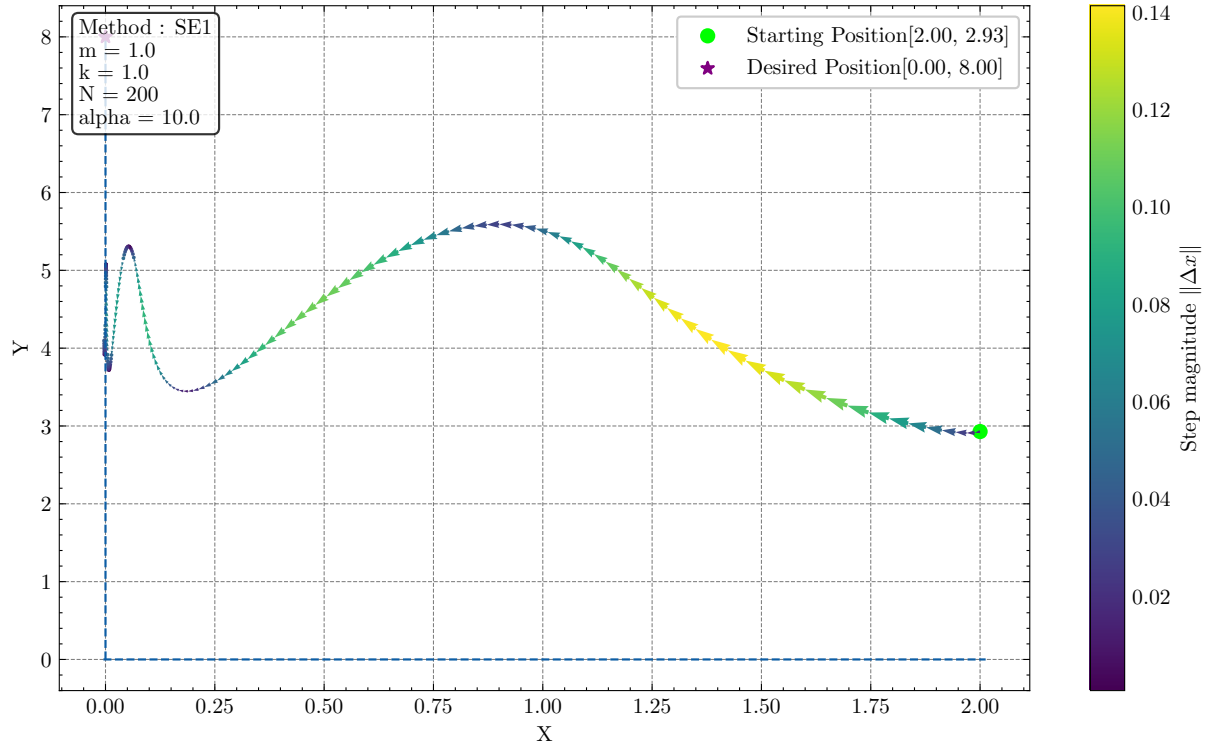
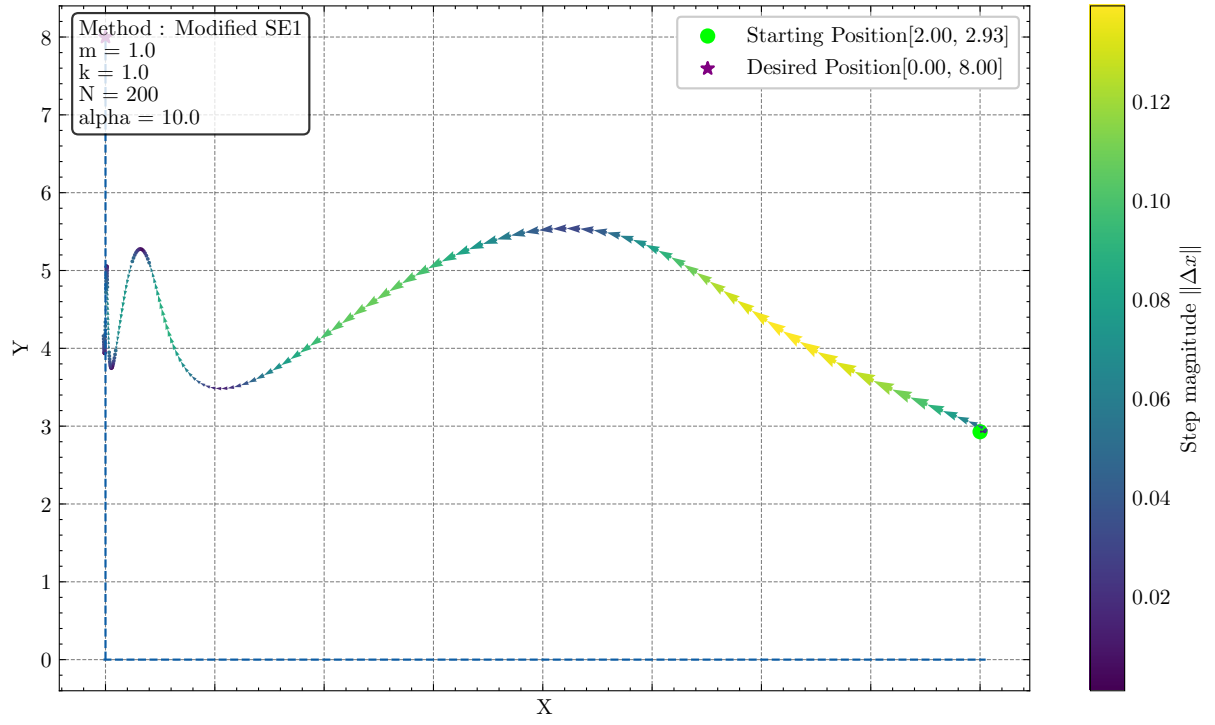
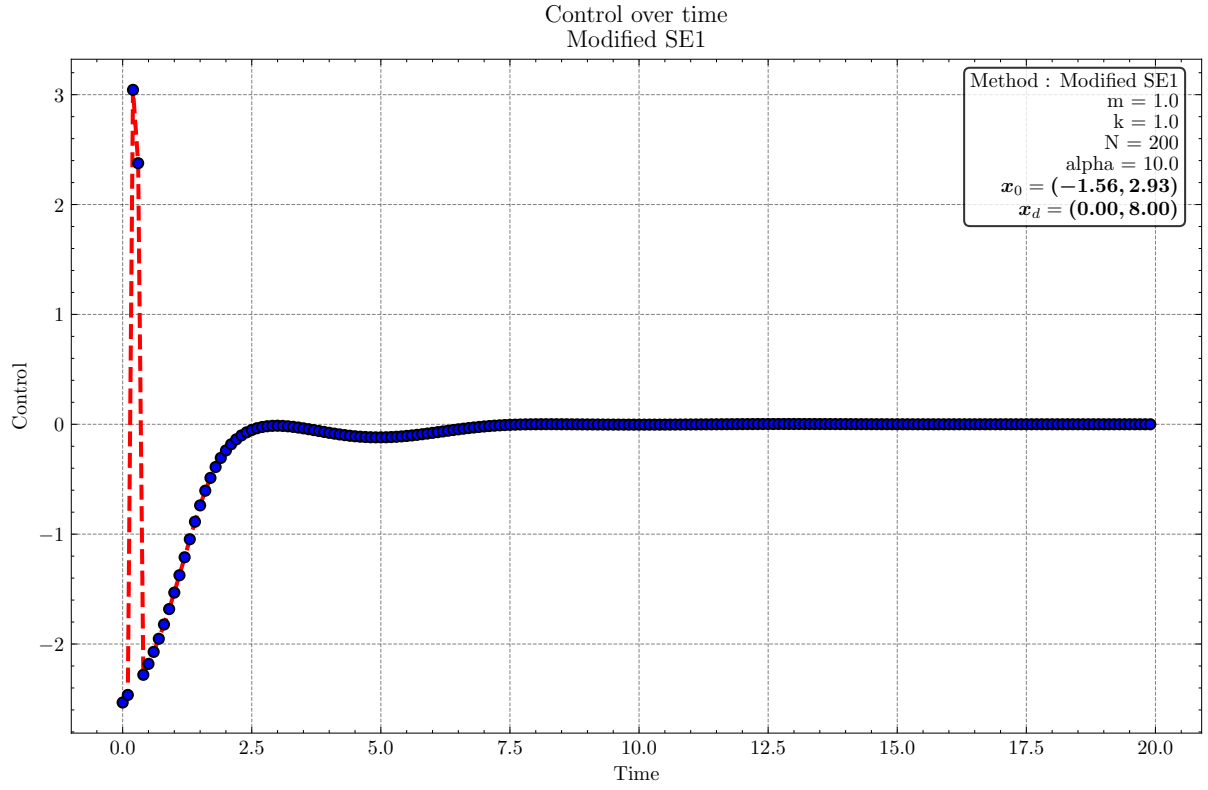
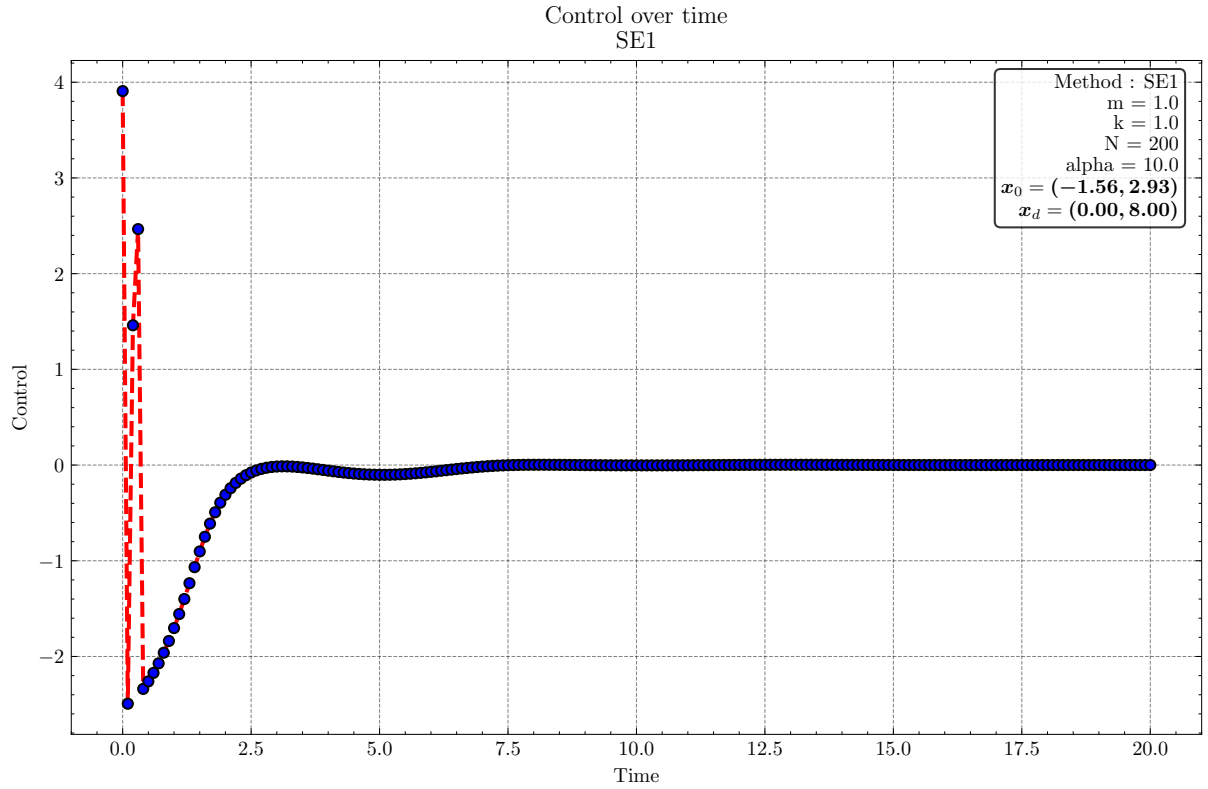


Figure 11: Trajectory comparison for $\mathbf{x}_0 = (2.00, 2.93)$ with $k = 1.0$, $h = 8.0$, $N = 200$, $m = 1.0$.



(a) Modified SE1



(b) SE1

Figure 12: Control trajectories for $\mathbf{x}_0 = (-1.56, 2.93)$ with $k = 1.0$, $h = 8.0$, $N = 200$, $m = 1.0$.

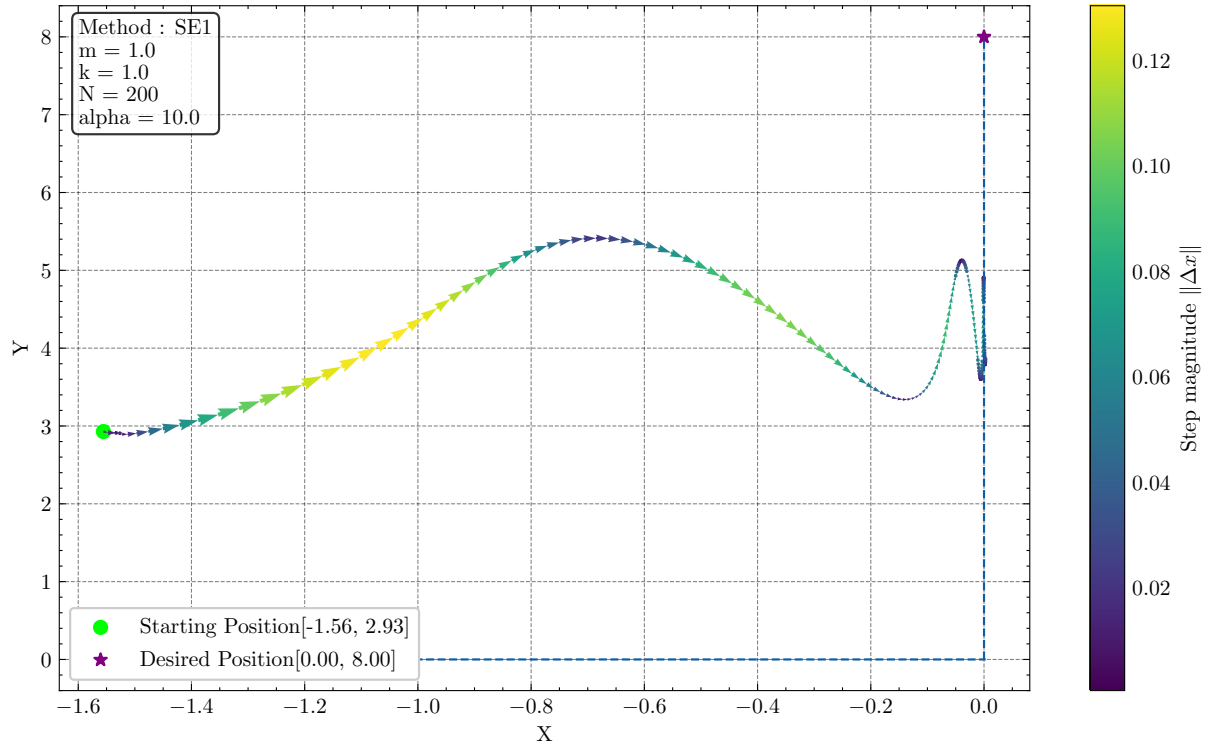
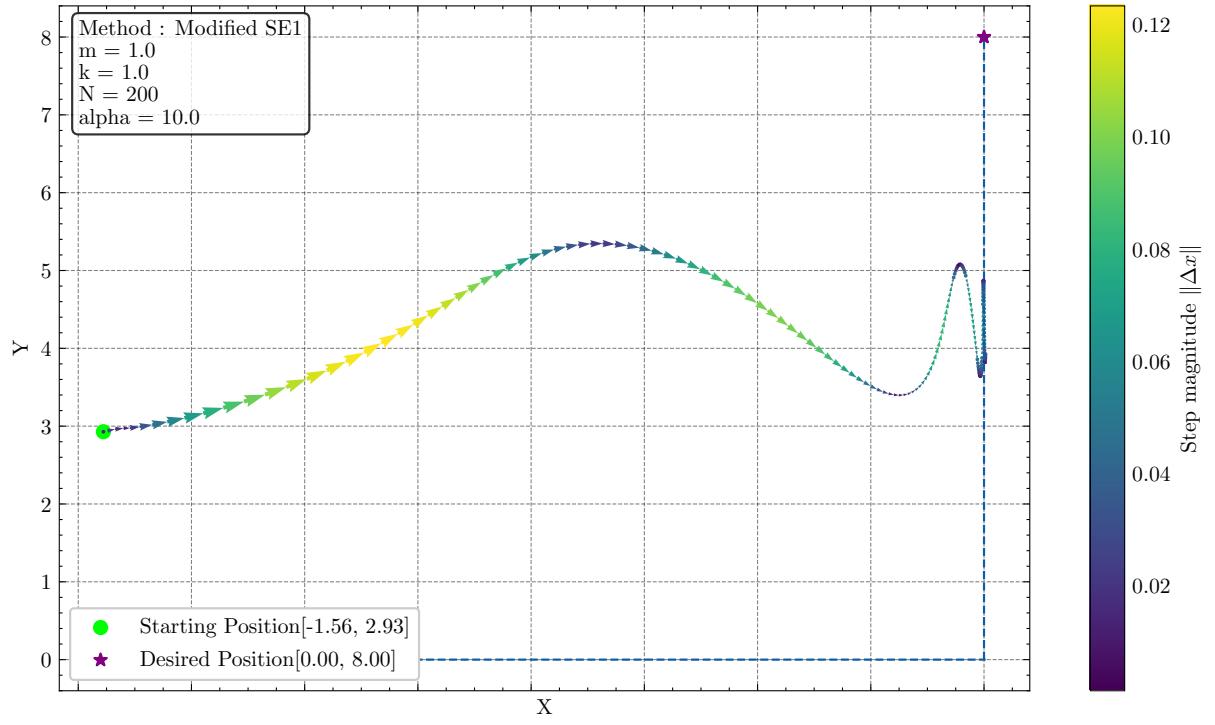


Figure 13: Trajectory comparison for $\mathbf{x}_0 = (-1.56, 2.93)$ with $k = 1.0$, $h = 8.0$, $N = 200$, $m = 1.0$.

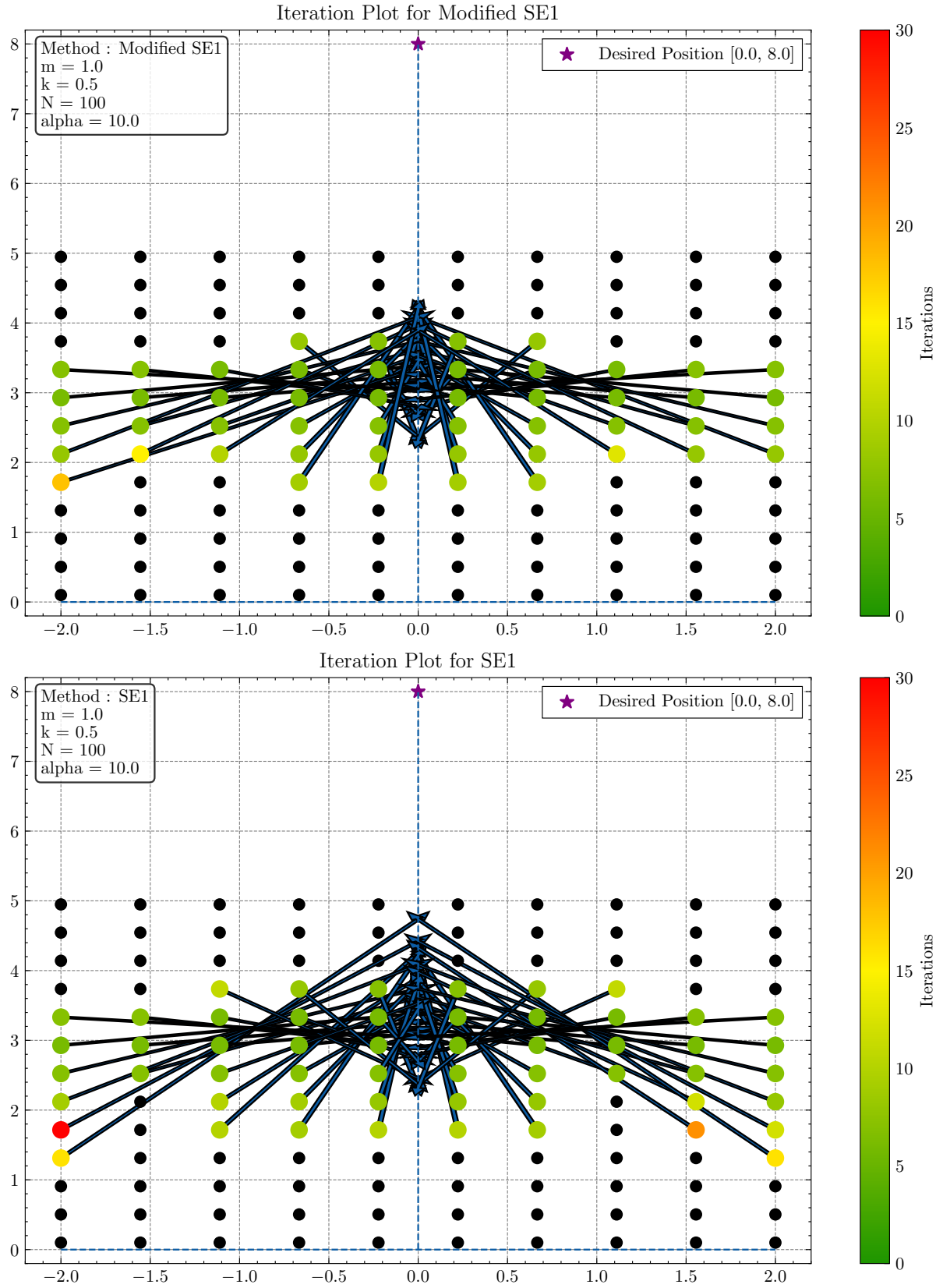
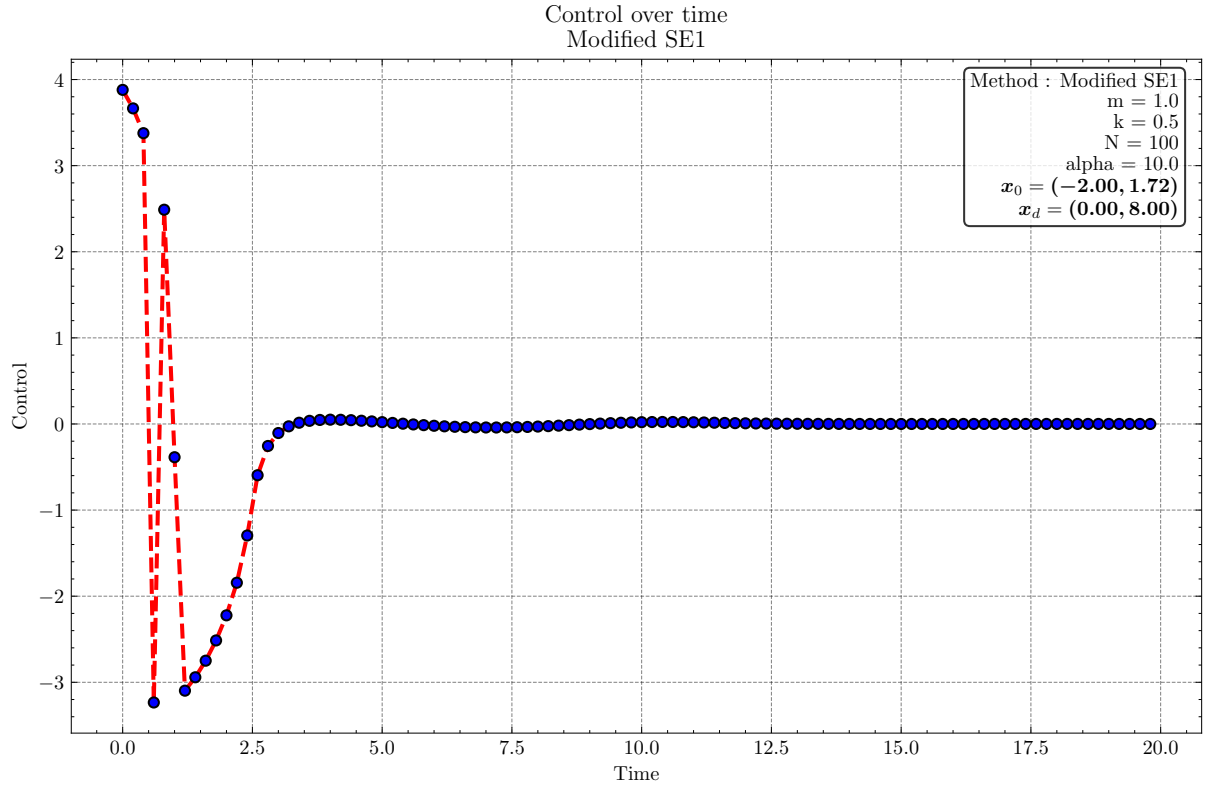
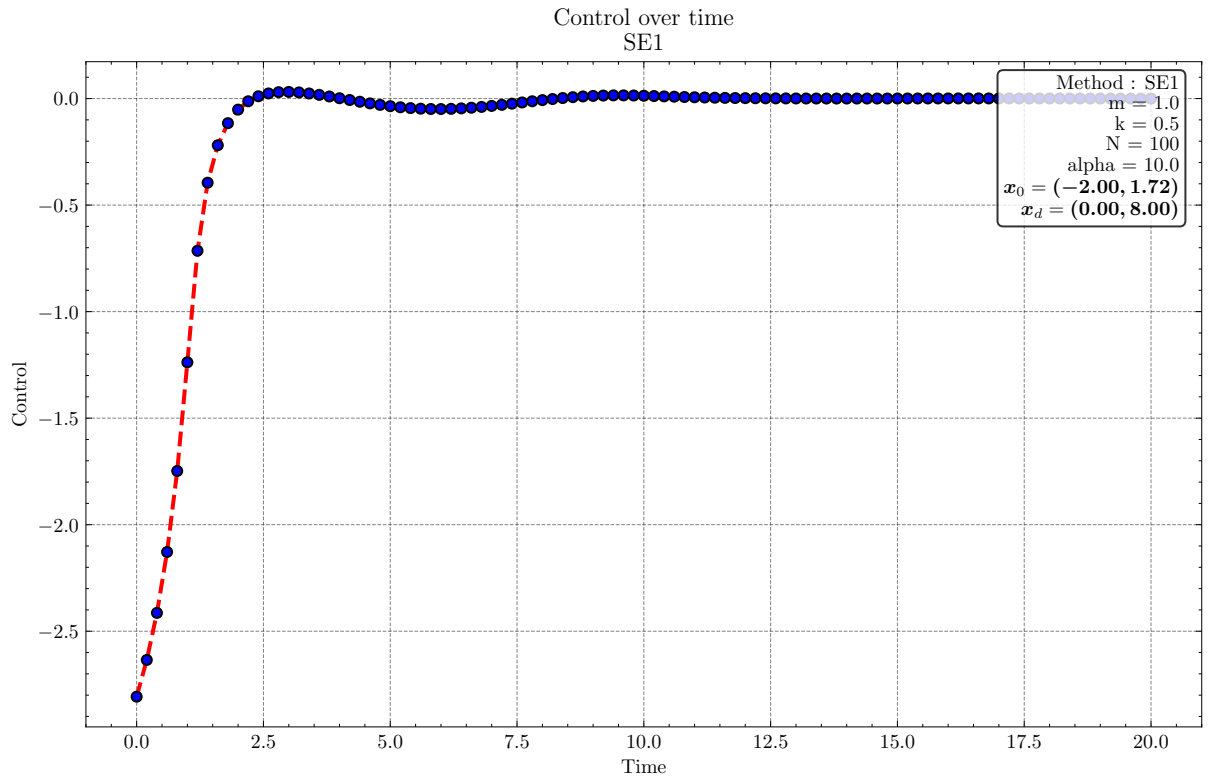


Figure 14: Iteration plots for SE1 and its modified variant with $k = 0.5$, $h = 8.0$, $N = 100$, $m = 1.0$.



(a) Modified SE1



(b) SE1

Figure 15: Control trajectories for $\mathbf{x}_0 = (-2.00, 1.72)$ with $k = 0.5$, $h = 8.0$, $N = 100$, $m = 1.0$.

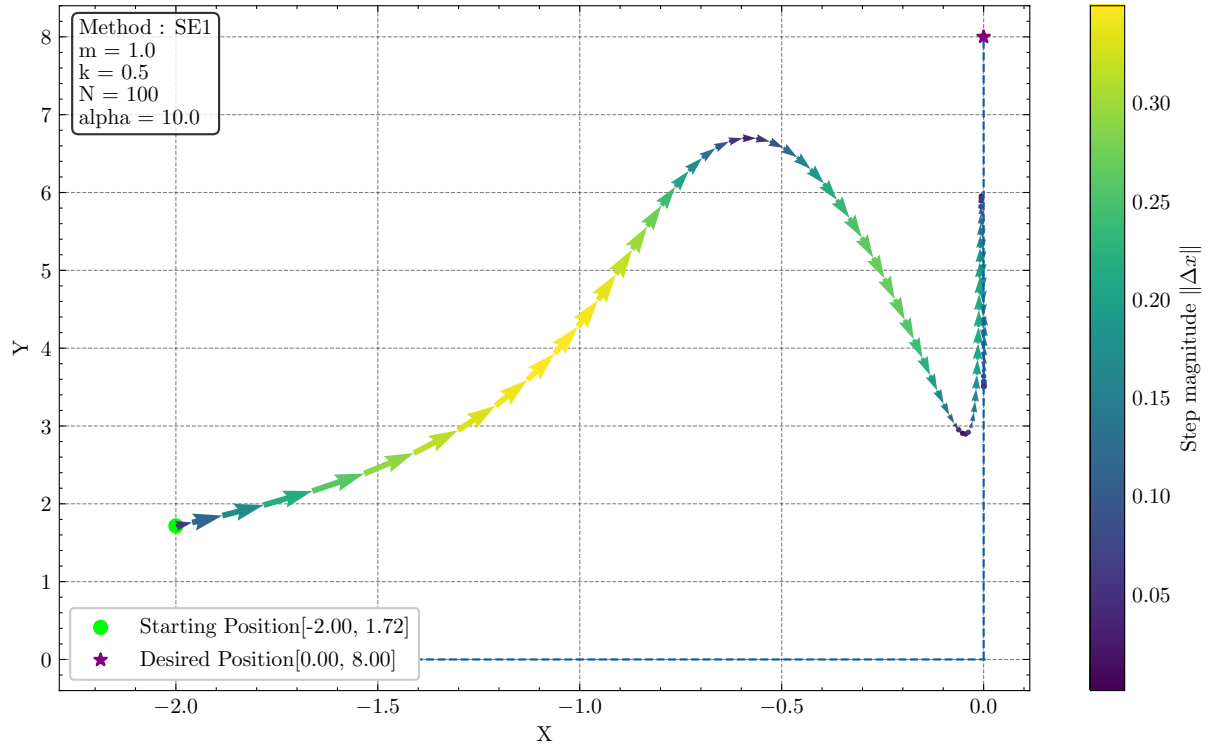
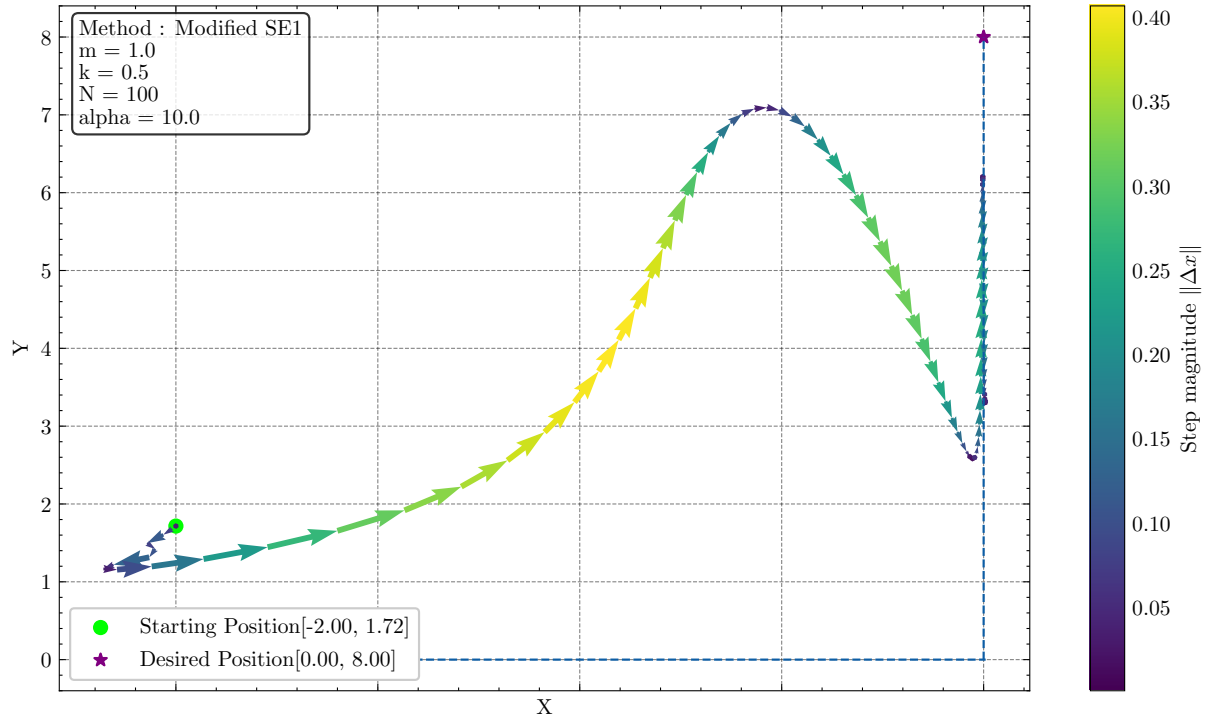


Figure 16: Trajectory comparison for $\mathbf{x}_0 = (-2.00, 1.72)$ with $k = 0.5$, $h = 8.0$, $N = 100$, $m = 1.0$.

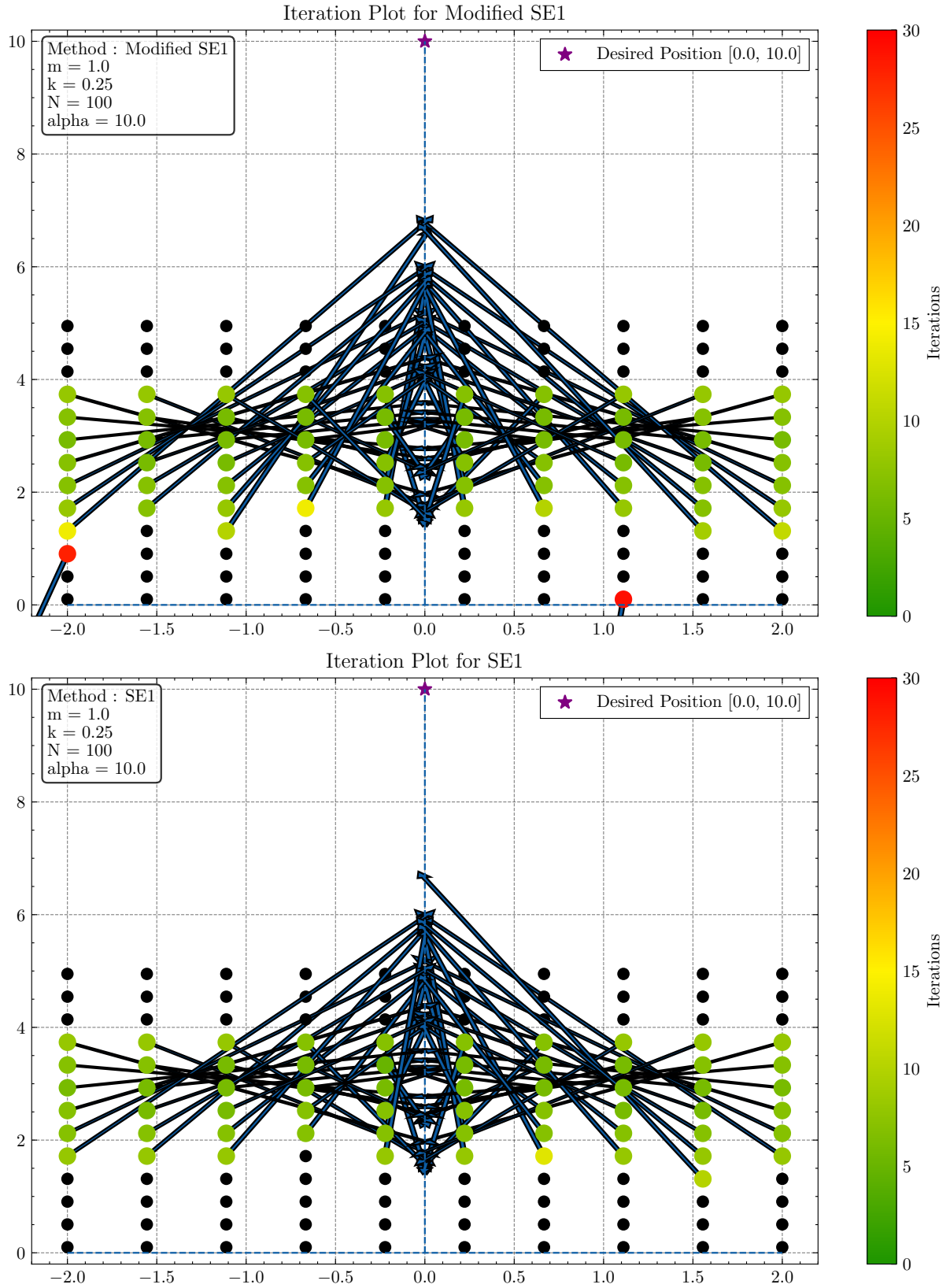
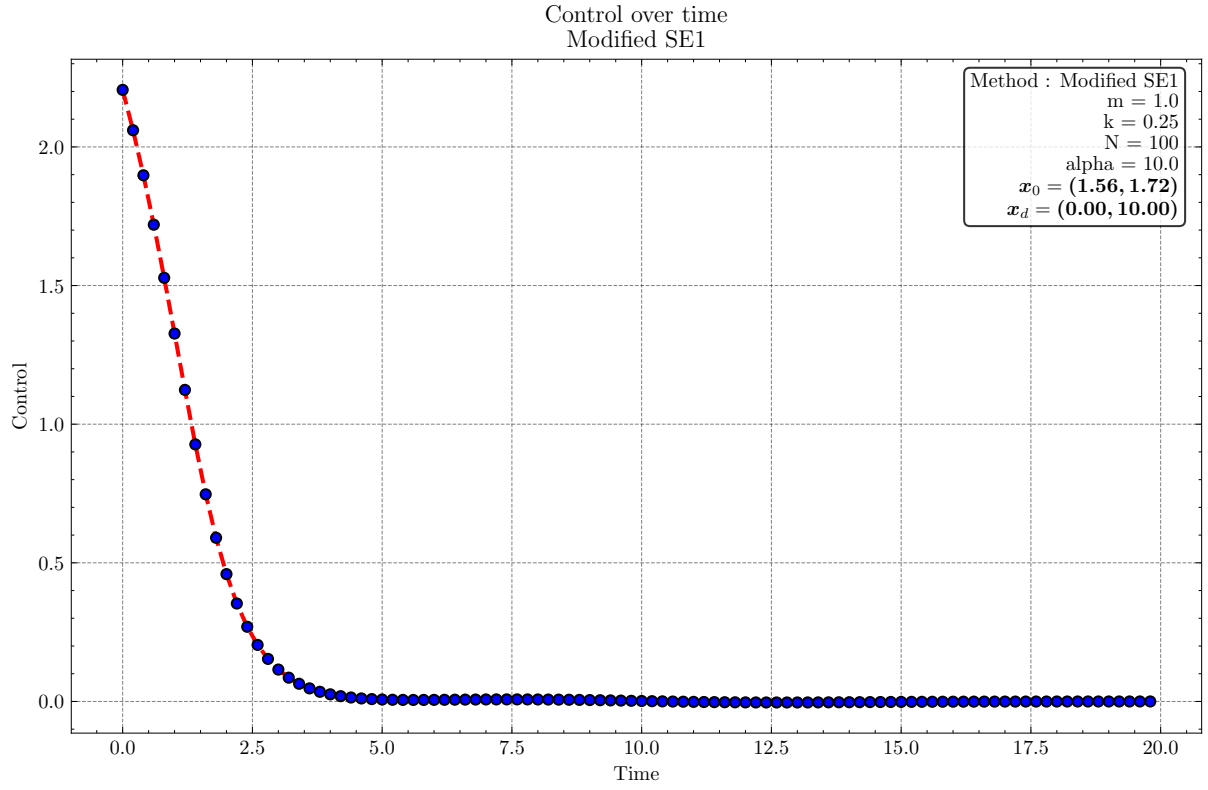
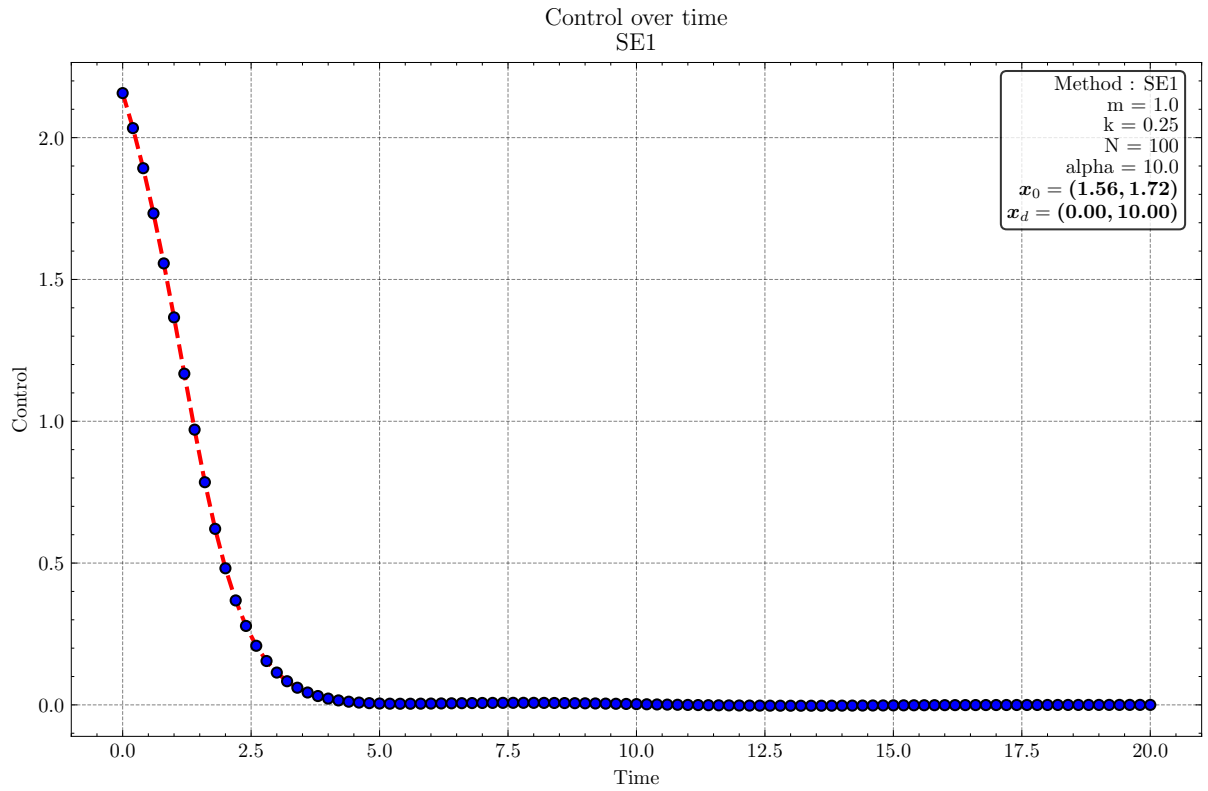


Figure 17: Iteration plots for SE1 and its modified variant with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.



(a) Modified SE1



(b) SE1

Figure 18: Control trajectories for $\mathbf{x}_0 = (1.56, 1.72)$ with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.

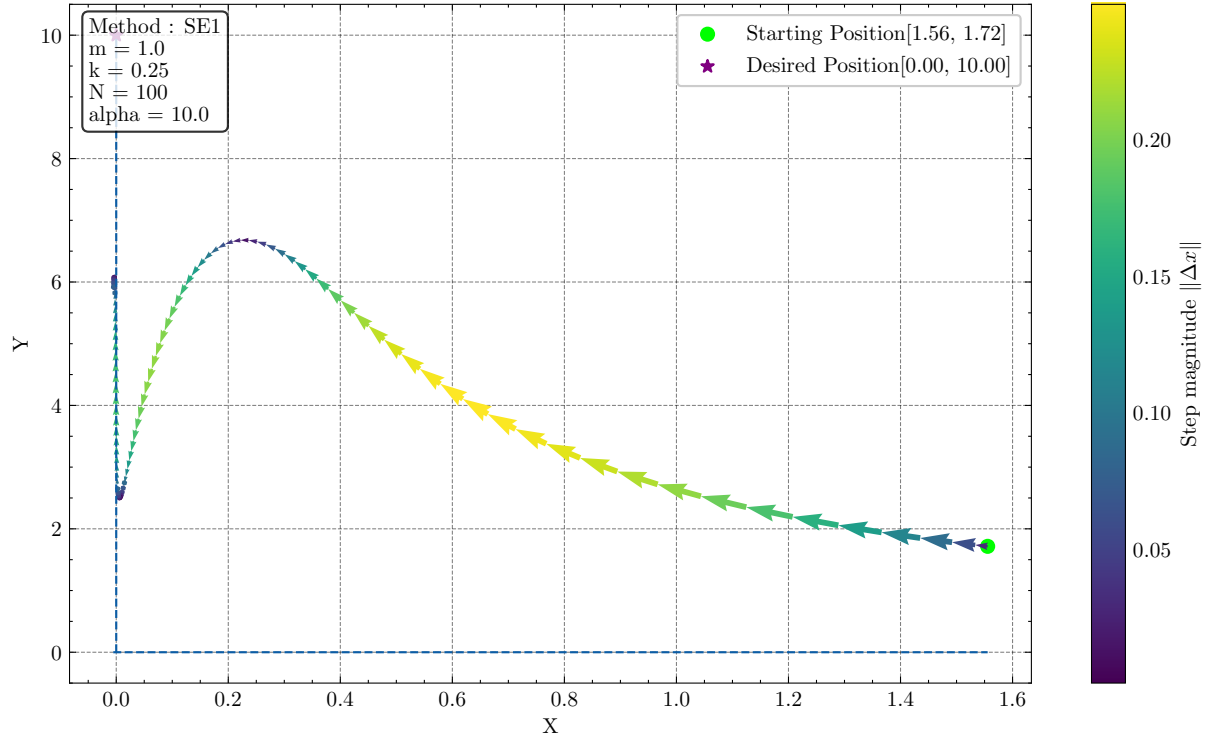
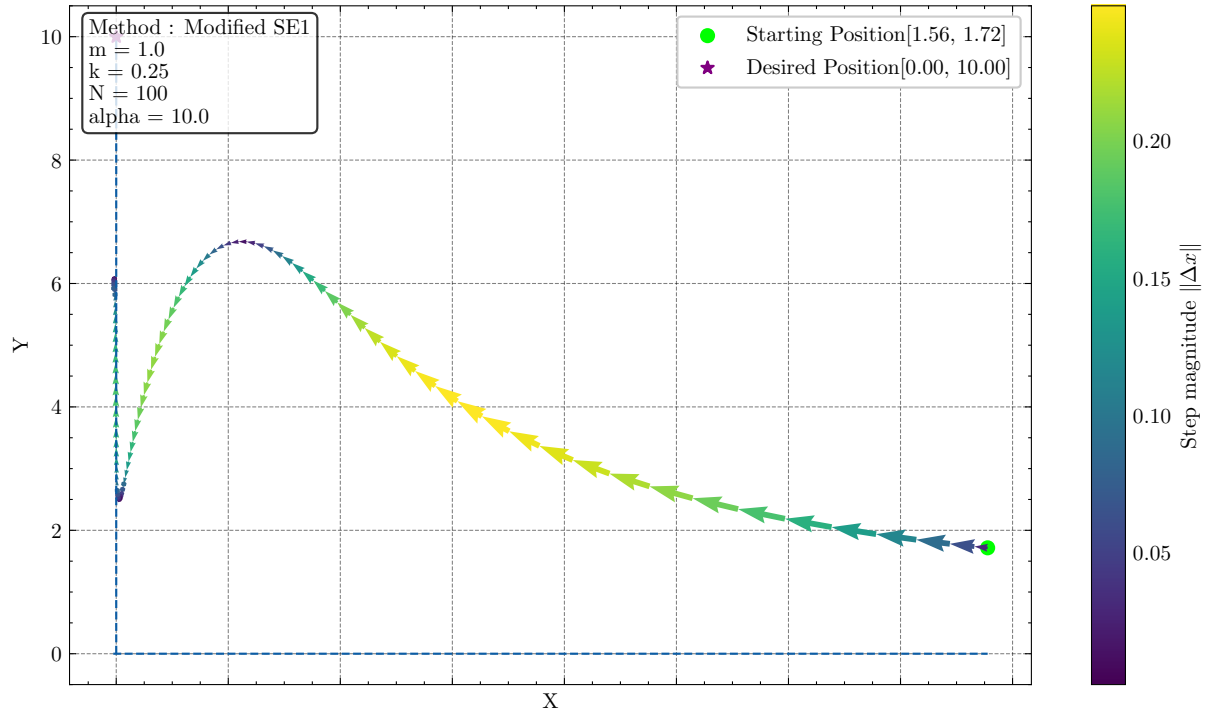


Figure 19: Trajectory comparison for $\mathbf{x}_0 = (1.56, 1.72)$ with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.

5.2 SE2

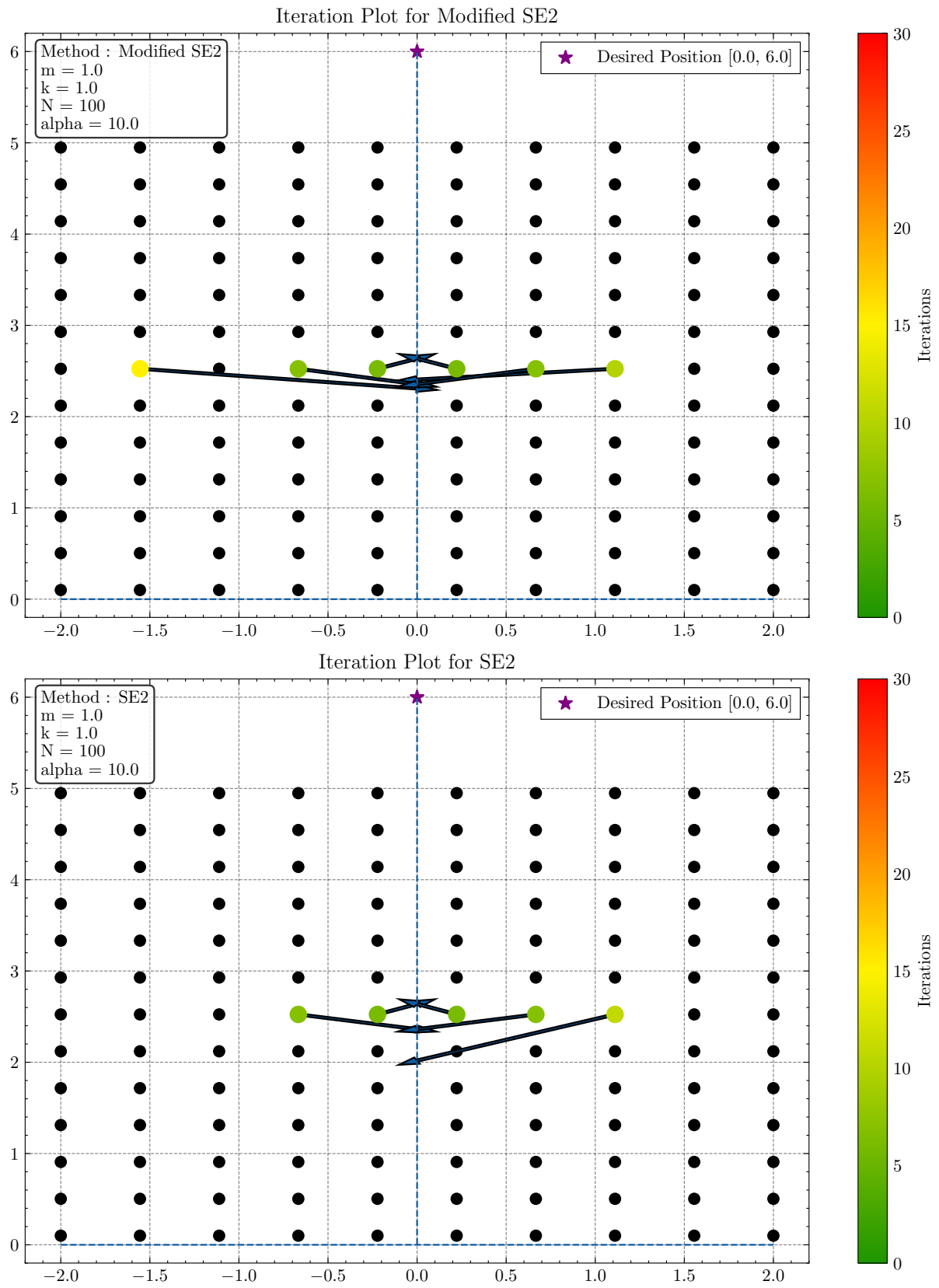
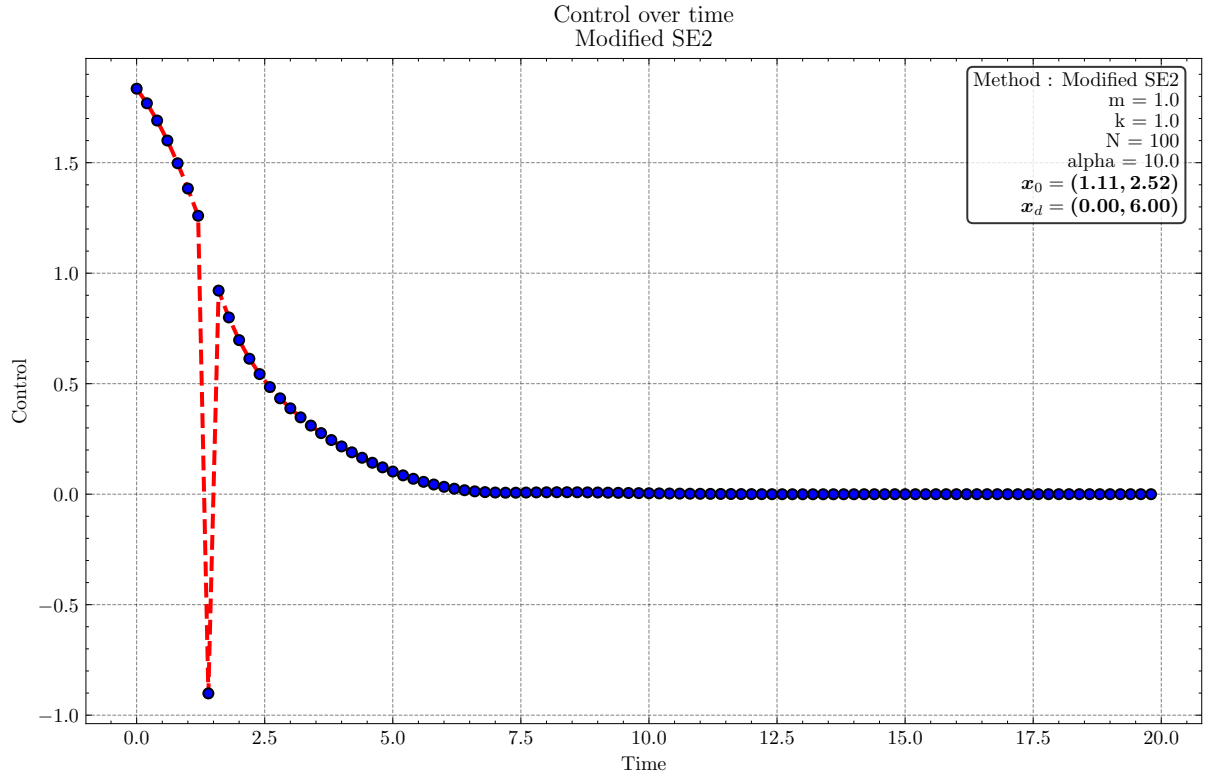
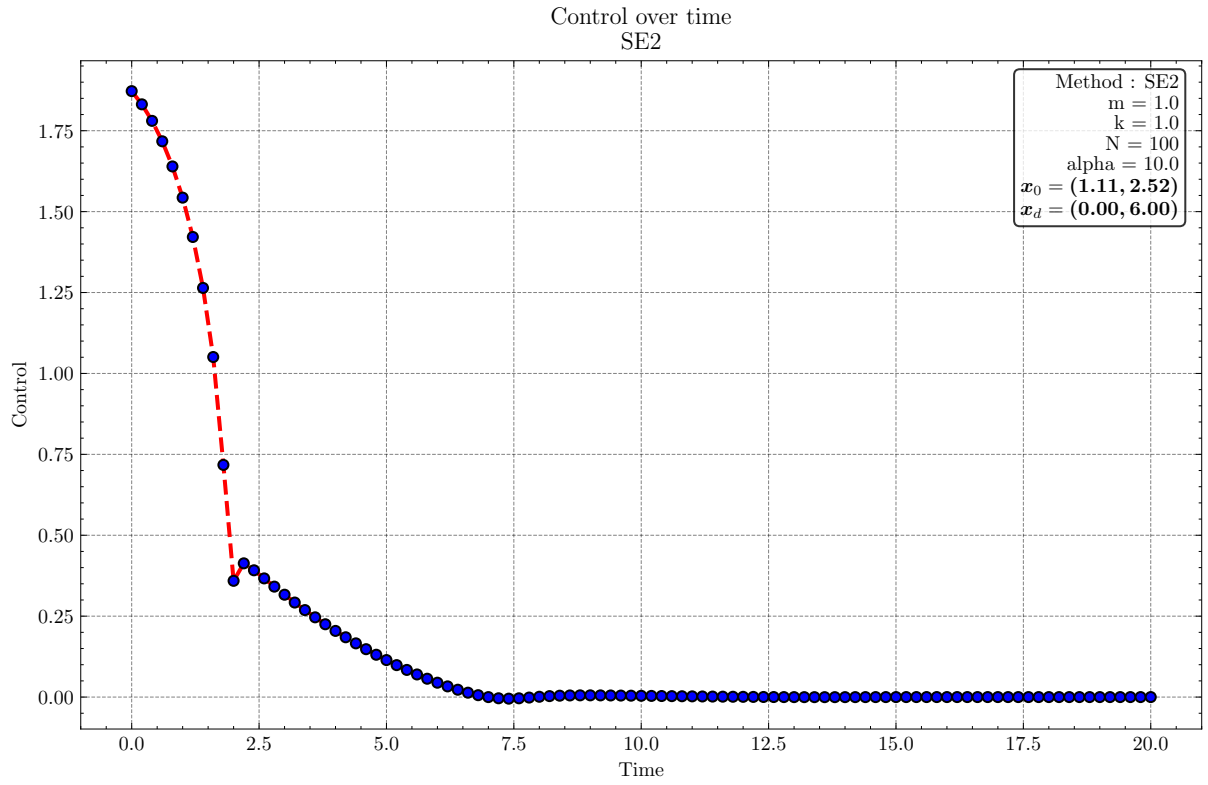


Figure 20: Iteration plots for SE2 and its modified variant with $k = 1.0$, $h = 6.0$, $N = 100$, $m = 1.0$.



(a) Modified SE2



(b) SE2

Figure 21: Control trajectories for $\mathbf{x}_0 = (1.11, 2.52)$ with $k = 1.0$, $h = 6.0$, $N = 100$, $m = 1.0$.

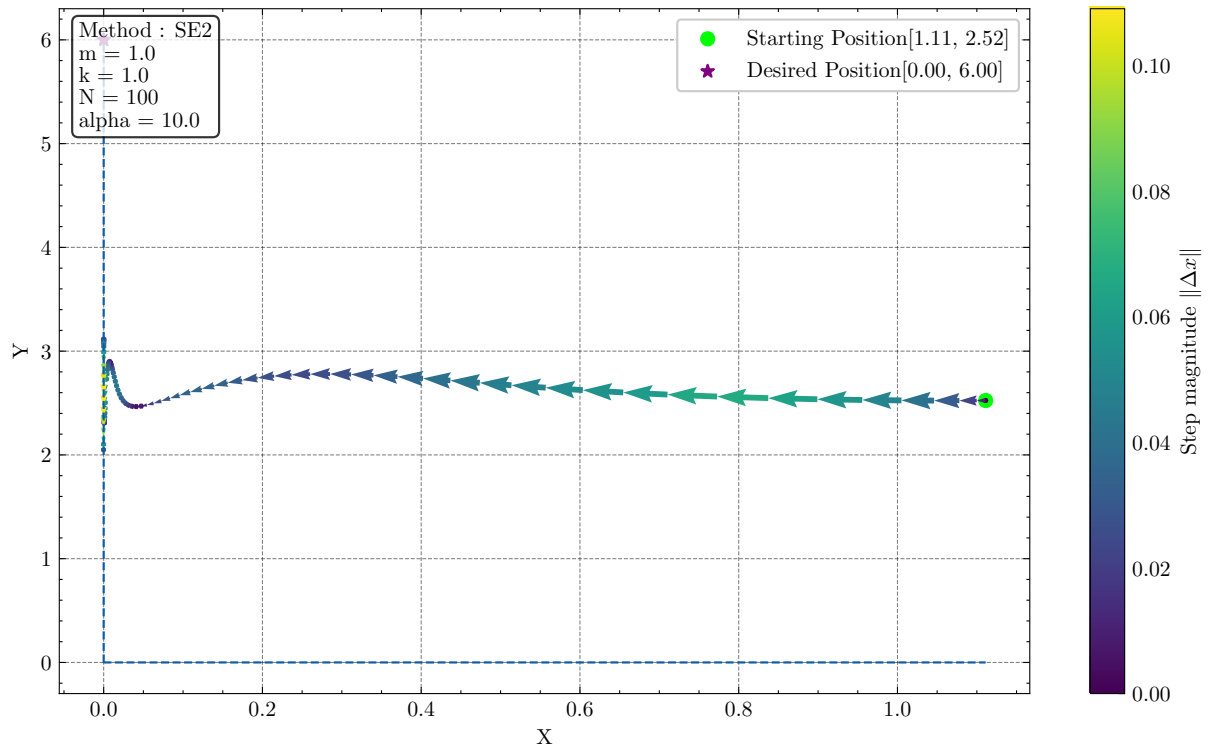
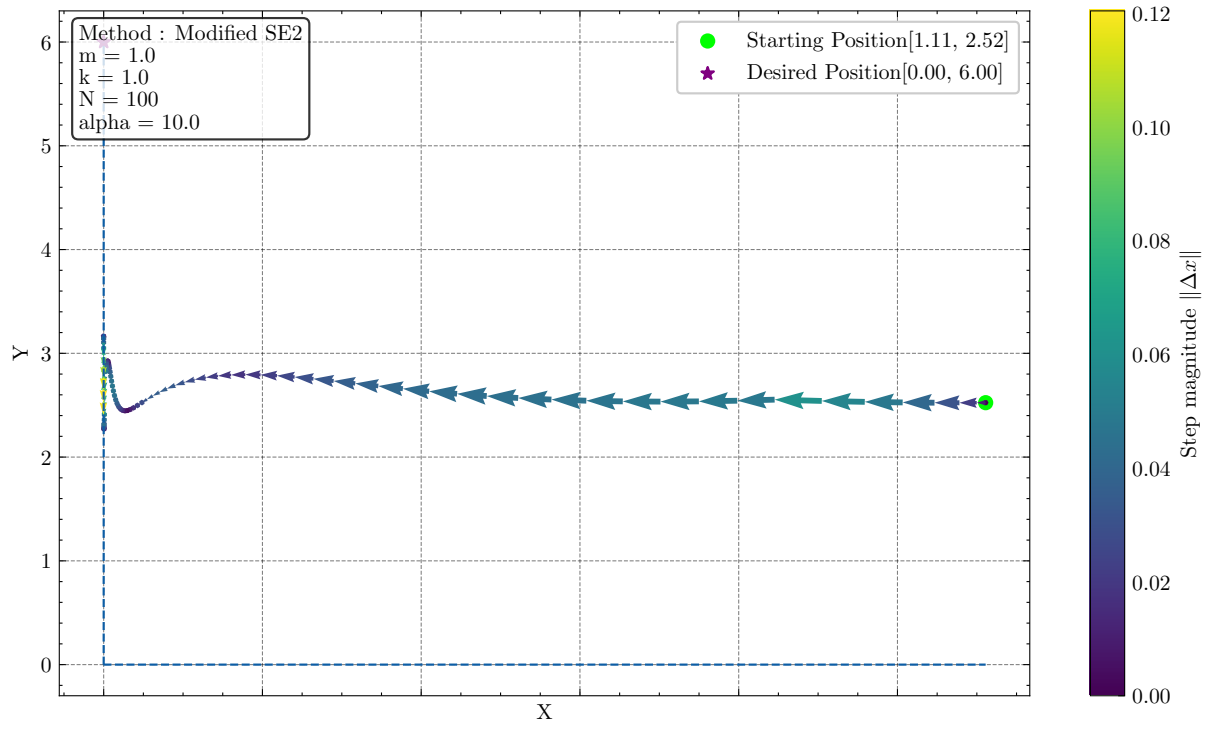


Figure 22: Trajectory comparison for $\mathbf{x}_0 = (1.11, 2.52)$ with $k = 1.0$, $h = 6.0$, $N = 100$, $m = 1.0$.

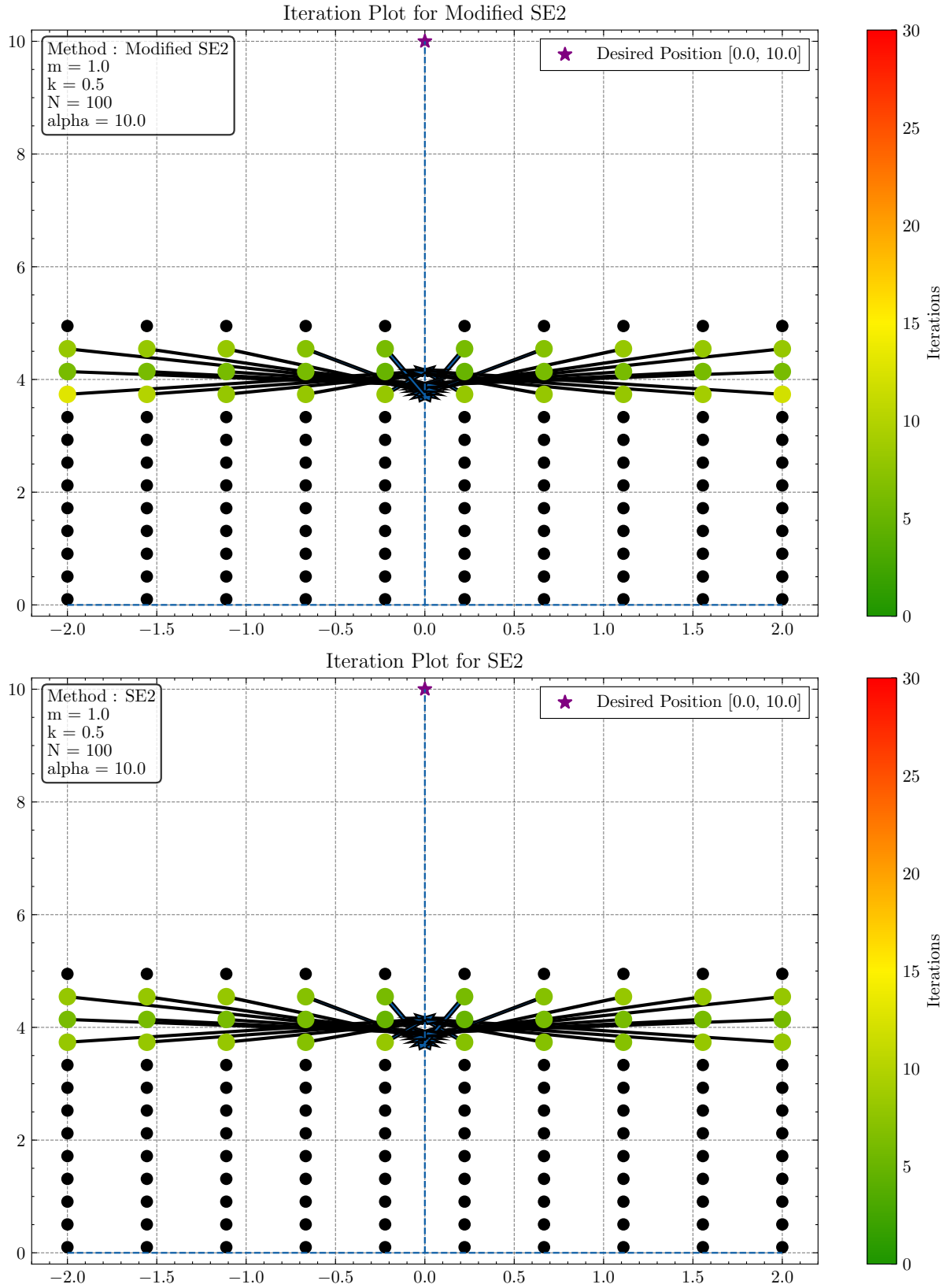
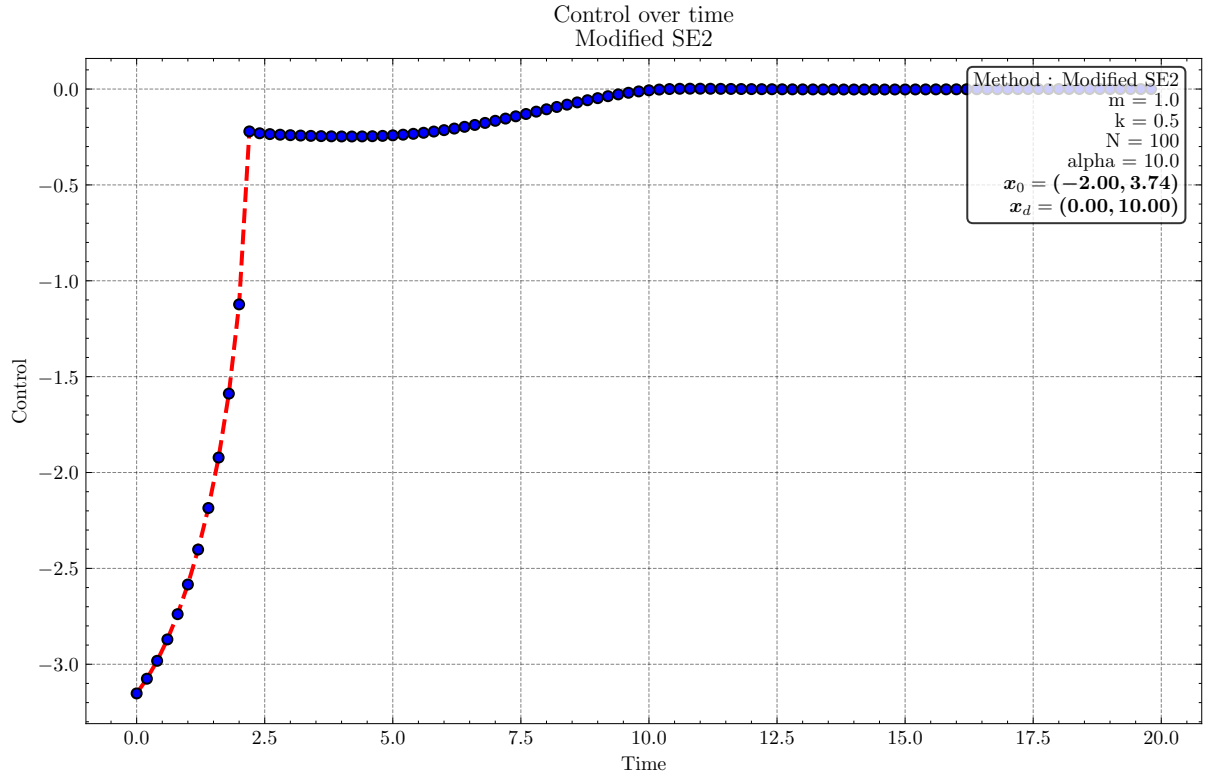
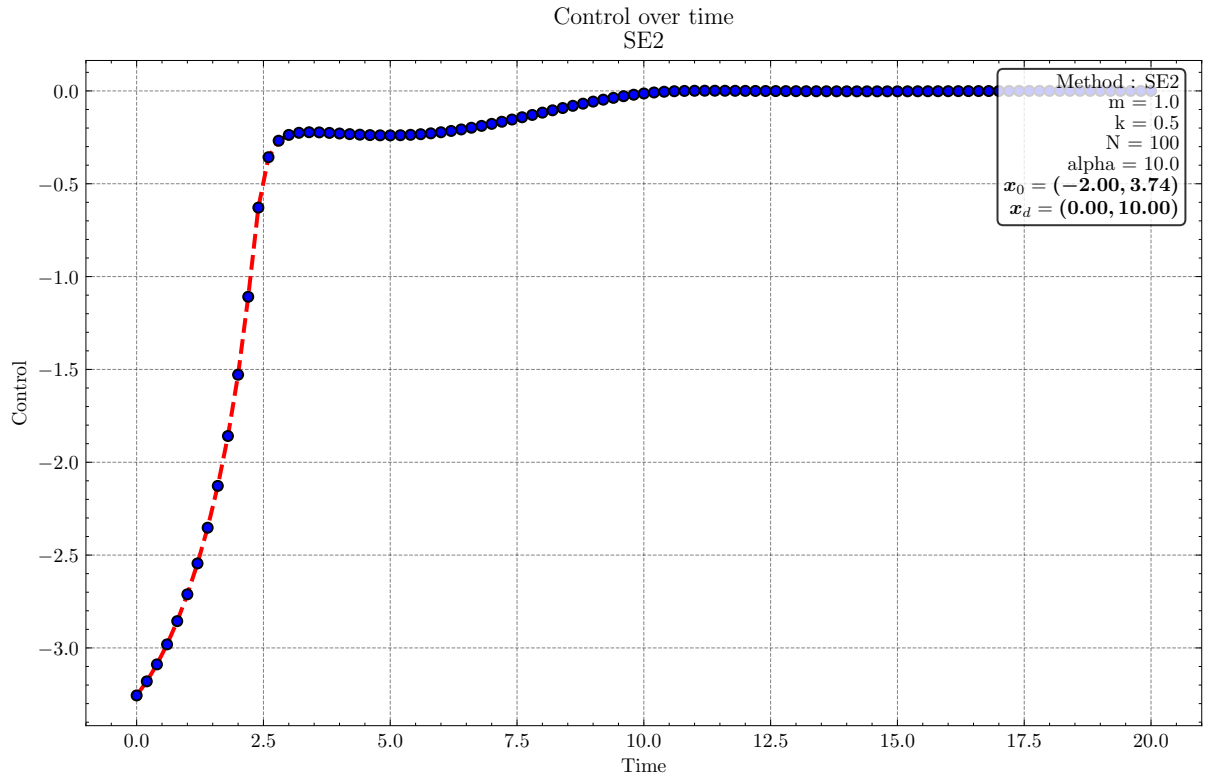


Figure 23: Iteration plots for SE2 and its modified variant with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.



(a) Modified SE2



(b) SE2

Figure 24: Control trajectories for $\mathbf{x}_0 = (-2.00, 3.74)$ with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.

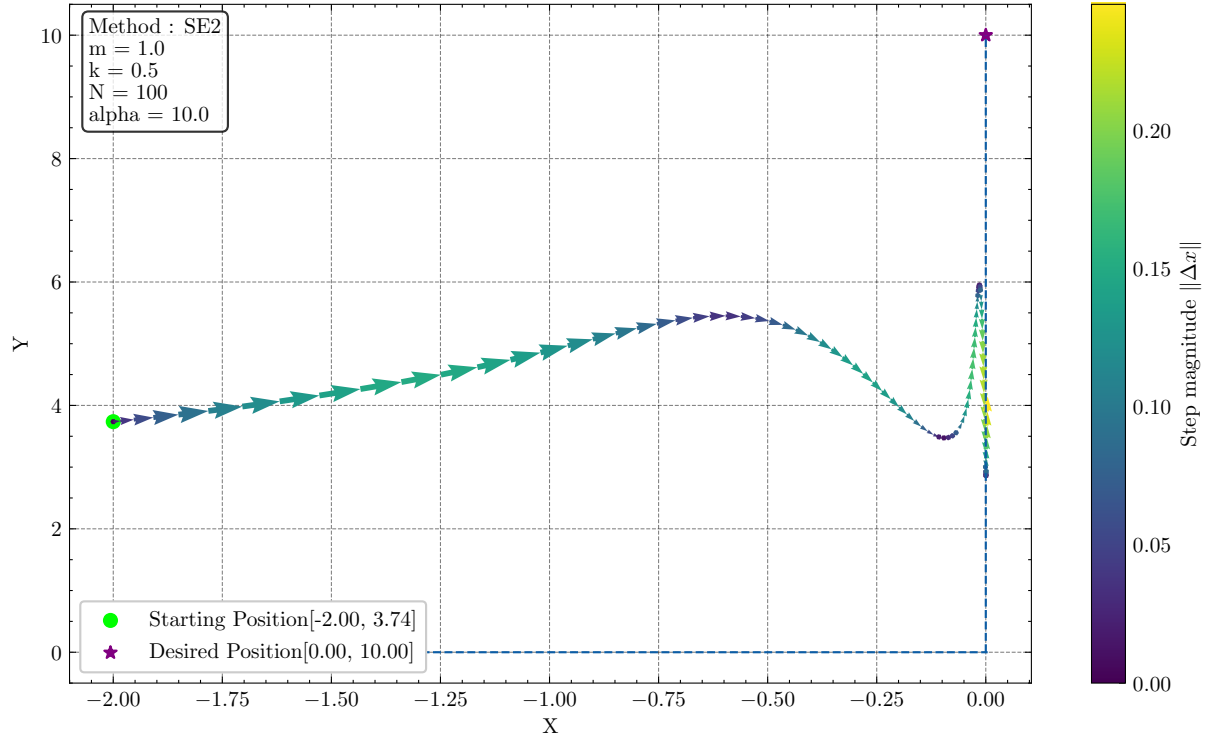
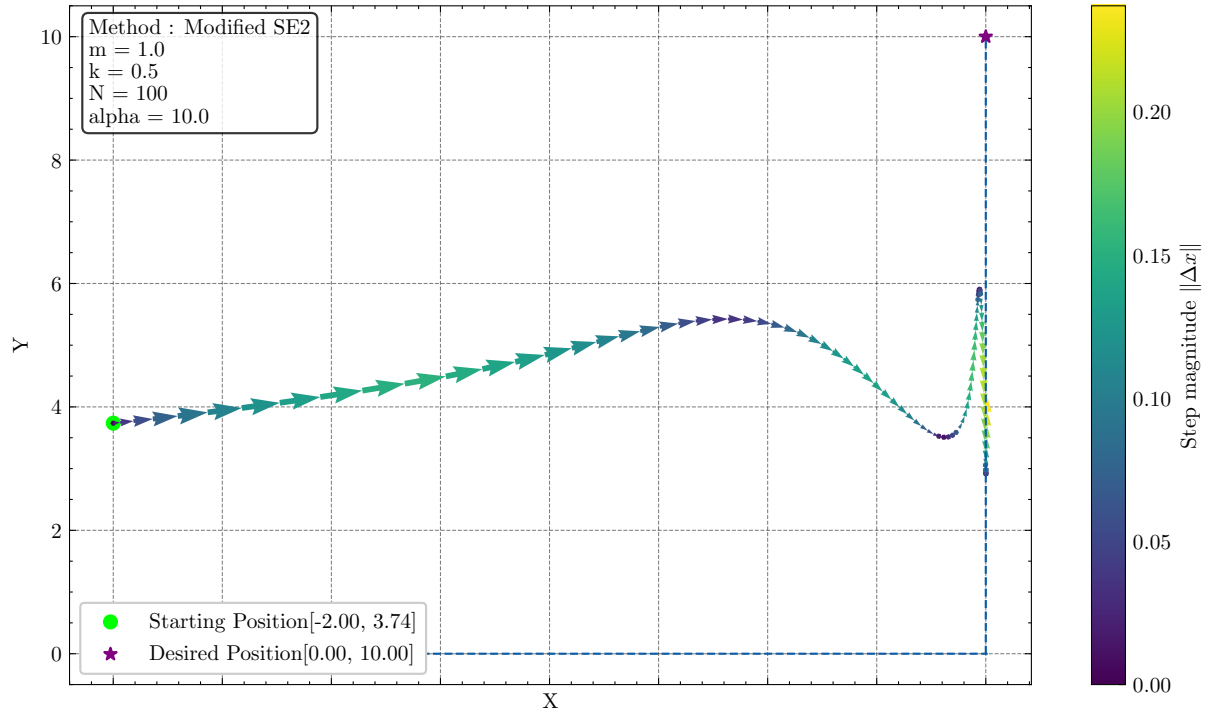
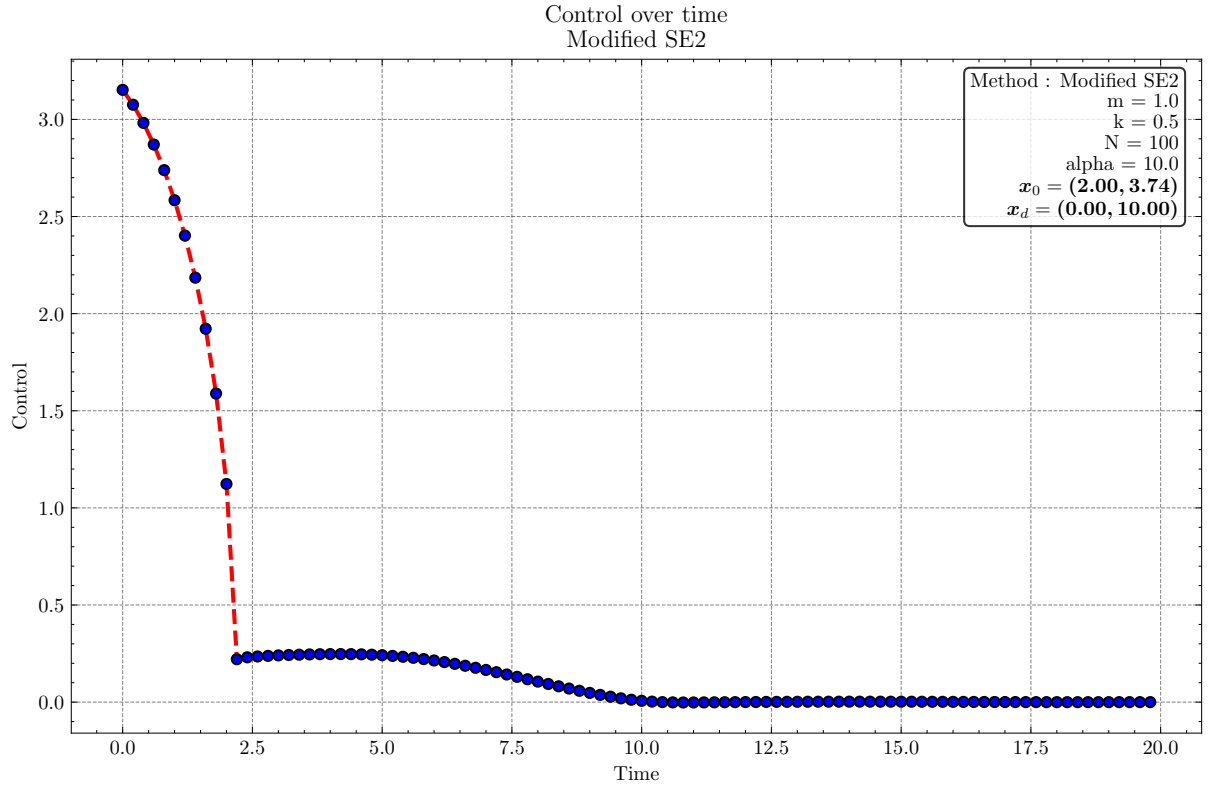
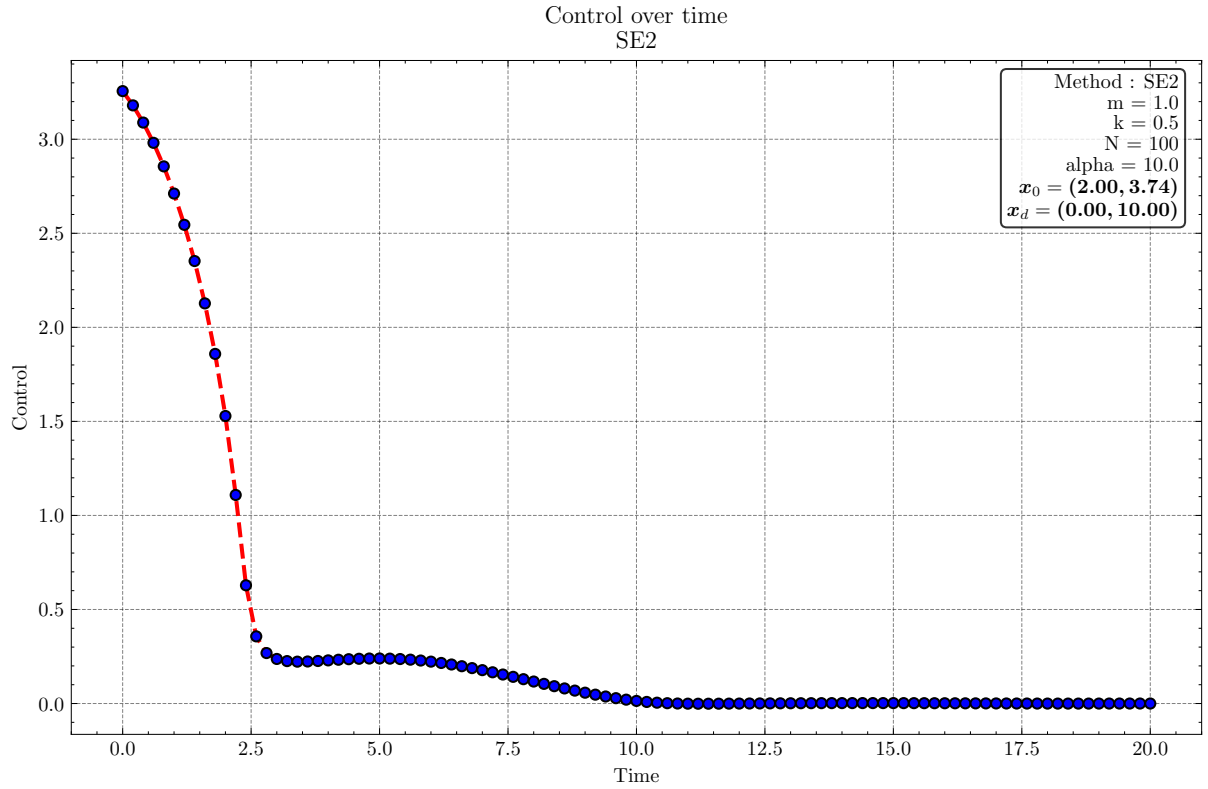


Figure 25: Trajectory comparison for $\mathbf{x}_0 = (-2.00, 3.74)$ with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.



(a) Modified SE2



(b) SE2

Figure 26: Control trajectories for $\mathbf{x}_0 = (2.00, 3.74)$ with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.

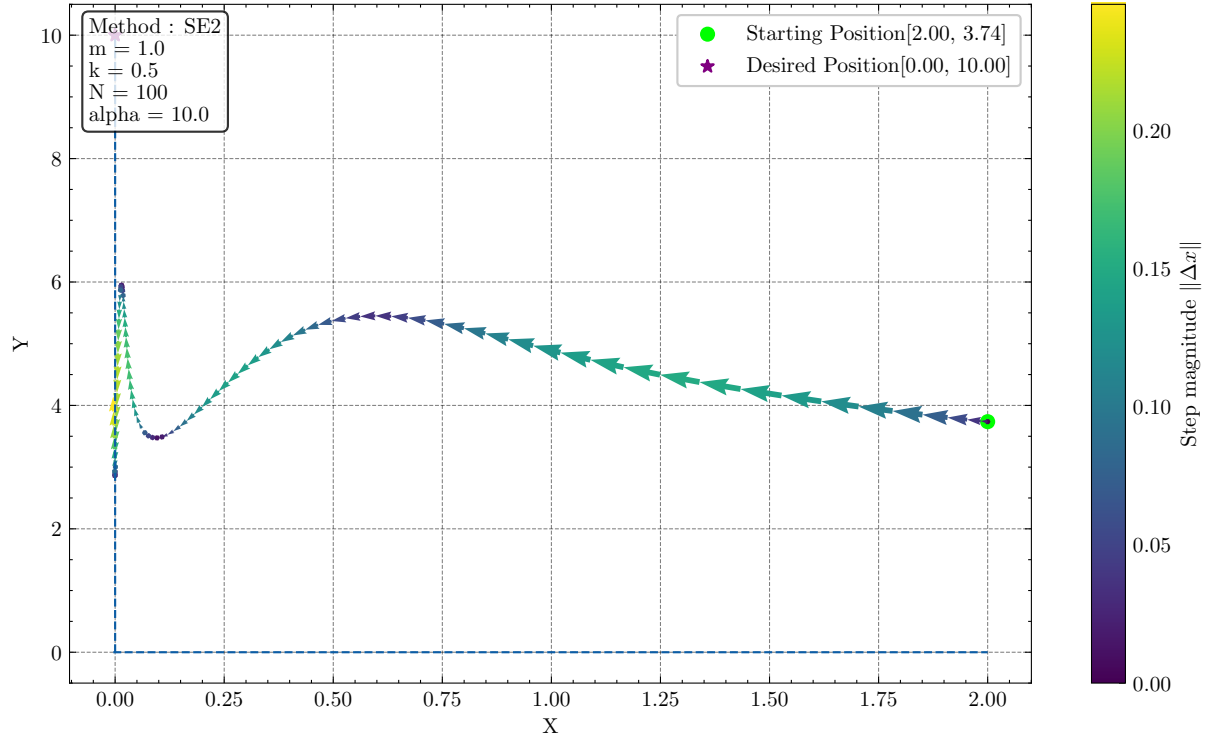
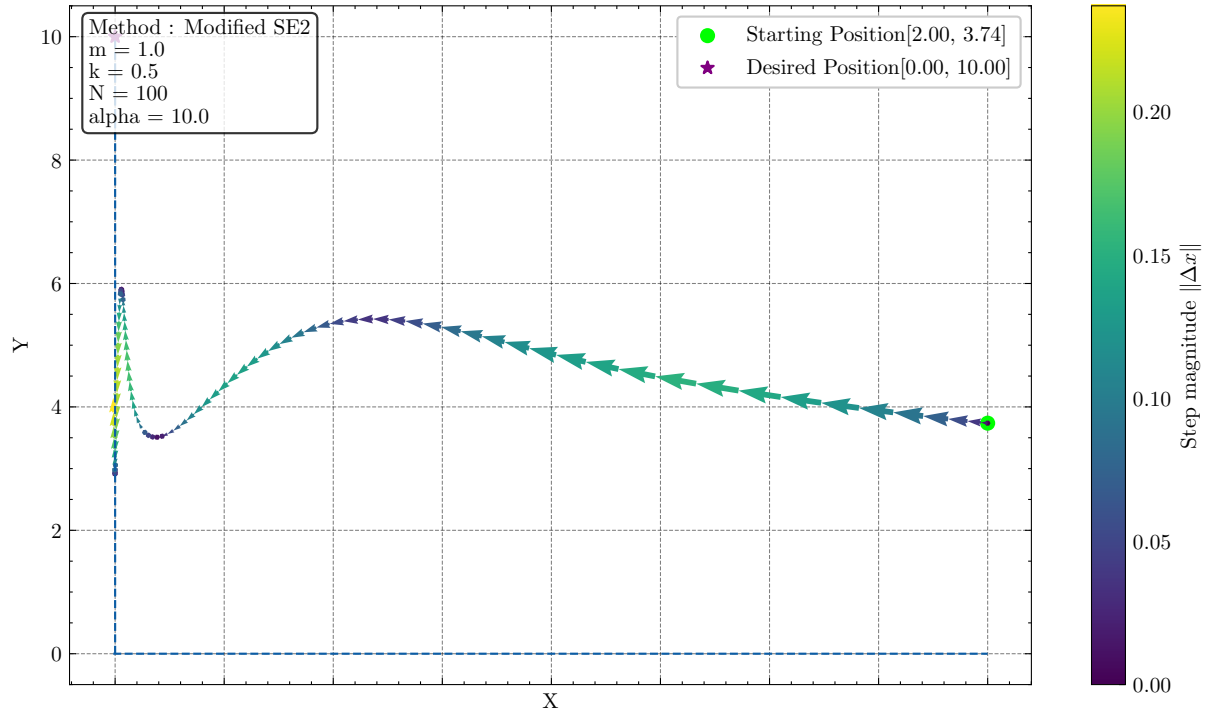


Figure 27: Trajectory comparison for $\mathbf{x}_0 = (2.00, 3.74)$ with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.

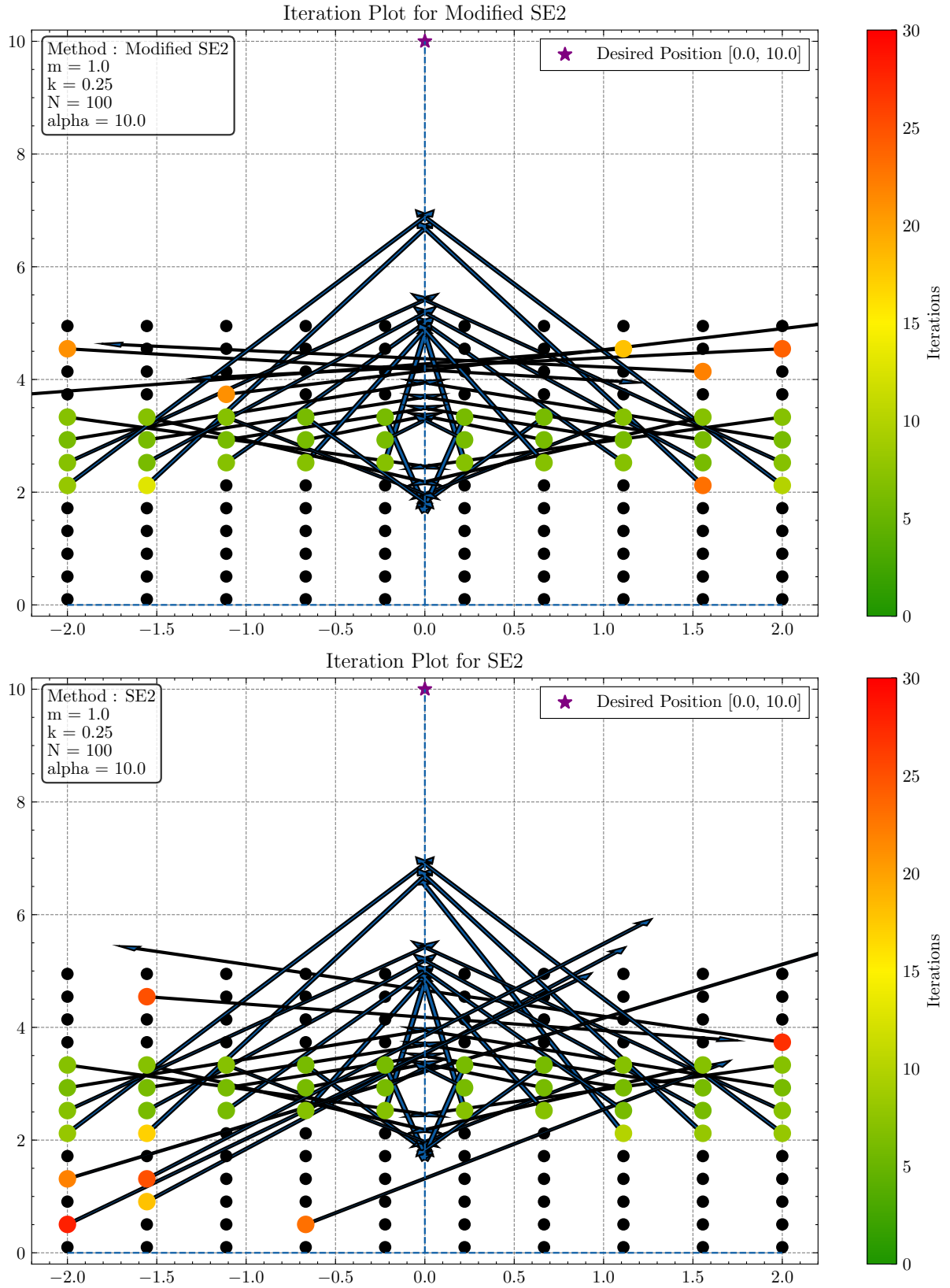
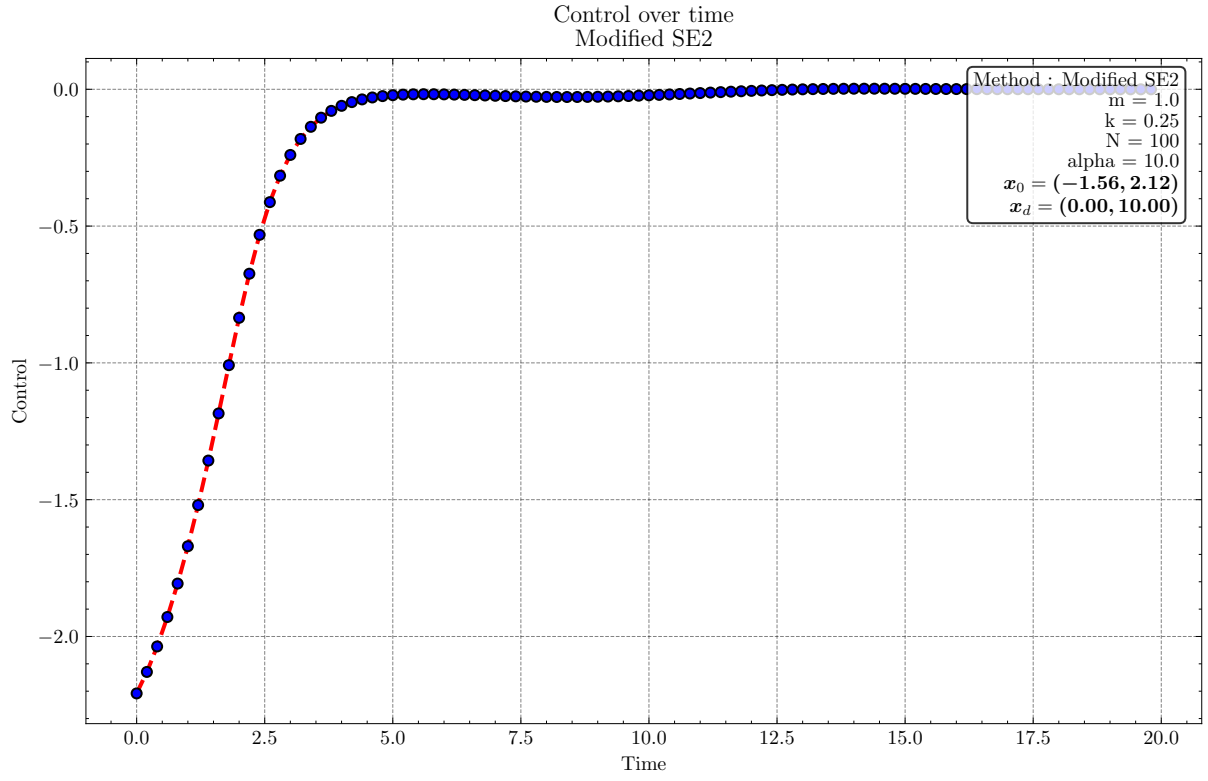
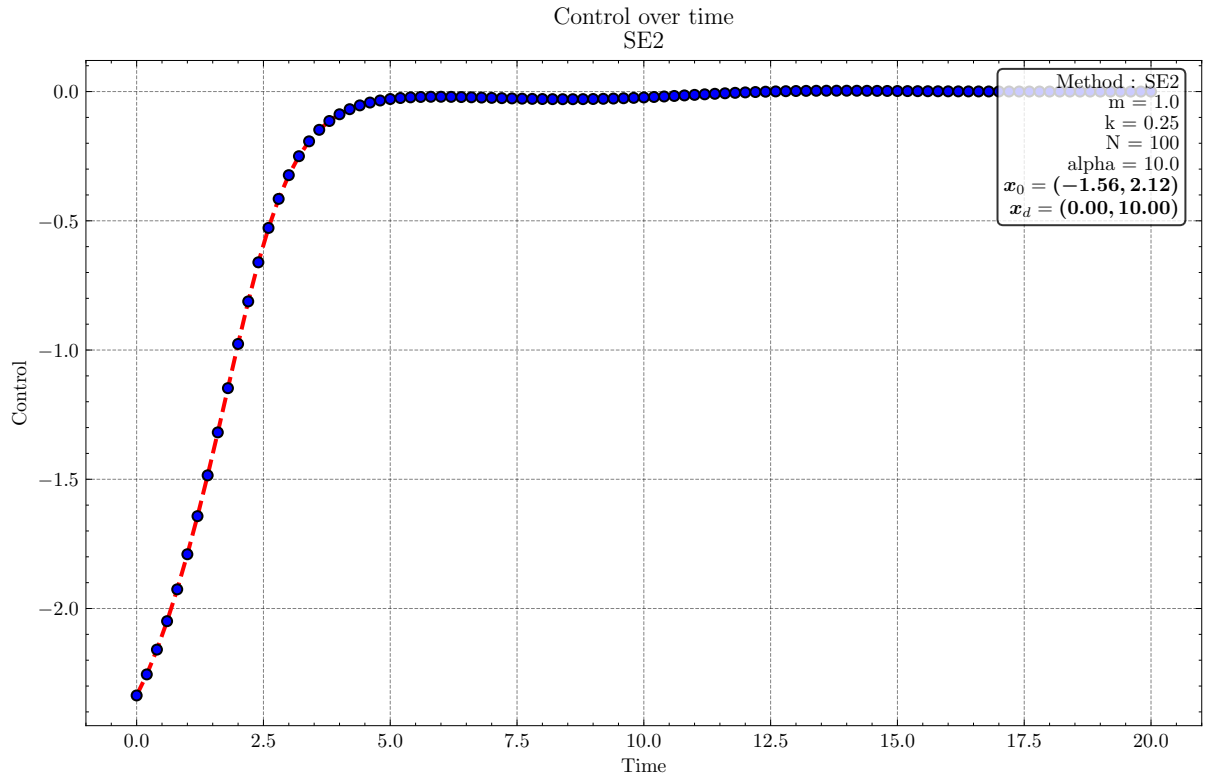


Figure 28: Iteration plots for SE2 and its modified variant with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.



(a) Modified SE2



(b) SE2

Figure 29: Control trajectories for $\mathbf{x}_0 = (-1.56, 2.12)$ with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.

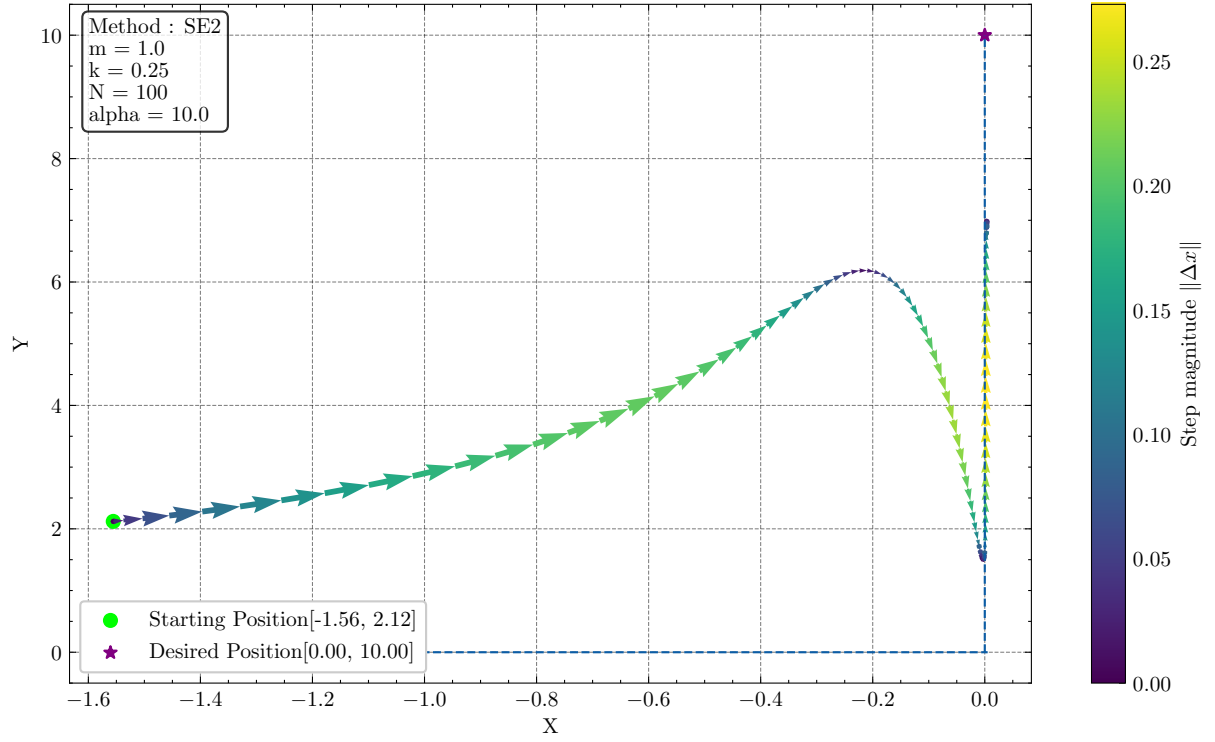
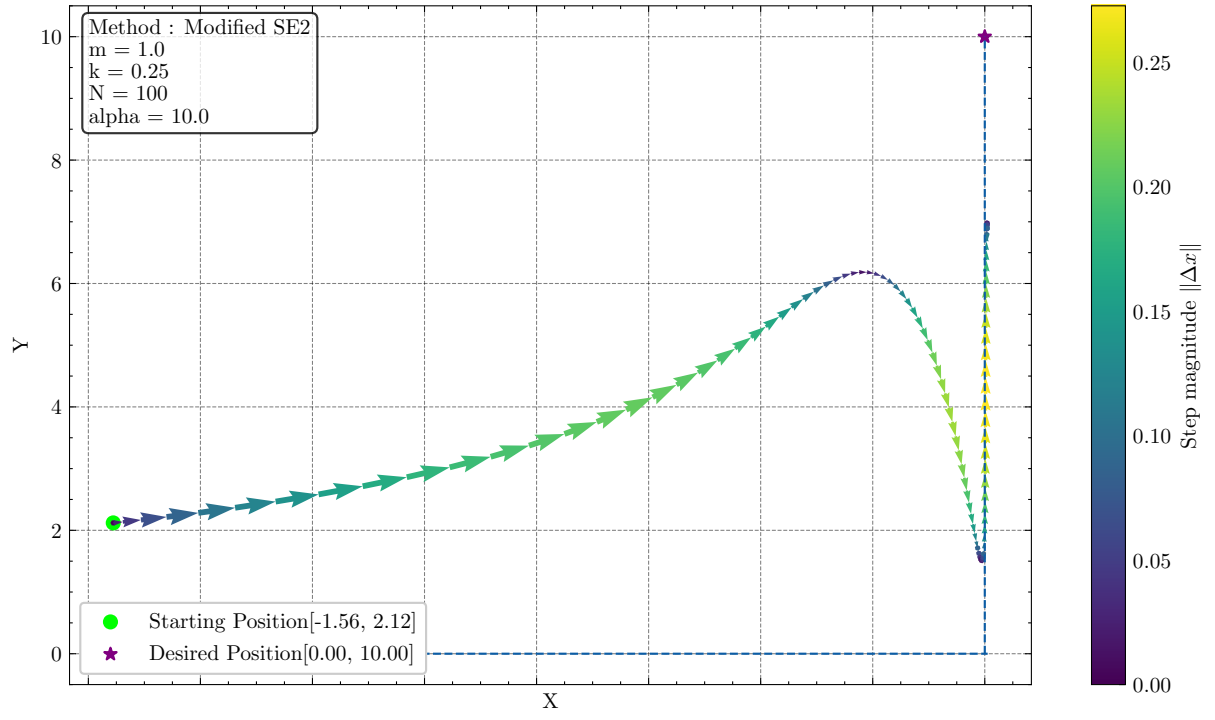


Figure 30: Trajectory comparison for $\mathbf{x}_0 = (-1.56, 2.12)$ with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.

5.3 MidPoint

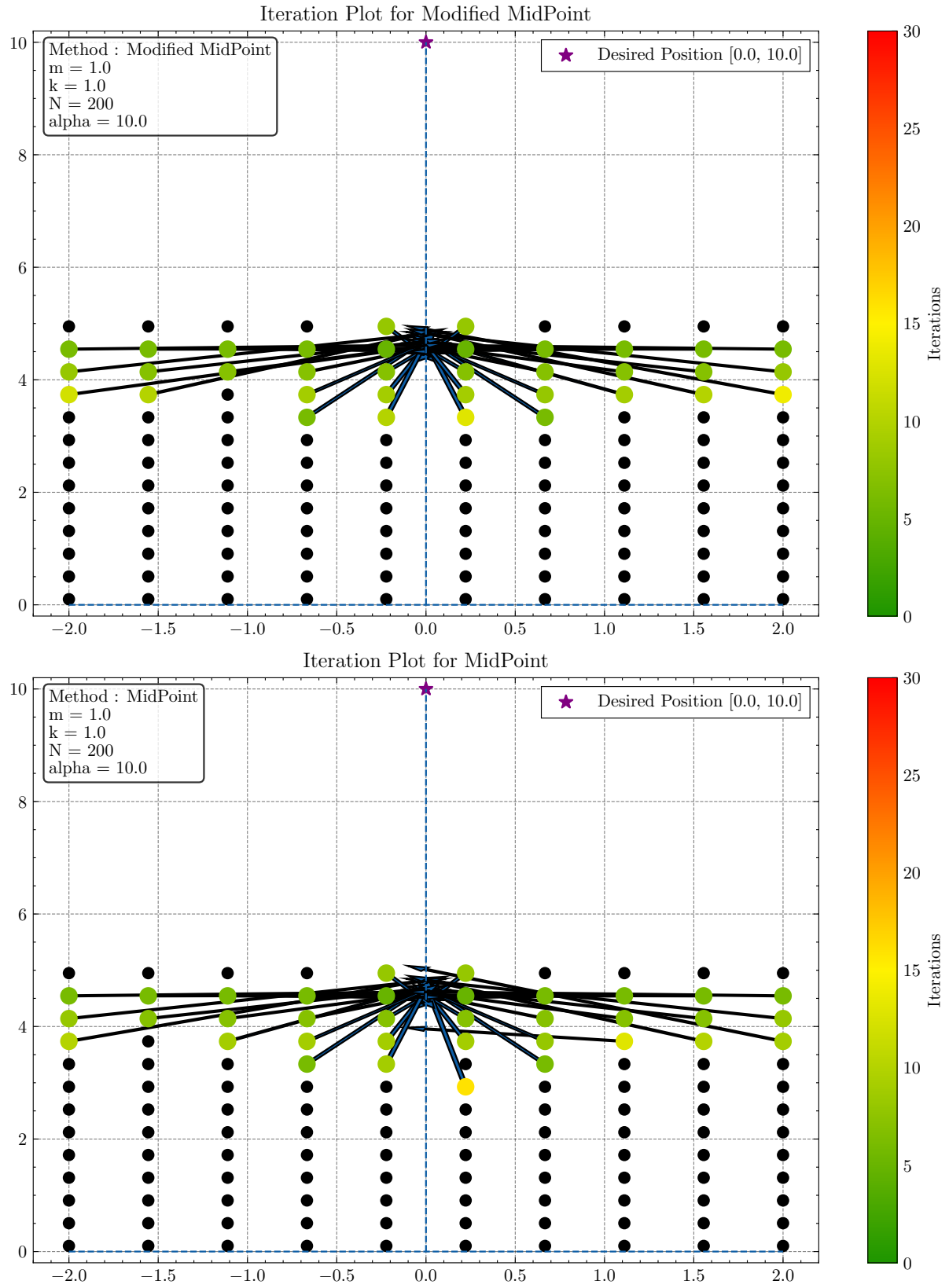


Figure 31: Iteration plots for MidPoint and its modified variant with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.

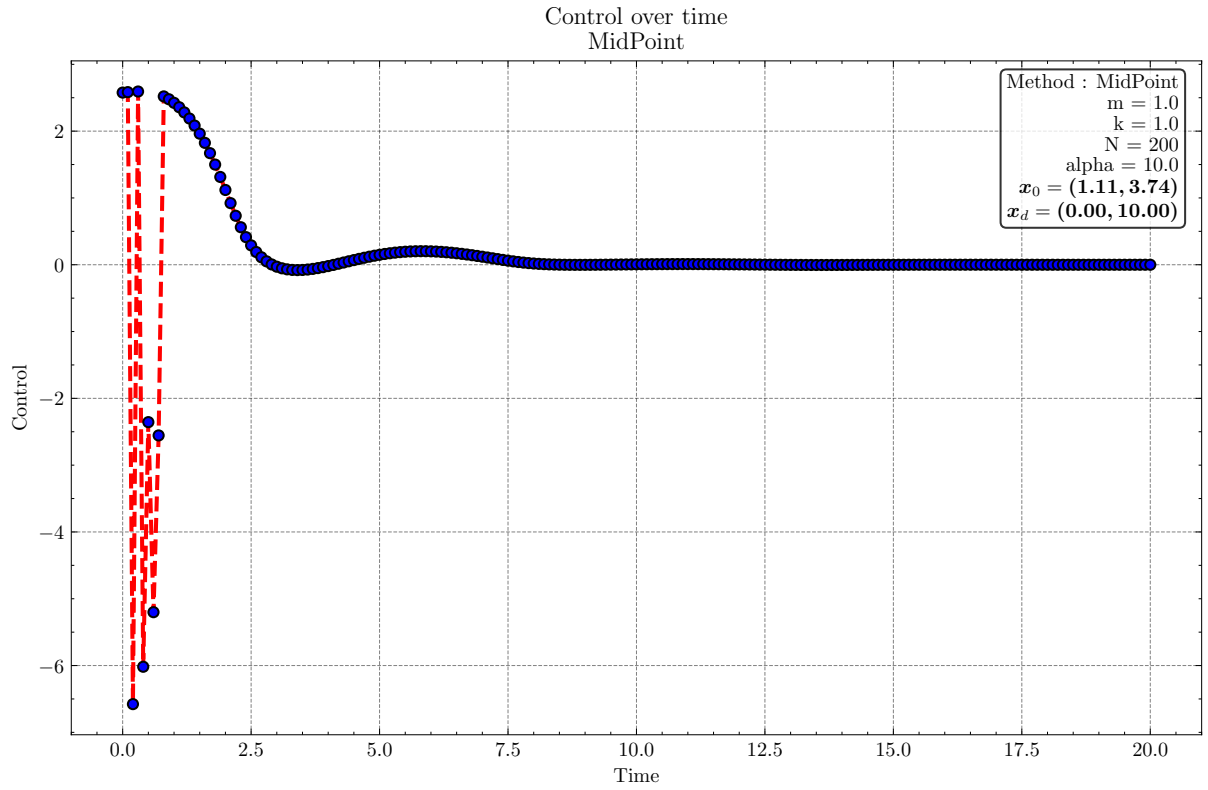
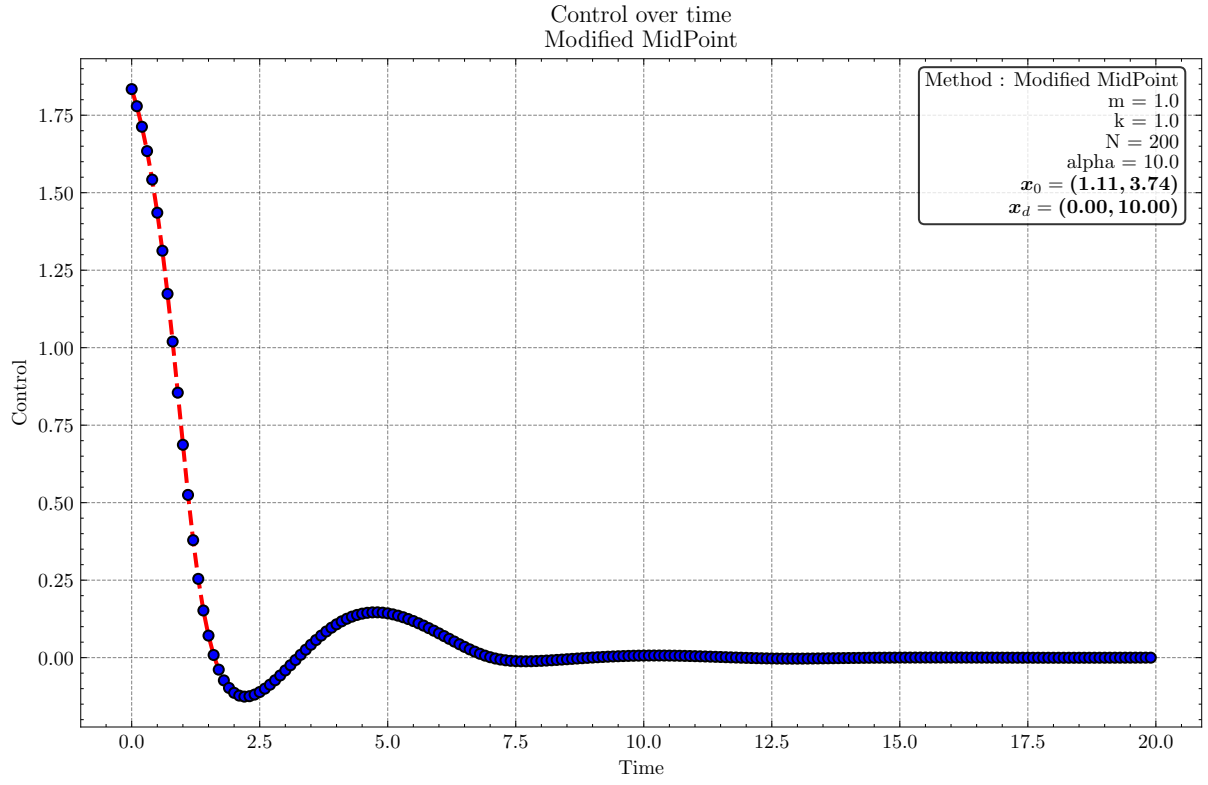


Figure 32: Control trajectories for $\mathbf{x}_0 = (1.11, 3.74)$ with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.

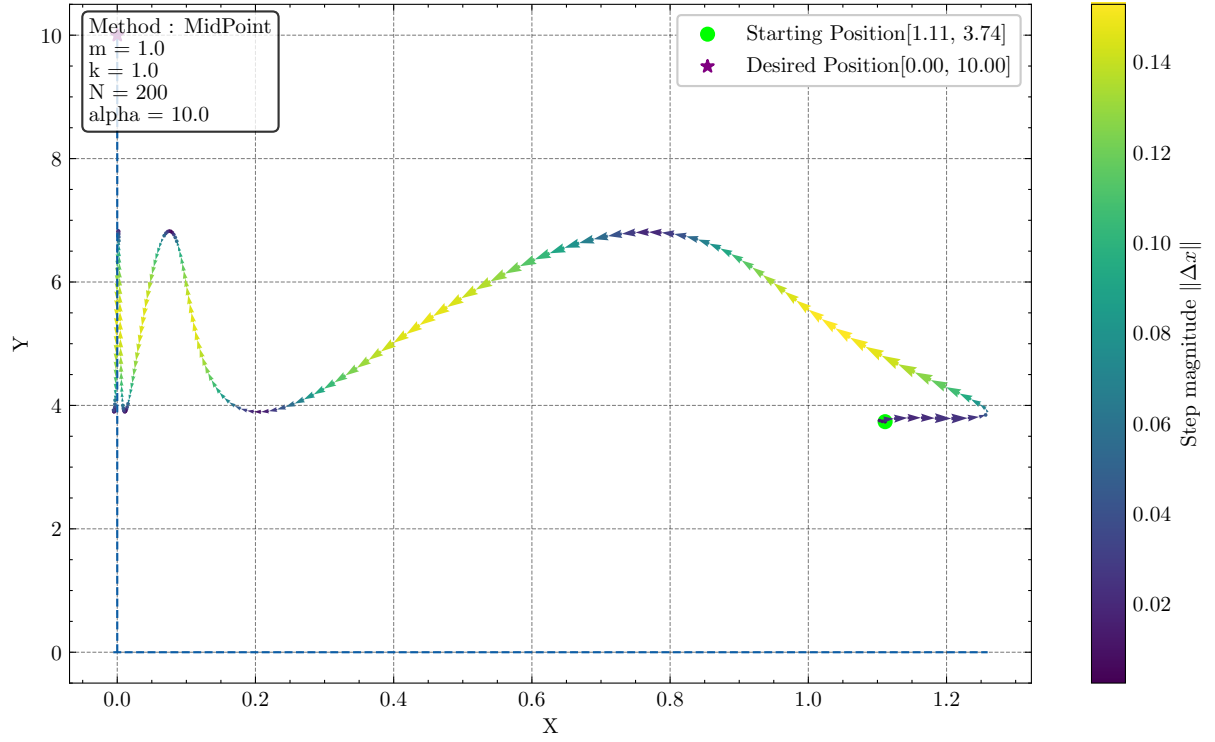
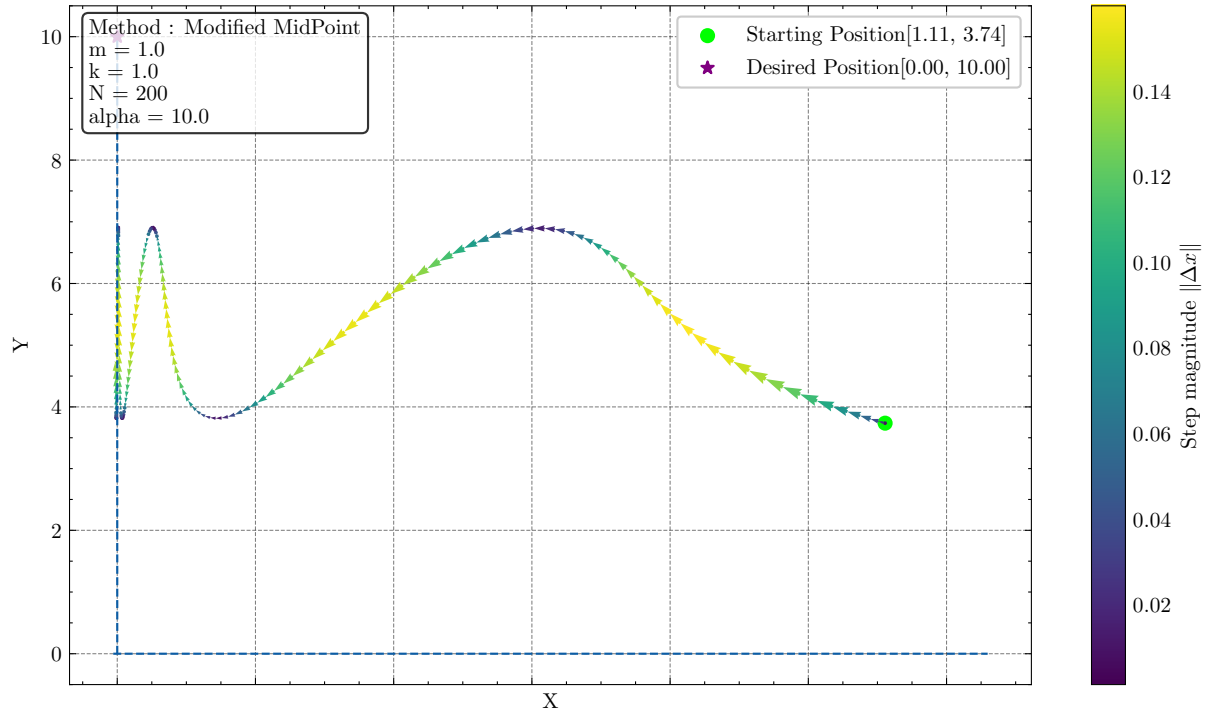
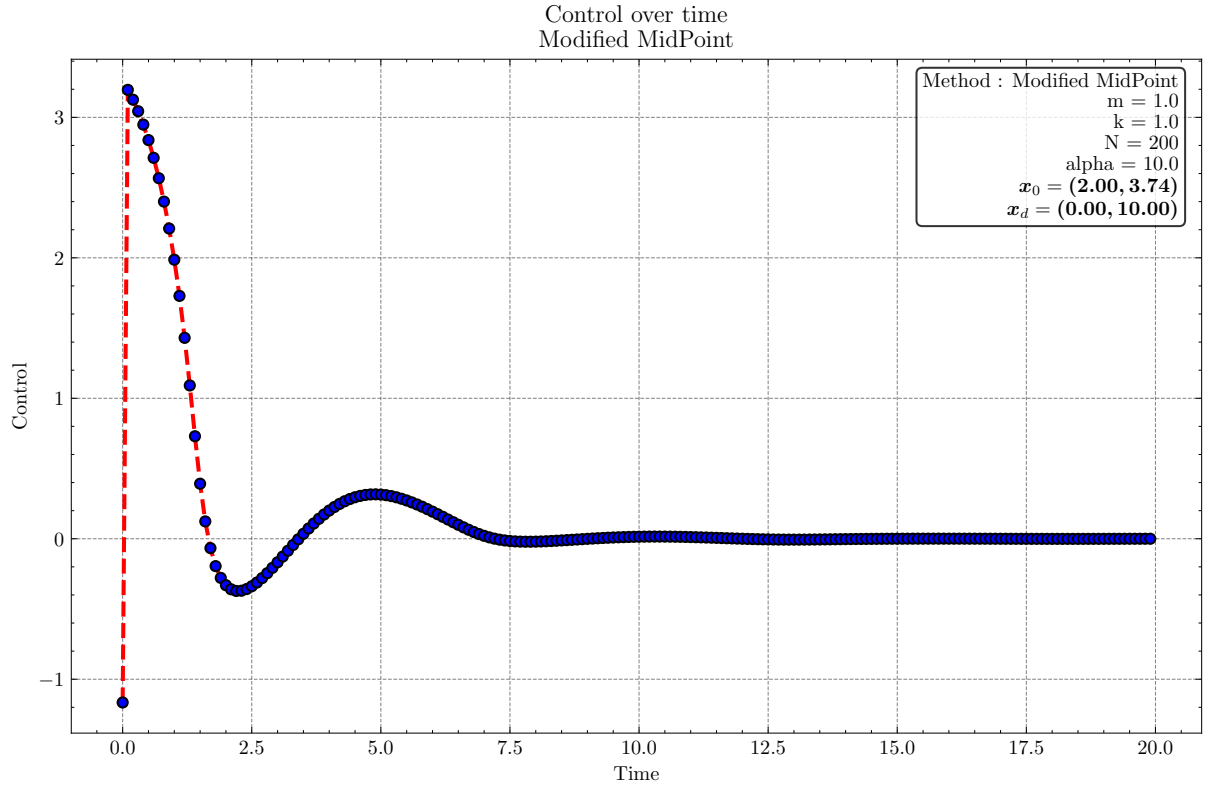
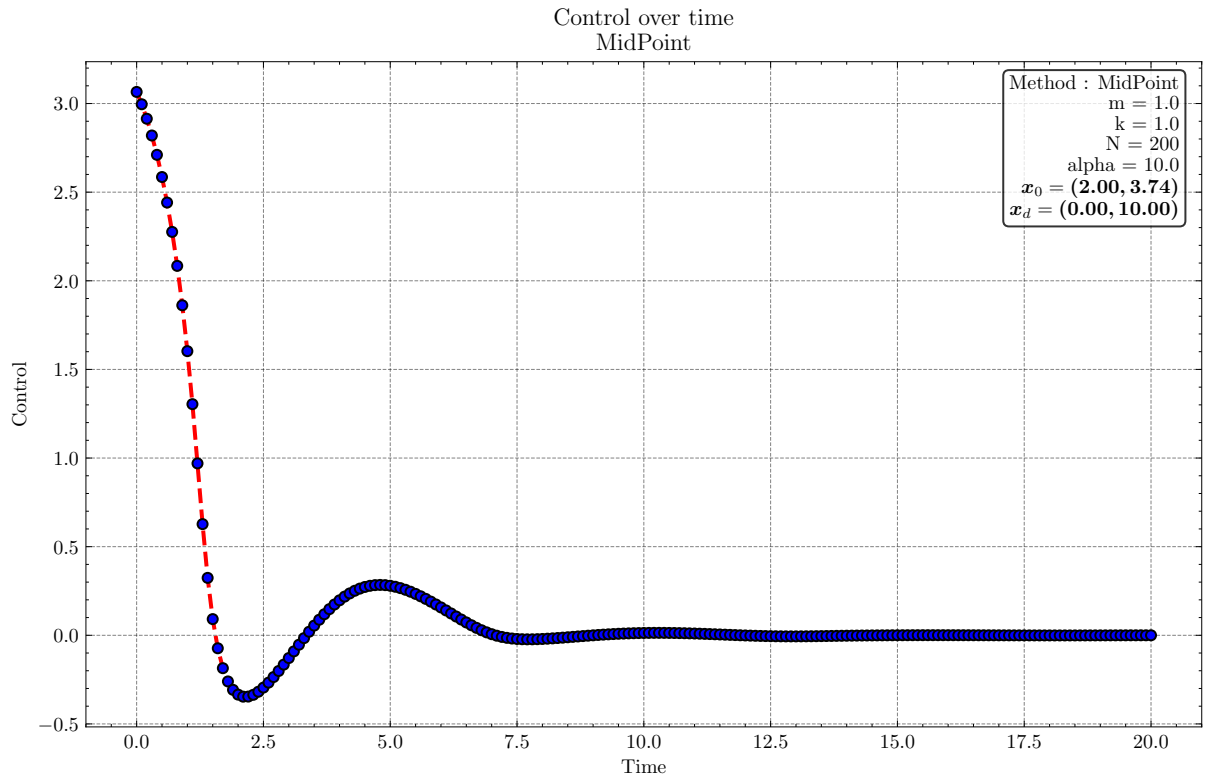


Figure 33: Trajectory comparison for $\mathbf{x}_0 = (1.11, 3.74)$ with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.



(a) Modified MidPoint



(b) MidPoint

Figure 34: Control trajectories for $\mathbf{x}_0 = (2.00, 3.74)$ with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.

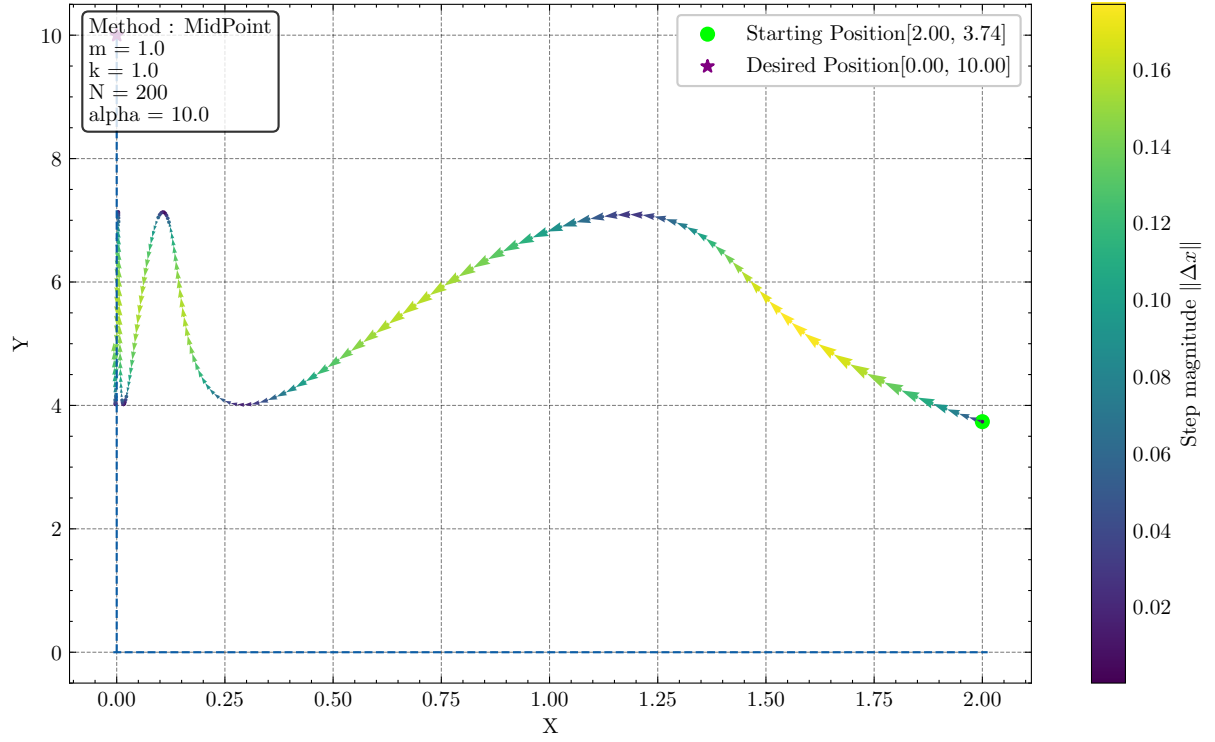
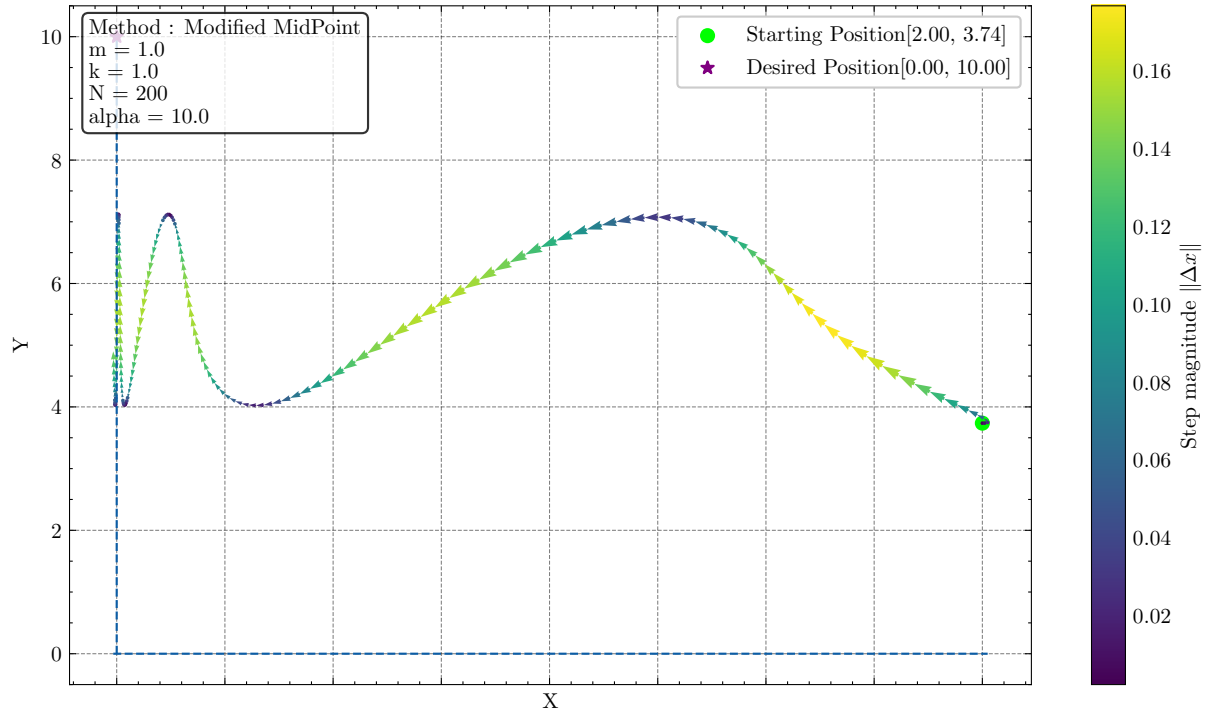
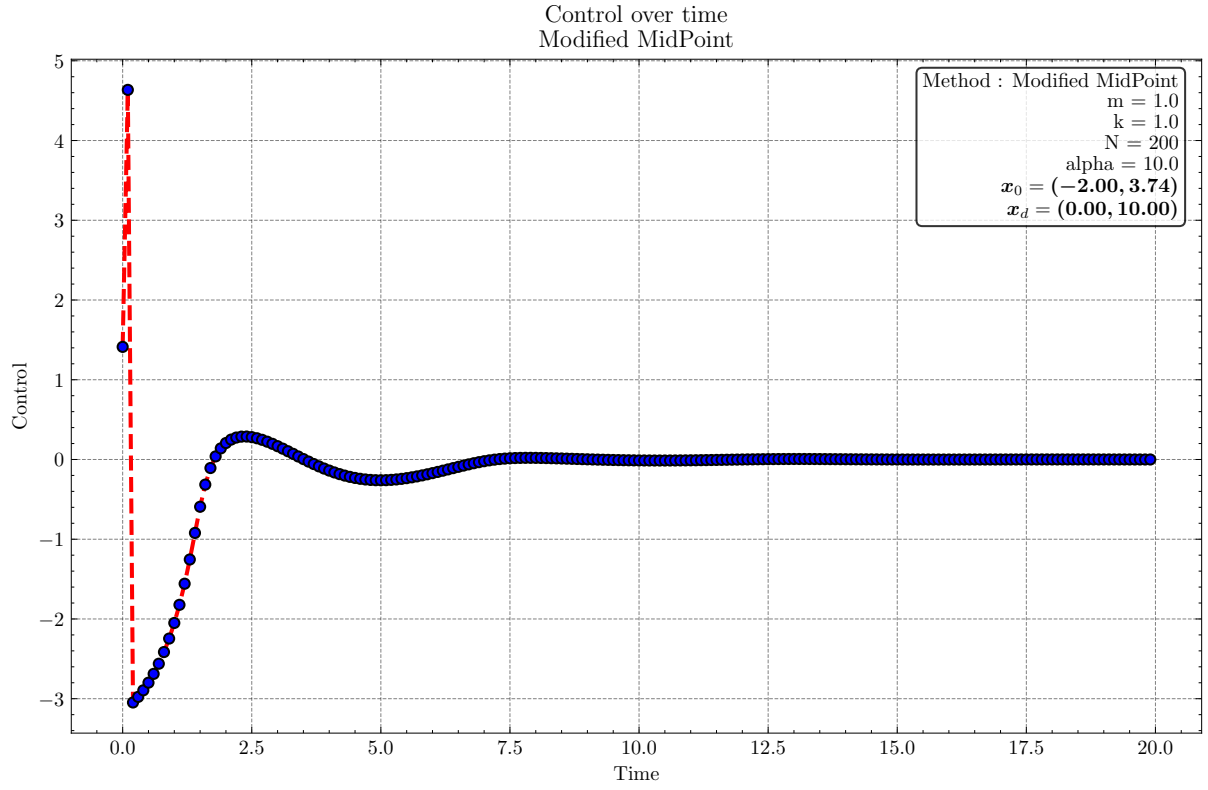
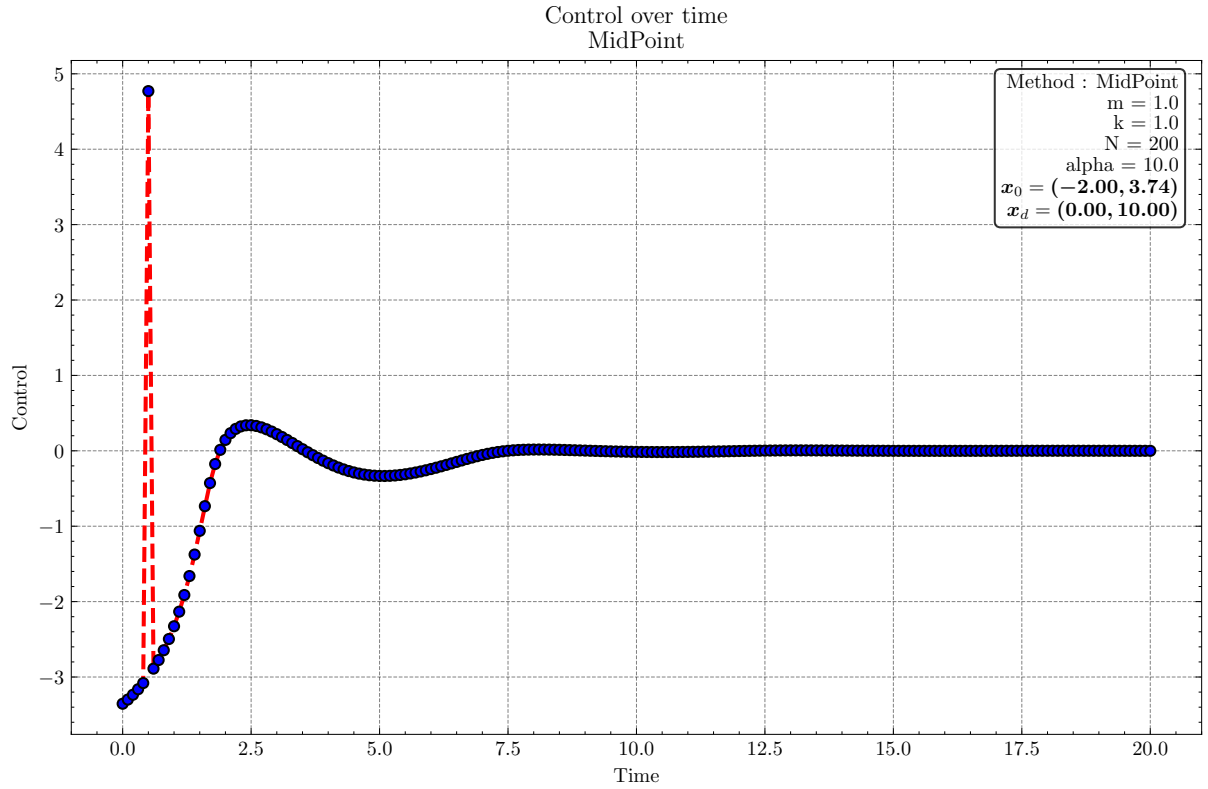


Figure 35: Trajectory comparison for $\mathbf{x}_0 = (2.00, 3.74)$ with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.



(a) Modified MidPoint



(b) MidPoint

Figure 36: Control trajectories for $\mathbf{x}_0 = (-2.00, 3.74)$ with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.

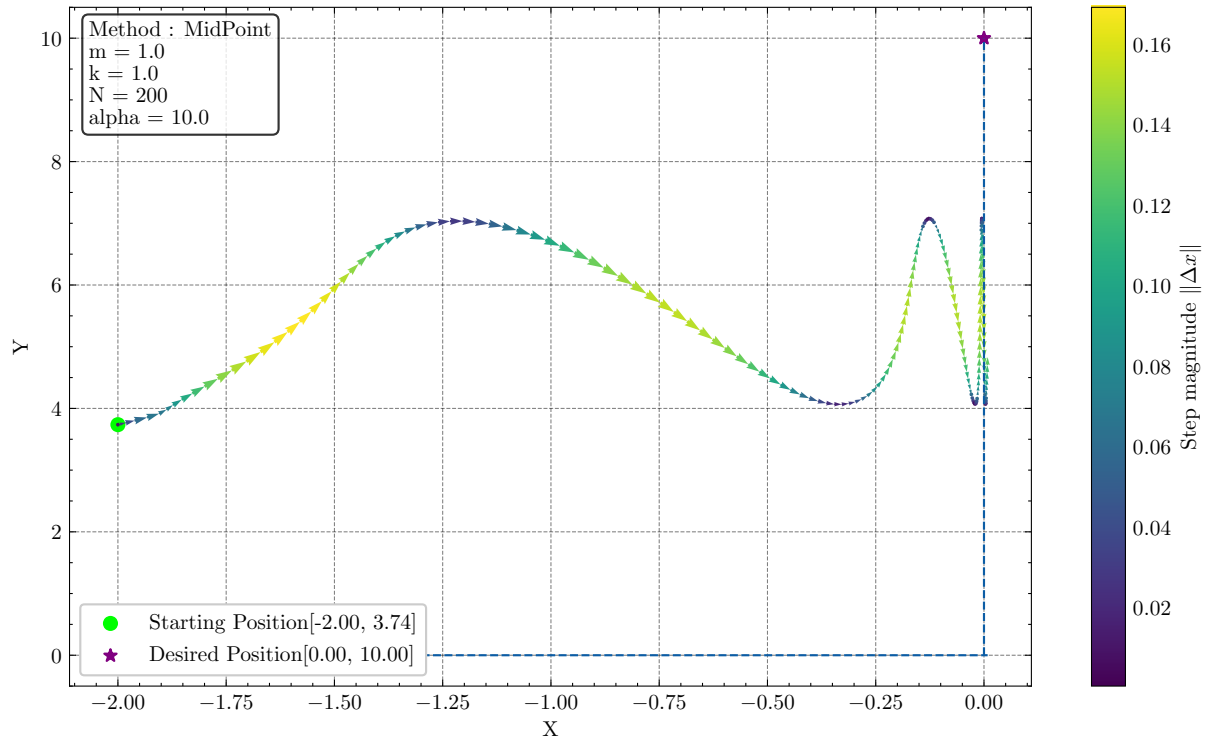
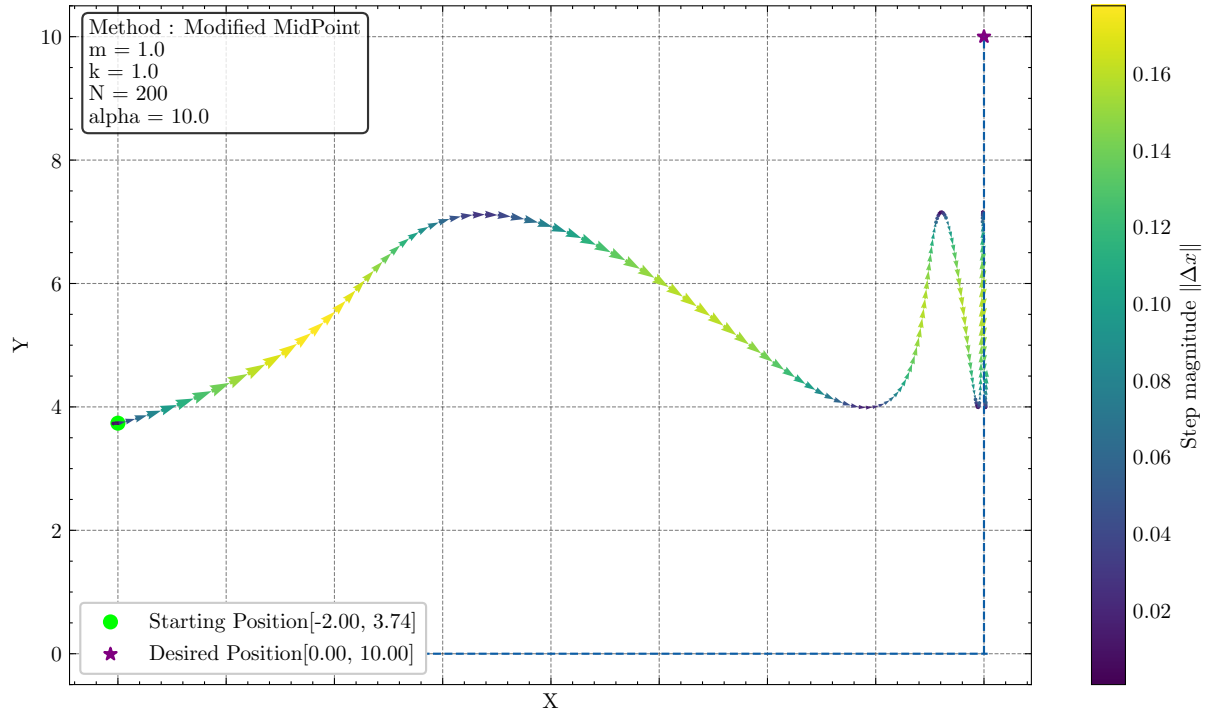


Figure 37: Trajectory comparison for $\mathbf{x}_0 = (-2.00, 3.74)$ with $k = 1.0$, $h = 10.0$, $N = 200$, $m = 1.0$.

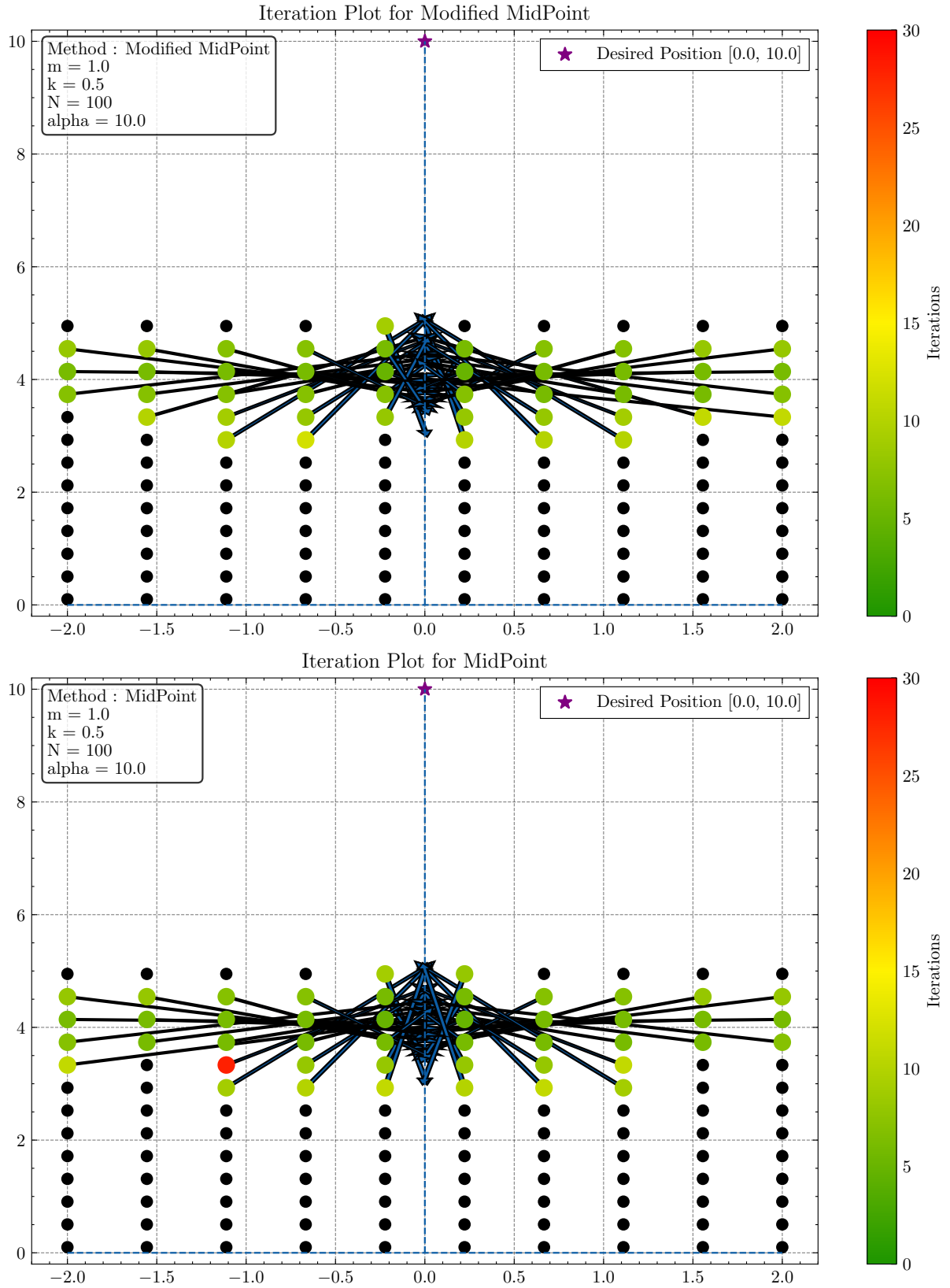
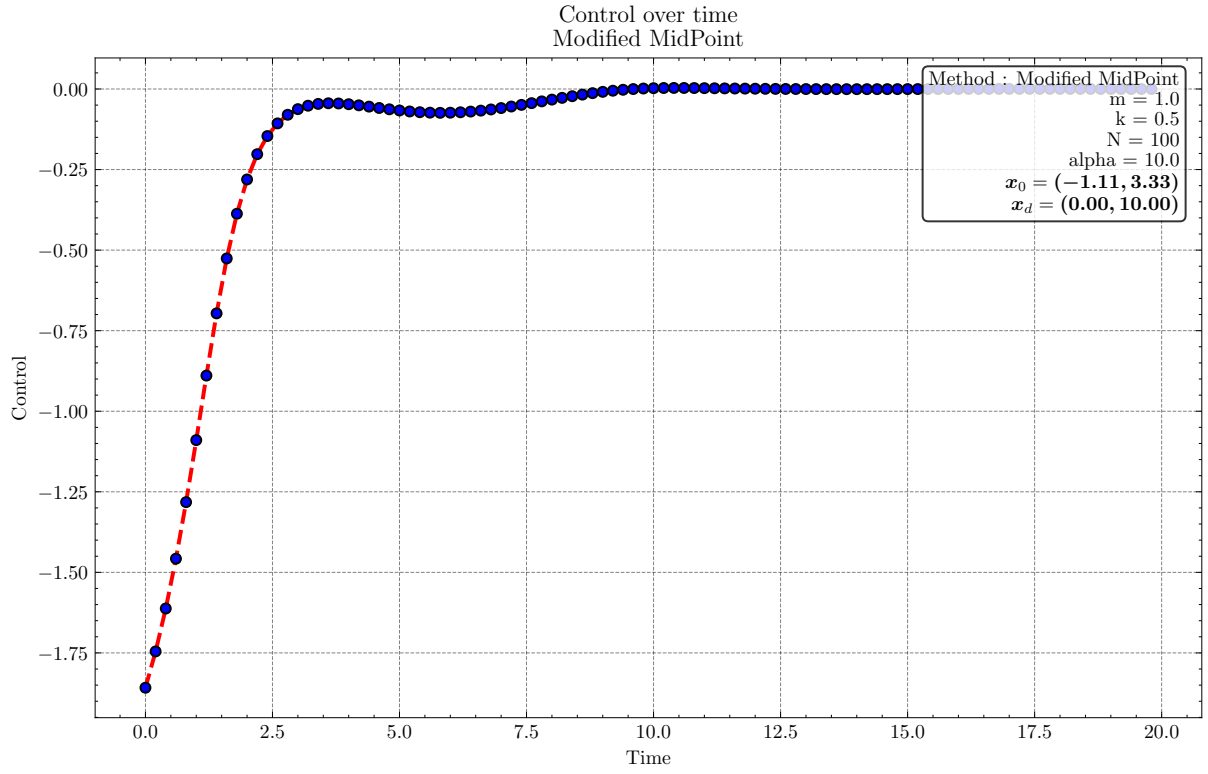
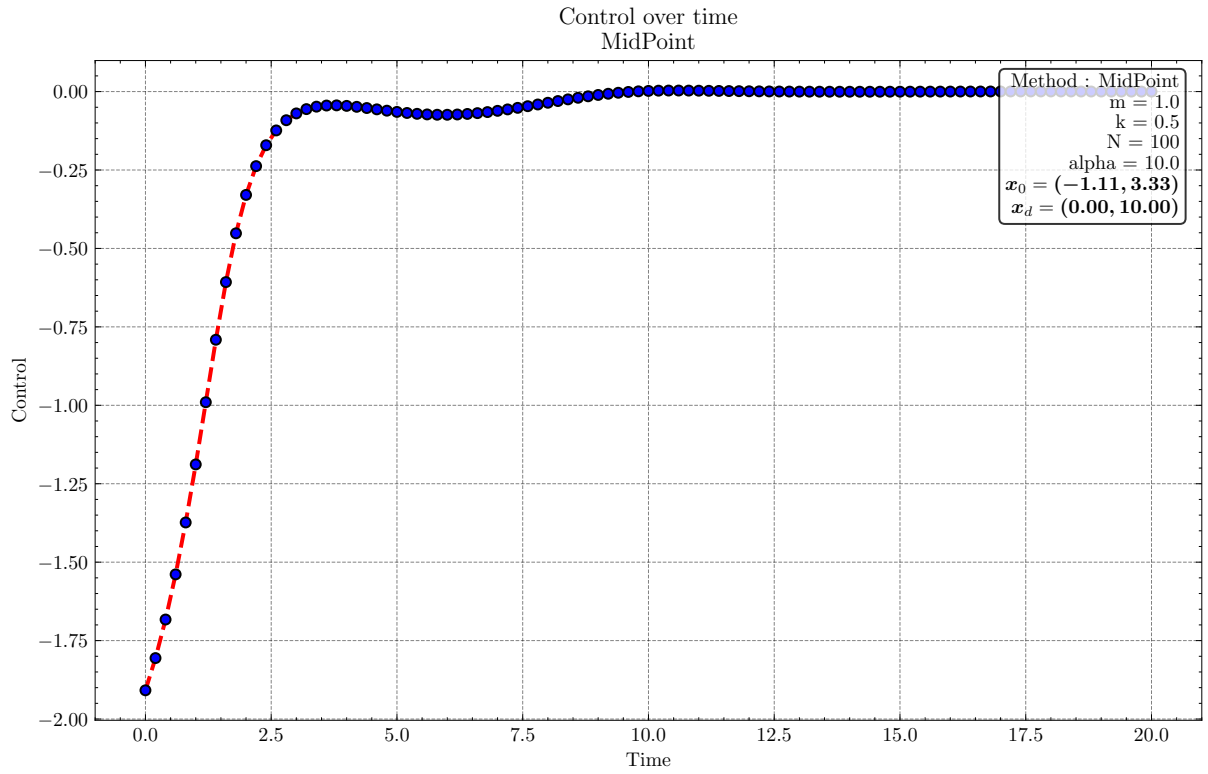


Figure 38: Iteration plots for MidPoint and its modified variant with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.



(a) Modified MidPoint



(b) MidPoint

Figure 39: Control trajectories for $\mathbf{x}_0 = (-1.11, 3.33)$ with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.

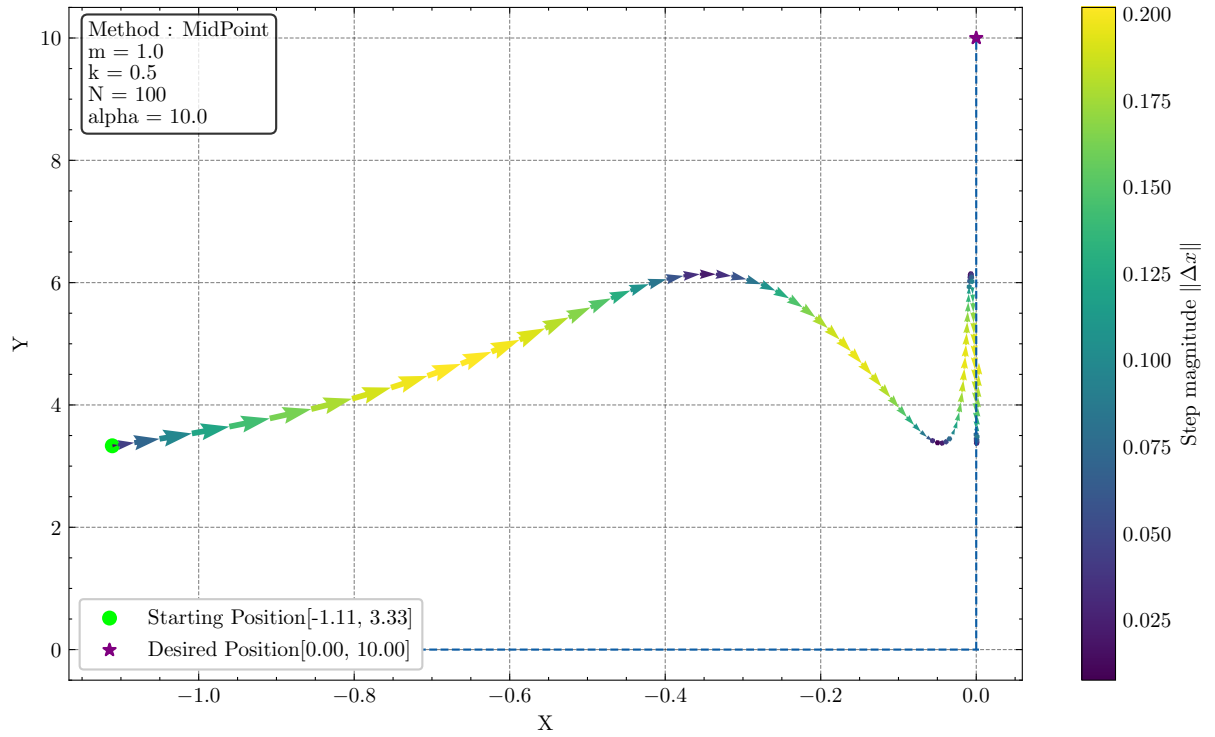
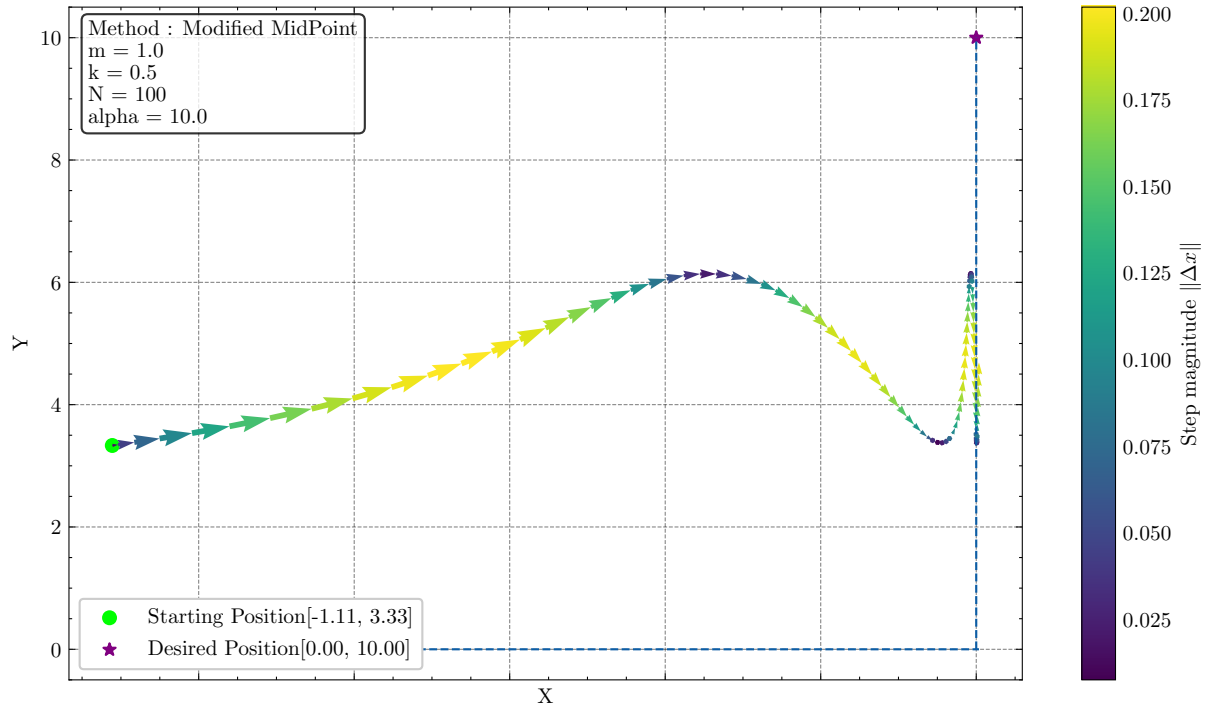


Figure 40: Trajectory comparison for $\mathbf{x}_0 = (-1.11, 3.33)$ with $k = 0.5$, $h = 10.0$, $N = 100$, $m = 1.0$.

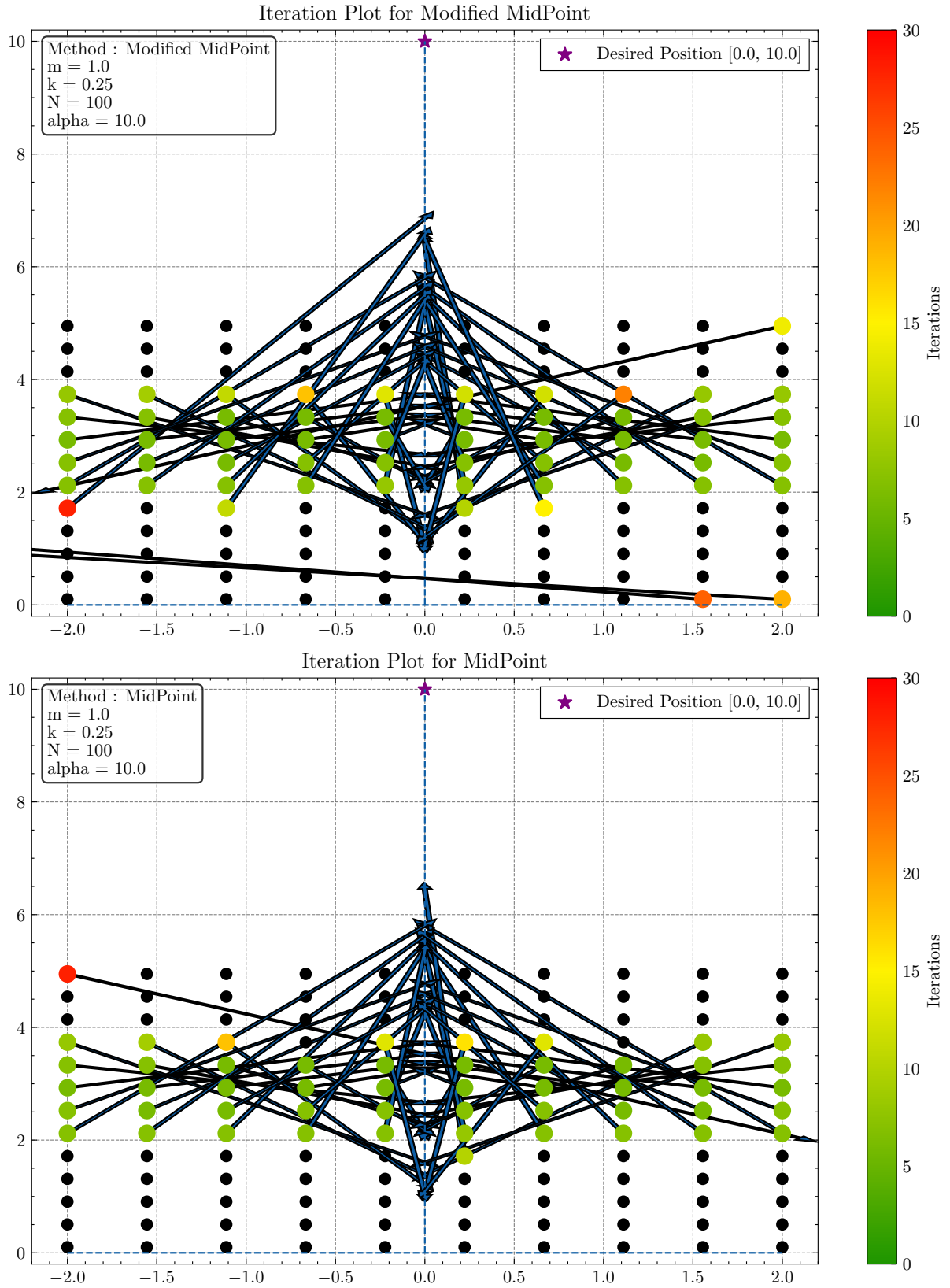
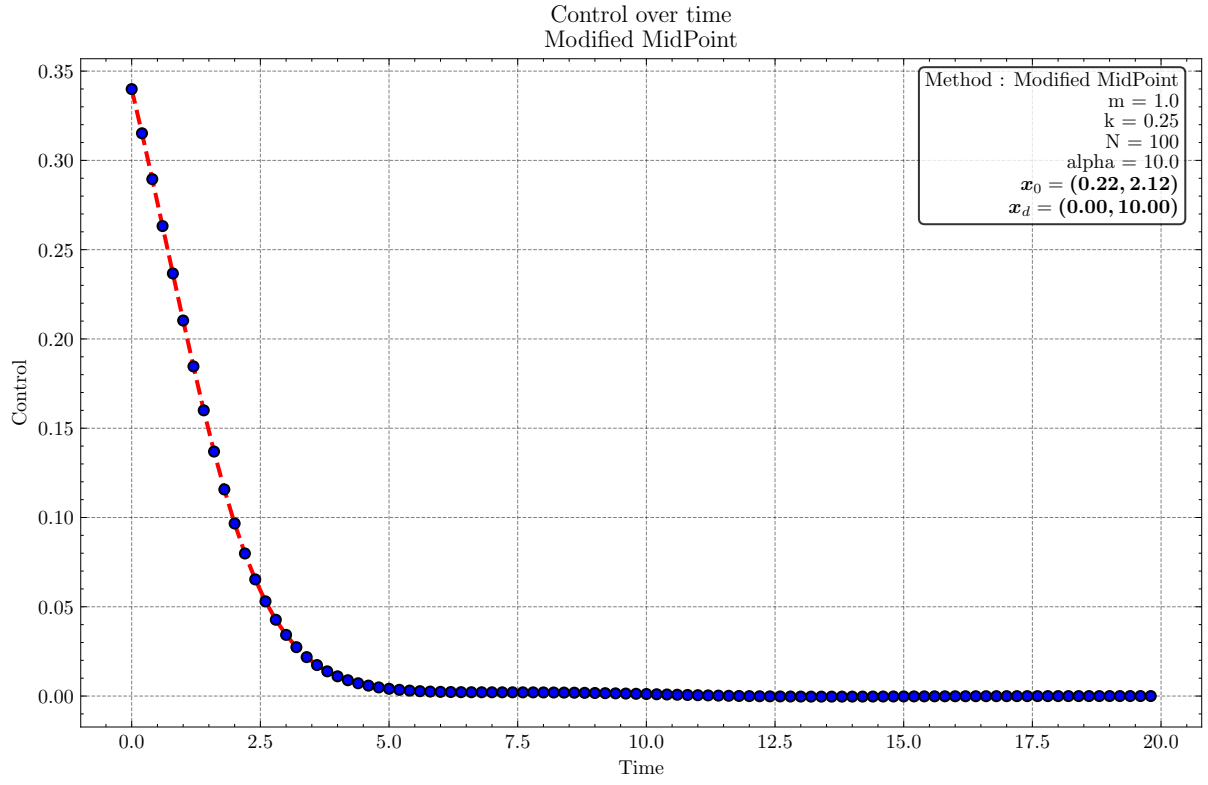
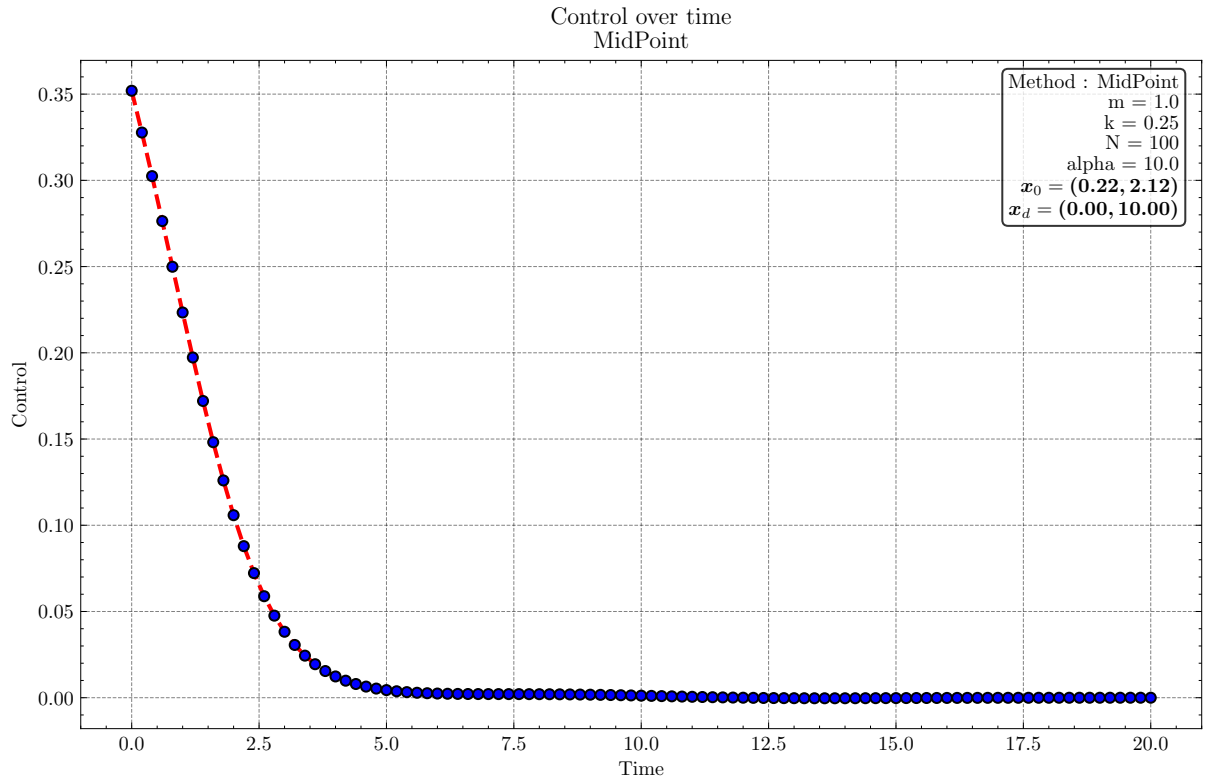


Figure 41: Iteration plots for MidPoint and its modified variant with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.



(a) Modified MidPoint



(b) MidPoint

Figure 42: Control trajectories for $\mathbf{x}_0 = (0.22, 2.12)$ with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.

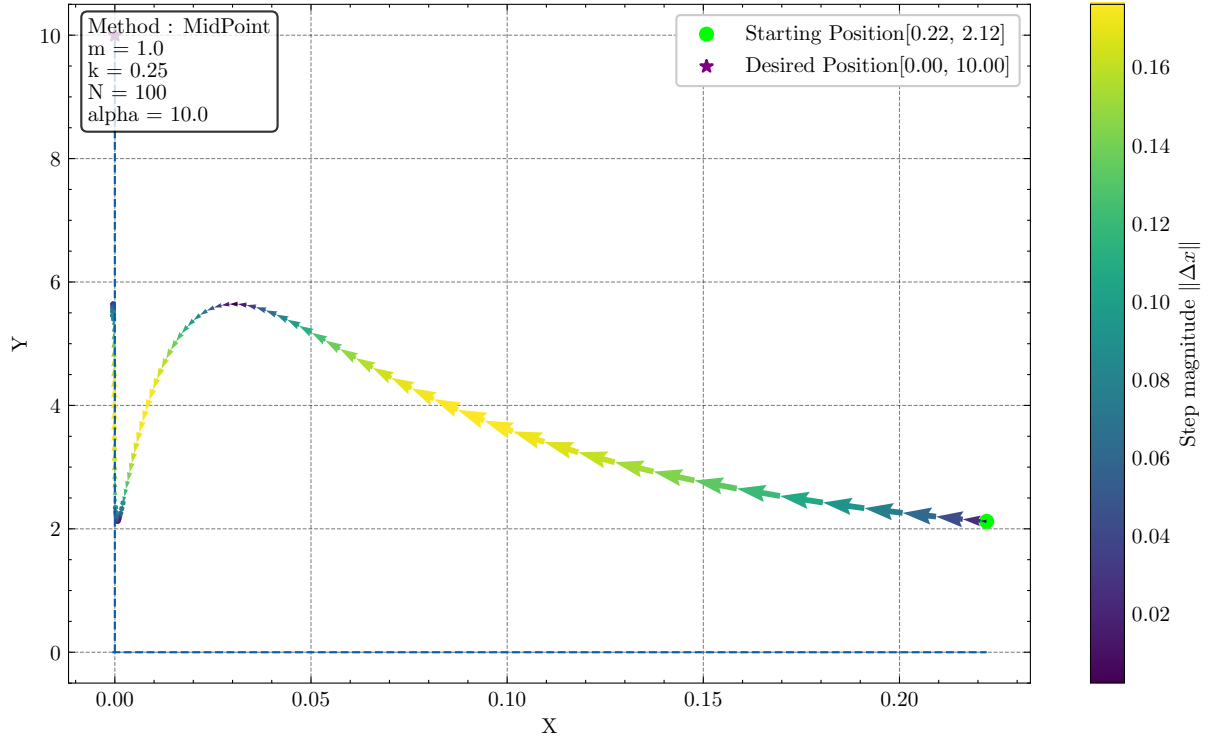
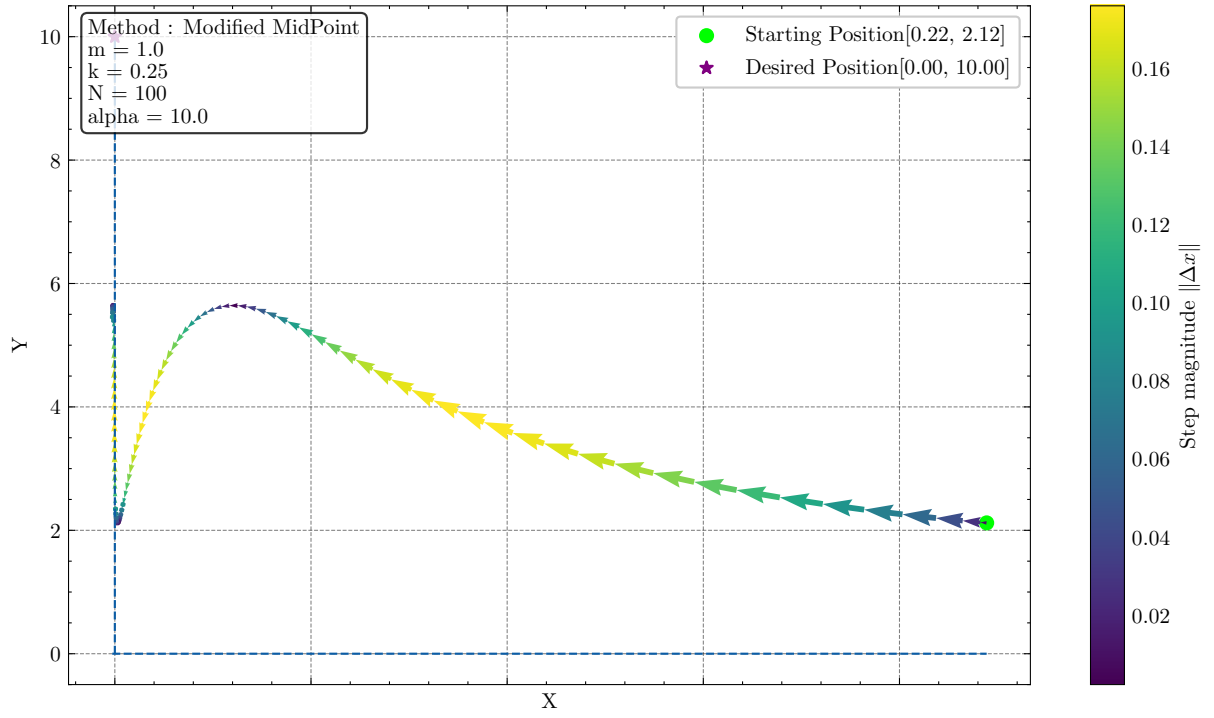


Figure 43: Trajectory comparison for $\mathbf{x}_0 = (0.22, 2.12)$ with $k = 0.25$, $h = 10.0$, $N = 100$, $m = 1.0$.

6 Forward-Backward Euler and RK Sweep Schemes

Newton's method, when supplied with a sufficiently good initial guess, converges to the solution of the discrete optimality conditions in only a handful of iterations. In our setting this is a substantial advantage: each Newton step operates monolithically on the full discretised two-point boundary value problem and is comparatively expensive, but the per-iteration progress is large. The bottleneck is therefore not the

cost of an individual step but the requirement that the iterate already lie inside the basin of attraction of Newton’s method. From a random or otherwise poorly chosen starting point, convergence is unreliable and frequently fails outright.

For the inverted-pendulum problem with $\alpha = 10$ we were able to construct an educated guess by hand. The intuition is straightforward: the regularisation is strong enough that the optimal control keeps the pendulum from falling, pushing the ball laterally so that it eventually reaches the y -axis, after which the pendulum ascends along a near-linear path toward the origin. Encoding this picture into an initial guess produced reliable Newton convergence for $\alpha = 10$ across most of the sampled initial states.

This heuristic, however, is fragile. As α decreases, the regularisation weakens and the optimal trajectory deviates substantially from the linear-ascent picture; the educated guess no longer lies in Newton’s basin and the method stalls or diverges. The same failure mode appears at fixed α when the initial state $\mathbf{x}_0$ corresponds to a hard configuration — a low height or an awkward angle — even when α is large.

To address this, we use the forward–backward gradient method to generate a “good enough” initial guess, which Newton’s method then refines. Forward–backward sweeps are cheap on a per-iteration basis: each sweep propagates the state forward, the adjoint backward, and updates the control without forming or factorising a Jacobian of the full discretised system. The trade-off is the opposite of Newton’s: many cheap iterations rather than a few expensive ones. Because the gradient method only needs to deliver an iterate inside Newton’s basin — not a tightly converged solution — it is well-suited to this preconditioning role.

We computed full forward–backward solutions with tolerance $\varepsilon = 10^{-12}$ using two variants:

- the classical forward–backward sweep with explicit Euler propagation of the state and adjoint, and
- an RK4 forward–backward sweep, which uses fourth-order Runge–Kutta integration for both equations.

All examples share $\mathbf{x}_d = (0.0, 10.0)^T$, $m = 1.0$, $k = 1.0$, $l_0 = 9.0$, and time horizon $[0, 10]$ s. We swept $\alpha \in \{10, 5, 1\}$ over the initial-state grid $\mathbf{x}_0 \in \{1\} \times \{1, 2, 3, 4\}$, with grid sizes $N \in \{100, 200, 400, 600, 800\}$. The study is currently being extended to $\alpha \in \{0.5, 0.1\}$.

For each configuration we report three views of the converged solution: the Hamiltonian $H(t_i)$, the control $u(t_i)$, and the residual cost as a function of iteration index. Together they expose how the solution refines as N increases: H becomes flatter, reflecting its first-integral character along the optimal trajectory; u becomes smoother; and the residual decays more cleanly.

Number of iterations and convergence cap. We recorded both the number of iterations and the wall time of each method. The number of iterations can be large — ranging from a few hundred for the easiest cases (e.g. $\alpha = 10$, $\mathbf{x}_0 = (1, 4)^T$, $N = 100$: 444 RK4-FB iterations, 749 classical-FB iterations) to several hundred thousand for the hardest — but each iteration is cheap, so the overall wall time remains tractable. To bound runtime we capped the gradient iteration at 500,000. A recorded number of iterations equal to 500,001 in the tables therefore signals that the run hit the cap rather than meeting the tolerance.

A non-trivial fraction of the runs in our grid fail to reach tolerance, predominantly at small α and large N . The non-convergence is visible both in the number of iterations — many runs hit the cap of 500,000 iterations — and in the diagnostic plots: the Hamiltonian and control profiles fail to settle to a clean shape, exhibiting noise and oscillations absent from the converged cases. The mechanism behind these failures is a collapse of the step-length rule used in the gradient update, which we describe in detail in Section 6.3, and the per-run outcome of every $(\alpha, N, \mathbf{x}_0)$ in our grid is tabulated in Table 6.

6.1 Discrete forward–backward sweep schemes

The numerical results of Section 7 are produced by two discrete forward–backward sweeps, identical in outer structure and differing only in how the state and adjoint ODEs are integrated. Both schemes target the following optimal-control problems

$$\min_{\mathbf{u}} J(\mathbf{u}) = \int_a^b L(t, \mathbf{y}, \mathbf{u}) dt \quad \text{s.t.} \quad \dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y}, \mathbf{u}), \quad \mathbf{y}(a) = \mathbf{y}_0, \quad (81)$$

which specialises to the inverted-pendulum problem (79) on identification of $\mathbf{y} := (\mathbf{v}, \mathbf{x})^T$ and $\mathbf{p} := (\boldsymbol{\lambda}, \boldsymbol{\mu})^T$. With Hamiltonian

$$H(t, \mathbf{y}, \mathbf{u}, \mathbf{p}) = L(t, \mathbf{y}, \mathbf{u}) + \mathbf{p}^T \mathbf{f}(t, \mathbf{y}, \mathbf{u}), \quad (82)$$

the continuous optimality system reads

$$\dot{\mathbf{y}} = \mathbf{f}, \quad \dot{\mathbf{p}} = -H_{\mathbf{y}} = -L_{\mathbf{y}} - \mathbf{f}_{\mathbf{y}}^T \mathbf{p}, \quad \mathbf{p}(b) = \mathbf{0}, \quad \nabla J(\mathbf{u})(t) = H_{\mathbf{u}} = L_{\mathbf{u}} + \mathbf{f}_{\mathbf{u}}^T \mathbf{p}. \quad (83)$$

The interval $[a, b]$ is divided into N equal subintervals of length $h = (b - a)/N$, with nodes $t_n = a + nh$, $n = 0, \dots, N$. We write $\mathbf{y}_n \approx \mathbf{y}(t_n)$, $\mathbf{u}_n \approx \mathbf{u}(t_n)$, $\mathbf{p}_n \approx \mathbf{p}(t_n)$, and use the superscript (k) to denote the gradient-descent iteration index.

6.1.1 Forward–backward sweep with explicit Euler

For the Euler variant the state is advanced forward by an explicit Euler step,

$$\mathbf{y}_0^{(k)} = \mathbf{y}_0, \quad \mathbf{y}_{n+1}^{(k)} = \mathbf{y}_n^{(k)} + h \mathbf{f}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)}), \quad n = 0, \dots, N-1. \quad (84)$$

Because the Euler update at step n depends on $\mathbf{u}_n$ only, the value $\mathbf{u}_N$ does not enter the state sweep; in practice the control is therefore stored at $t_0, \dots, t_{N-1}$.

For the adjoint, $\dot{\mathbf{p}} = -H_{\mathbf{y}}$ is discretised backward in time. Writing the backward Euler step as $\mathbf{p}_{n-1} = \mathbf{p}_n - h \dot{\mathbf{p}}(t_n) = \mathbf{p}_n + h H_{\mathbf{y}}(t_n, \mathbf{y}_n, \mathbf{u}_n, \mathbf{p}_n)$ — the plus sign coming from $\dot{\mathbf{p}} = -H_{\mathbf{y}}$ — the discrete adjoint sweep is

$$\mathbf{p}_N^{(k)} = \mathbf{0}, \quad \mathbf{p}_{n-1}^{(k)} = \mathbf{p}_n^{(k)} + h [L_{\mathbf{y}}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)}) + \mathbf{f}_{\mathbf{y}}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)})^T \mathbf{p}_n^{(k)}], \quad n = N, \dots, 1. \quad (85)$$

The discrete gradient at node t_n is

$$\mathbf{g}_n^{(k)} := H_{\mathbf{u}}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)}, \mathbf{p}_n^{(k)}) = L_{\mathbf{u}}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)}) + \mathbf{f}_{\mathbf{u}}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)})^T \mathbf{p}_n^{(k)}, \quad (86)$$

and the control is updated by gradient descent,

$$\mathbf{u}_n^{(k+1)} = \mathbf{u}_n^{(k)} - \alpha_k \mathbf{g}_n^{(k)} \quad (87)$$

with step length α_k chosen as in Section 6.2. The iteration is started from an initial guess $\{\mathbf{u}_n^{(0)}\}_{n=0}^{N-1}$ and stopped when either

$$\|\mathbf{u}^{(k+1)} - \mathbf{u}^{(k)}\| < \varepsilon \quad \text{or} \quad \|\mathbf{g}^{(k)}\| < \varepsilon, \quad (88)$$

where $\mathbf{u}^{(k)}$ and $\mathbf{g}^{(k)}$ are the stacked control and gradient vectors at iteration k (over all nodes), and $\|\cdot\|$ is the Euclidean norm. The role of the *or* is discussed in Section 6.3.

6.1.2 Forward–backward sweep with RK4

The RK4 variant replaces both Euler integrations by classical fourth-order Runge–Kutta. Midpoint quantities are required both for the forward sweep at $t_{n+\frac{1}{2}} = t_n + h/2$ and for the backward sweep at $t_{n-\frac{1}{2}} = t_n - h/2$; we approximate the control and state at midpoints by linear interpolation between the surrounding nodes,

$$\mathbf{u}_{n+\frac{1}{2}} \approx \frac{\mathbf{u}_n + \mathbf{u}_{n+1}}{2}, \quad \mathbf{u}_{n-\frac{1}{2}} \approx \frac{\mathbf{u}_{n-1} + \mathbf{u}_n}{2}, \quad \mathbf{y}_{n-\frac{1}{2}} \approx \frac{\mathbf{y}_{n-1} + \mathbf{y}_n}{2}. \quad (89)$$

Unlike the Euler variant, the RK4 forward update at step n involves $\mathbf{u}_{n+1}$ through $\mathbf{k}_4$ (see below), so the control is stored at *every* node $t_0, \dots, t_N$. The backward sweep additionally reuses the state midpoint $\mathbf{y}_{n-\frac{1}{2}}$ from the converged forward trajectory, rather than recomputing intermediate stages.

Forward sweep. For $n = 0, \dots, N-1$:

$$\mathbf{k}_1 = \mathbf{f}(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)}), \quad (90)$$

$$\mathbf{k}_2 = \mathbf{f}(t_n + \frac{h}{2}, \mathbf{y}_n^{(k)} + \frac{h}{2} \mathbf{k}_1, \mathbf{u}_{n+\frac{1}{2}}^{(k)}), \quad (91)$$

$$\mathbf{k}_3 = \mathbf{f}(t_n + \frac{h}{2}, \mathbf{y}_n^{(k)} + \frac{h}{2} \mathbf{k}_2, \mathbf{u}_{n+\frac{1}{2}}^{(k)}), \quad (92)$$

$$\mathbf{k}_4 = \mathbf{f}(t_n + h, \mathbf{y}_n^{(k)} + h \mathbf{k}_3, \mathbf{u}_{n+1}^{(k)}), \quad (93)$$

$$\mathbf{y}_{n+1}^{(k)} = \mathbf{y}_n^{(k)} + \frac{h}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad (94)$$

with $\mathbf{y}_0^{(k)} = \mathbf{y}_0$.

Backward sweep. Defining

$$g(t, \mathbf{y}, \mathbf{u}, \mathbf{p}) := -L_{\mathbf{y}}(t, \mathbf{y}, \mathbf{u}) - \mathbf{f}_{\mathbf{y}}(t, \mathbf{y}, \mathbf{u})^T \mathbf{p} \quad (95)$$

so that $\dot{\mathbf{p}} = g$, the RK4 stages for $n = N, N-1, \dots, 1$ read

$$\ell_1 = g(t_n, \mathbf{y}_n^{(k)}, \mathbf{u}_n^{(k)}, \mathbf{p}_n^{(k)}), \quad (96)$$

$$\ell_2 = g(t_n - \frac{h}{2}, \mathbf{y}_{n-\frac{1}{2}}^{(k)}, \mathbf{u}_{n-\frac{1}{2}}^{(k)}, \mathbf{p}_n^{(k)} - \frac{h}{2}\ell_1), \quad (97)$$

$$\ell_3 = g(t_n - \frac{h}{2}, \mathbf{y}_{n-\frac{1}{2}}^{(k)}, \mathbf{u}_{n-\frac{1}{2}}^{(k)}, \mathbf{p}_n^{(k)} - \frac{h}{2}\ell_2), \quad (98)$$

$$\ell_4 = g(t_{n-1}, \mathbf{y}_{n-1}^{(k)}, \mathbf{u}_{n-1}^{(k)}, \mathbf{p}_n^{(k)} - h\ell_3), \quad (99)$$

$$\mathbf{p}_{n-1}^{(k)} = \mathbf{p}_n^{(k)} - \frac{h}{6}(\ell_1 + 2\ell_2 + 2\ell_3 + \ell_4), \quad (100)$$

with $\mathbf{p}_N^{(k)} = \mathbf{0}$.

The gradient is evaluated at every node $n = 0, \dots, N$ via (86); the control update (87) is applied at every node $n = 0, \dots, N$; and the stopping rule (88) is identical to the Euler variant.

6.2 Step length: stabilised Barzilai–Borwein

The step length α_k in (87) is critical to convergence: too small and the iteration crawls, too large and the cost diverges. Throughout this work α_k is set by the stabilised Barzilai–Borwein (BB) line search of Bijalwan and Muñoz (Computer Methods in Applied Mechanics and Engineering 420, 2024, Algorithm 2), which combines the short BB step with a safeguard against overlong increments.

Stacking the control and gradient over all nodes into the vectors $\mathbf{u}^k = (\mathbf{u}_n^{(k)})_n$ and $\mathbf{d}^k = (\mathbf{g}_n^{(k)})_n$, let $\Delta\mathbf{u} = \mathbf{u}^k - \mathbf{u}^{k-1}$ and $\Delta\mathbf{d} = \mathbf{d}^k - \mathbf{d}^{k-1}$. The short BB step is

$$\theta_s^k = -\frac{\Delta\mathbf{u}^T \Delta\mathbf{d}}{\|\Delta\mathbf{d}\|^2}. \quad (101)$$

For non-convex objectives θ_s^k may be negative; the geometric-mean fallback

$$\theta_m^k = \frac{\|\Delta\mathbf{u}\|}{\|\Delta\mathbf{d}\|} \quad (102)$$

is then used instead. To prevent overlong increments, an upper bound $\theta_{\max} > 0$ on the displacement $\|\mathbf{u}^{k+1} - \mathbf{u}^k\|$ is imposed, which yields the threshold

$$\theta_{\text{th}}^k = \frac{\theta_{\max}}{\|\mathbf{d}^k\|}. \quad (103)$$

The final BB step length used in (87) is

$$\alpha_k = \theta^k = \begin{cases} \min(\theta_s^k, \theta_{\text{th}}^k), & \theta_s^k > 0, \\ \min(\theta_m^k, \theta_{\text{th}}^k), & \theta_s^k < 0, \end{cases} \quad (104)$$

with $\theta_{\max}$ a fixed positive parameter of the algorithm. The Stabilised BB rule (104) requires the previous increments $\Delta\mathbf{u}, \Delta\mathbf{d}$, which are only defined for $k \geq 1$; we therefore supply α_0 as an externally specified input parameter and apply (104) from $k = 1$ onward.

6.3 Premature termination from BB collapse

The stabilised BB rule of Section 6.2 bounds α_k from above but *not* from below. Near a stationary point the increments $\Delta\mathbf{u}$ and $\Delta\mathbf{d}$ shrink together at comparable rates, so the ratios in (101)–(102) approach 0/0 and eventually return NaN in floating-point arithmetic. Once $\alpha_k = \text{NaN}$, the next control update (87) destroys the iterate.

The collapse is most severe at small α — the regularisation weight in the cost, not to be confused with the BB step α_k — where the gradient $H_{\mathbf{u}} = \alpha \mathbf{u} + \dots$ responds weakly to changes in $\mathbf{u}$, and at large N , where per-component magnitudes shrink with the time step. In those regimes the BB step length collapses toward zero faster than $\|H_{\mathbf{u}}\|$ falls below the tolerance ε .

To avoid silent corruption of the iterate we relaxed the convergence test from the conjunction

$$\|\mathbf{u}^{(k+1)} - \mathbf{u}^{(k)}\| < \varepsilon \quad \text{and} \quad \|H_{\mathbf{u}}^{(k)}\| < \varepsilon$$

to the disjunction in (88): the iteration stops as soon as *either* the control update or the gradient becomes small. The price is that some runs terminate while $\|H_{\mathbf{u}}\|$ is still well above ε ; these are the rows shaded in Table 6. The final BB step lengths θ_{last}^k in those rows are several orders of magnitude smaller than in the converged rows at the same $(\alpha, N, \mathbf{x}_0)$, which is precisely the BB-collapse signature.

7 Forward–Backward Numerical Results

We present the converged forward–backward solutions across the $(\Delta t, \alpha, \mathbf{x}_0)$ grid described in Section 6. For each configuration we report two views of the converged solution: the Hamiltonian $H_i := H(\mathbf{y}_i, u_i, \mathbf{p}_i)$ along the discretised trajectory and the optimal control u_i . Each plot compares the two forward–backward variants on the same axes: classical Euler forward–backward (blue) and RK4 forward–backward (orange).

We organise the results by α . For each $(\alpha, \mathbf{x}_0)$ shown we display the Hamiltonian and the control as Δt is decreased from 0.1 down to 0.0125. We restrict the figures to configurations in which *every* Δt in the sweep converged according to the criterion of Section 6.3; the full convergence outcome of every $(\alpha, N, \mathbf{x}_0)$ in our grid is in Table 6. With this restriction, only $\alpha = 10$ shows the complete $\mathbf{x}_0 \in \{(1, 1), (1, 2), (1, 3), (1, 4)\}^T$ sweep; at $\alpha = 5$ only $\mathbf{x}_0 = (1, 2)^T$ converged across every Δt ; and at $\alpha = 1$ no initial state did. Two trends are universal across the converged cases: as Δt decreases, the Hamiltonian flattens toward the constant value expected of a first integral of the optimality system, and the control profile sharpens around the boundary layers and stabilises in the bulk. The discrepancies between Euler and RK4 forward–backward shrink with Δt as well, since both variants are recovering the same continuous-time optimum.

7.1 Convergence and non-convergence

In a non-trivial fraction of configurations the gradient method failed to reach tolerance. The diagnostic plots make these runs unmistakable: H_i is far from constant and the control profile fails to settle, instead exhibiting persistent oscillations and discontinuous jumps. The mechanism behind the failure is the BB step-length collapse analysed in Section 6.3: the iteration is stopped by the $\|\Delta \mathbf{u}\| < \varepsilon$ branch of (88) while $\|H_{\mathbf{u}}\|$ is still well above ε , because the stabilised BB rule has driven θ^k to floating-point underflow. Such runs are not used as Newton seeds.

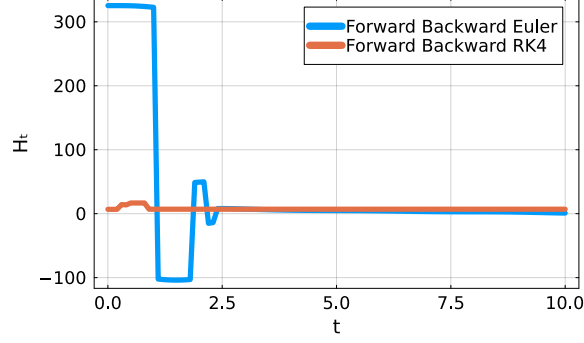
Table 6 reports, for every $(\alpha, N, \mathbf{x}_0)$ in our grid, the number of iterations, wall time, final $\|H_{\mathbf{u}}\|$, and final BB step length θ_{last}^k for both Euler and RK4 forward–backward. Runs flagged as failed — those where $\|H_{\mathbf{u}}\| > 10^{-5}$ at termination — are shaded; their θ_{last}^k entries are several orders of magnitude smaller than the converged rows, which is the BB-collapse signature described in Section 6.3. A dash “—” in the number-of-iterations column means the run hit the 500,000-iteration cap before any stopping criterion fired. The figures below restrict to configurations in which every Δt converged.

Table 6: Forward–backward results across the $(N, \alpha, \mathbf{x}_0)$ grid. The first three columns are the problem parameters: regularisation weight α , time-grid resolution N , and initial state $\mathbf{x}_0$. For each of the two forward–backward variants — RK4 (“FB RK4”) and explicit Euler (“FB Euler”) — we report five quantities at termination: *iters* — the number of gradient iterations until the stopping rule (88) fired; *time* — wall time of the run; $\|H_u\|$ — the Euclidean norm of the stacked discrete control gradient $\mathbf{g}^{(k)}$ (the historical control-equation diagnostic); $\|r\|$ — the Euclidean norm of the full optimality-system residual, i.e. the state, adjoint and control conditions of the chosen discretisation stacked into one vector, which is zero iff the iterate is an exact discrete solution; and θ_{last} — the final stabilised Barzilai–Borwein step length. All norms are Euclidean. — in the *iters* column indicates the 500,000-iteration cap was reached without meeting tolerance. Rows shaded in red are runs whose $\|r\| > 10^{-5}$ — i.e. the iterate is not a discrete solution of the optimality system; their θ_{last} is several orders of magnitude smaller than the converged rows at the same $(\alpha, N, \mathbf{x}_0)$, which is the BB-collapse signature analysed in Section 6.3.

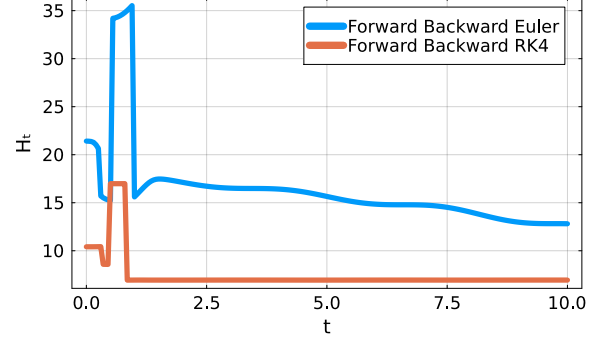
			FB RK4					FB Euler				
α	N	$\mathbf{x}_0$	iters	time	$\ H_u\ $	$\ r\ $	θ_{last}	iters	time	$\ H_u\ $	$\ r\ $	θ_{last}
1.0	100	[1.0, 1.0]	4,541	3.2s	1.46×10^8	1.46×10^8	3.47×10^{-23}	—	2m 15.3s	6.08×10^2	6.08×10^2	1.00×10^{-6}
1.0	100	[1.0, 2.0]	5,838	5.2s	3.03×10^{-11}	3.03×10^{-11}	3.08×10^{-4}	—	2m 50.5s	2.43×10^2	2.43×10^2	1.00×10^{-6}
1.0	100	[1.0, 3.0]	41,767	45.9s	1.04×10^2	1.04×10^2	1.63×10^{-18}	1,831	0.8s	2.90×10^{-11}	2.90×10^{-11}	5.49×10^{-5}
1.0	100	[1.0, 4.0]	3,131	2.8s	4.71×10^{-10}	4.71×10^{-10}	1.20×10^{-5}	—	2m 45.5s	1.11×10^3	1.11×10^3	1.00×10^{-6}
1.0	200	[1.0, 1.0]	38,720	58.6s	3.70×10^7	3.70×10^7	3.71×10^{-23}	—	4m 10.6s	1.55×10^3	1.55×10^3	1.00×10^{-6}
1.0	200	[1.0, 2.0]	31,305	52.8s	4.73×10^5	4.73×10^5	2.19×10^{-21}	—	5m 14.5s	2.00×10^3	2.00×10^3	1.00×10^{-6}
1.0	200	[1.0, 3.0]	98,850	2m 18.1s	1.06×10^2	1.06×10^2	8.58×10^{-17}	—	4m 40.0s	1.57×10^3	1.57×10^3	1.00×10^{-6}
1.0	200	[1.0, 4.0]	1,902	5.1s	8.50×10^{-11}	8.50×10^{-11}	1.08×10^{-4}	1,518	1.8s	1.03×10^{-10}	1.03×10^{-10}	5.05×10^{-5}
1.0	400	[1.0, 1.0]	9,511	29.0s	7.21×10^6	7.21×10^6	5.06×10^{-22}	—	9m 36.4s	2.52×10^3	2.52×10^3	1.00×10^{-6}
1.0	400	[1.0, 2.0]	15,370	47.7s	3.33×10^4	3.33×10^4	2.40×10^{-19}	—	9m 36.6s	5.96×10^2	5.96×10^2	1.20×10^{-6}
1.0	400	[1.0, 3.0]	20,758	1m 5.2s	2.22×10^2	2.22×10^2	3.87×10^{-17}	—	9m 37.2s	9.76×10^2	9.76×10^2	1.00×10^{-6}
1.0	400	[1.0, 4.0]	1,681	9.3s	3.97×10^{-11}	3.97×10^{-11}	1.68×10^{-4}	2,201	3.5s	2.06×10^{-11}	2.06×10^{-11}	7.66×10^{-5}
1.0	600	[1.0, 1.0]	19,975	1m 1.5s	7.53×10^2	7.53×10^2	8.12×10^{-18}	—	9m 51.0s	2.62×10^3	2.62×10^3	1.00×10^{-6}
1.0	600	[1.0, 2.0]	8,229	28.8s	2.28×10^3	2.28×10^3	1.59×10^{-19}	—	9m 58.0s	3.10×10^3	3.10×10^3	1.00×10^{-6}
1.0	600	[1.0, 3.0]	45,000	1m 54.6s	2.12×10^2	2.12×10^2	7.00×10^{-18}	—	6m 41.2s	2.36×10^3	2.36×10^3	1.00×10^{-6}
1.0	600	[1.0, 4.0]	3,119	18.4s	3.80×10^{-11}	3.81×10^{-11}	8.44×10^{-5}	2,527	5.1s	4.54×10^{-11}	4.54×10^{-11}	6.25×10^{-5}
1.0	800	[1.0, 1.0]	11,624	1m 3.2s	1.14×10^3	1.14×10^3	6.90×10^{-17}	338,728	10m 40.4s	6.70×10^{-9}	6.70×10^{-9}	1.41×10^{-4}
1.0	800	[1.0, 2.0]	15,388	1m 4.3s	5.73×10^2	5.73×10^2	1.71×10^{-15}	493,336	14m 38.5s	2.87×10^{-9}	2.87×10^{-9}	3.29×10^{-4}
1.0	800	[1.0, 3.0]	7,091	31.1s	1.95×10^2	1.95×10^2	4.16×10^{-15}	—	14m 48.6s	1.30×10^3	1.30×10^3	1.00×10^{-6}
1.0	800	[1.0, 4.0]	1,814	15.4s	8.57×10^{-10}	8.57×10^{-10}	8.31×10^{-4}	2,410	7.0s	9.65×10^{-10}	9.65×10^{-10}	8.86×10^{-4}
5.0	100	[1.0, 1.0]	82,037	1m 3.7s	7.35×10^5	7.35×10^5	1.12×10^{-20}	—	2m 21.6s	1.72×10^3	1.72×10^3	1.00×10^{-6}
5.0	100	[1.0, 2.0]	2,132	2.8s	3.01×10^{-10}	3.01×10^{-10}	2.84×10^{-5}	1,102	0.7s	4.08×10^{-10}	4.08×10^{-10}	2.41×10^{-5}
5.0	100	[1.0, 3.0]	4,823	6.7s	1.13×10^9	1.13×10^9	6.83×10^{-24}	1,526	1.3s	1.37×10^{-10}	1.37×10^{-10}	2.69×10^{-5}
5.0	100	[1.0, 4.0]	568	1.3s	8.27×10^{-12}	8.27×10^{-12}	2.41×10^{-4}	1,062	0.6s	4.53×10^{-11}	4.53×10^{-11}	2.10×10^{-4}
5.0	200	[1.0, 1.0]	7,611	12.7s	7.90×10^{-8}	7.90×10^{-8}	1.20×10^{-7}	—	5m 7.3s	1.42×10^3	1.42×10^3	1.00×10^{-6}
5.0	200	[1.0, 2.0]	2,648	6.9s	1.10×10^{-9}	1.10×10^{-9}	1.63×10^{-7}	2,557	2.1s	4.24×10^{-9}	4.24×10^{-9}	1.00×10^{-6}
5.0	200	[1.0, 3.0]	99,944	3m 20.1s	6.14×10^4	6.14×10^4	1.07×10^{-20}	1,709	1.5s	2.81×10^{-11}	2.81×10^{-11}	8.99×10^{-5}
5.0	200	[1.0, 4.0]	690	1.6s	3.11×10^{-11}	3.11×10^{-11}	2.41×10^{-4}	1,243	1.1s	7.85×10^{-11}	7.85×10^{-11}	7.38×10^{-5}
5.0	400	[1.0, 1.0]	8,811	26.0s	2.95×10^{-8}	2.95×10^{-8}	1.05×10^{-7}	9,252	8.9s	7.24×10^{-9}	7.24×10^{-9}	1.00×10^{-6}
5.0	400	[1.0, 2.0]	3,696	12.9s	3.22×10^{-9}	3.22×10^{-9}	1.14×10^{-6}	4,100	5.1s	7.06×10^{-10}	7.06×10^{-10}	1.30×10^{-5}
5.0	400	[1.0, 3.0]	122,658	5m 7.6s	3.20×10^2	3.20×10^2	1.84×10^{-17}	2,592	3.1s	4.55×10^{-10}	4.55×10^{-10}	4.49×10^{-6}
5.0	400	[1.0, 4.0]	211,088	11m 20.6s	1.75×10^3	1.75×10^3	3.37×10^{-18}	21,949	33.4s	4.45×10^{-11}	4.45×10^{-11}	2.16×10^{-4}
5.0	600	[1.0, 1.0]	8,662	39.2s	7.95×10^{-8}	7.95×10^{-8}	5.95×10^{-8}	9,038	12.9s	1.33×10^{-9}	1.33×10^{-9}	5.47×10^{-6}
5.0	600	[1.0, 2.0]	4,575	19.9s	1.57×10^{-9}	1.57×10^{-9}	3.15×10^{-6}	4,653	7.8s	1.73×10^{-9}	1.73×10^{-9}	1.00×10^{-6}
5.0	600	[1.0, 3.0]	180,313	9m 20.4s	2.22×10^2	2.22×10^2	1.29×10^{-17}	253,748	5m 12.8s	4.92×10^{-10}	4.92×10^{-10}	1.37×10^{-5}
5.0	600	[1.0, 4.0]	1,097	6.5s	2.08×10^{-11}	2.08×10^{-11}	2.41×10^{-4}	23,963	53.6s	2.21×10^{-10}	2.21×10^{-10}	1.31×10^{-5}
5.0	800	[1.0, 1.0]	9,088	1m 5.5s	4.10×10^{-7}	4.10×10^{-7}	9.33×10^{-7}	8,824	27.1s	1.61×10^{-7}	1.61×10^{-7}	5.27×10^{-6}
5.0	800	[1.0, 2.0]	4,751	38.7s	3.01×10^{-8}	3.01×10^{-8}	2.87×10^{-5}	4,869	14.9s	1.50×10^{-8}	1.50×10^{-8}	2.96×10^{-5}
5.0	800	[1.0, 3.0]	62,552	3m 32.6s	2.84×10^2	2.84×10^2	2.67×10^{-15}	—	14m 48.2s	2.16×10^3	2.16×10^3	1.00×10^{-6}
5.0	800	[1.0, 4.0]	6,758	47.8s	5.40×10^2	5.40×10^2	9.41×10^{-16}	15,816	45.5s	2.70×10^{-9}	2.70×10^{-9}	2.29×10^{-4}
10.0	100	[1.0, 1.0]	4,800	4.6s	1.55×10^{-8}	1.55×10^{-8}	5.84×10^{-8}	299,125	1m 43.9s	6.85×10^{-11}	6.85×10^{-11}	1.36×10^{-4}
10.0	100	[1.0, 2.0]	2,453	2.6s	1.49×10^{-8}	1.49×10^{-8}	5.96×10^{-7}	2,152	1.0s	3.62×10^{-9}	3.62×10^{-9}	2.05×10^{-6}
10.0	100	[1.0, 3.0]	990	1.1s	1.52×10^{-10}	1.52×10^{-10}	6.14×10^{-5}	1,886	0.8s	4.22×10^{-9}	4.22×10^{-9}	1.79×10^{-6}
10.0	100	[1.0, 4.0]	424	0.6s	1.16×10^{-11}	1.16×10^{-11}	2.26×10^{-4}	840	0.5s	9.12×10^{-11}	9.12×10^{-11}	1.08×10^{-4}
10.0	200	[1.0, 1.0]	4,717	6.8s	1.61×10^{-7}	1.61×10^{-7}	1.55×10^{-8}	5,105	2.5s	8.99×10^{-9}	8.99×10^{-9}	1.00×10^{-6}
10.0	200	[1.0, 2.0]	3,253	5.7s	2.61×10^{-10}	2.61×10^{-10}	1.17×10^{-5}	3,809	2.3s	2.56×10^{-9}	2.56×10^{-9}	3.12×10^{-6}
10.0	200	[1.0, 3.0]	1,235	2.6s	1.36×10^{-10}	1.36×10^{-10}	6.13×10^{-5}	1,989	1.5s	5.33×10^{-11}	5.33×10^{-11}	3.57×10^{-5}
10.0	200	[1.0, 4.0]	531	0.9s	1.39×10^{-11}	1.39×10^{-11}	2.26×10^{-4}	688	0.5s	5.26×10^{-11}	5.26×10^{-11}	1.67×10^{-4}
10.0	400	[1.0, 1.0]	6,878	21.7s	2.55×10^{-8}	2.55×10^{-8}	1.38×10^{-7}	4,347	4.3s	1.98×10^{-9}	1.98×10^{-9}	1.96×10^{-6}
10.0	400	[1.0, 2.0]	3,893	13.5s	9.38×10^{-8}	9.38×10^{-8}	2.72×10^{-8}	4,475	5.1s	1.94×10^{-9}	1.94×10^{-9}	2.48×10^{-6}
10.0	400	[1.0, 3.0]	1,484	5.0s	4.01×10^{-11}	4.01×10^{-11}	6.13×10^{-5}	1,807	1.9s	2.09×10^{-10}	2.09×10^{-10}	4.67×10^{-5}
10.0	400	[1.0, 4.0]	698	3.2s	3.93×10^{-11}	3.93×10^{-11}	2.51×10^{-4}	721	1.5s	4.62×10^{-11}	4.62×10^{-11}	1.97×10^{-4}
10.0	600	[1.0, 1.0]	7,635	37.6s	4.83×10^{-8}	4.83×10^{-8}	5.28×10^{-8}	9,508	15.4s	9.95×10^{-9}	9.95×10^{-9}	1.00×10^{-6}
10.0	600	[1.0, 2.0]	4,692	22.0s	4.83×10^{-9}	4.83×10^{-9}	2.01×10^{-6}	4,493	7.8s	3.99×10^{-9}	3.99×10^{-9}	2.35×10^{-6}
10.0	600	[1.0, 3.0]	1,860	9.0s	1.29×10^{-10}	1.29×10^{-10}	6.13×10^{-5}	2,048	3.2s	2.53×10^{-10}	2.53×10^{-10}	2.71×10^{-5}
10.0	600	[1.0, 4.0]	761	5.3s	3.06×10^{-11}	3.07×10^{-11}	2.26×10^{-4}	892	2.7s	4.53×10^{-11}	4.53×10^{-11}	2.08×10^{-4}
10.0	800	[1.0, 1.0]	10,947	1m 14.1s	4.00×10^{-7}	4.00×10^{-7}	2.09×10^{-6}	8,102	23.1s	8.04×10^{-9}	8.04×10^{-9}	2.02×10^{-6}
10.0	800	[1.0, 2.0]	4,907	40.2s	8.32×10^{-8}	8.32×10^{-8}	1.17×10^{-5}	4,946	15.2s	5.39×10^{-8}	5.39×10^{-8}	1.04×10^{-5}

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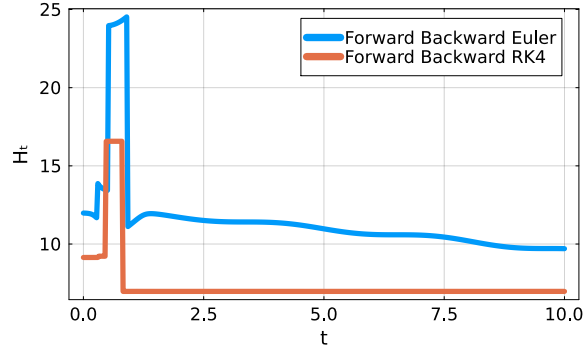
α	N	$\mathbf{x}_0$	FB RK4					FB Euler				
			iters	time	$\ H_u\ $	$\ r\ $	θ_{last}	iters	time	$\ H_u\ $	$\ r\ $	θ_{last}
10.0	800	[1.0, 3.0]	1,897	17.5s	1.50×10^{-8}	1.50×10^{-8}	6.13×10^{-5}	2,014	6.2s	3.30×10^{-9}	3.30×10^{-9}	5.46×10^{-5}
10.0	800	[1.0, 4.0]	860	7.8s	3.16×10^{-9}	3.16×10^{-9}	2.26×10^{-4}	896	2.9s	1.96×10^{-9}	1.96×10^{-9}	2.11×10^{-4}



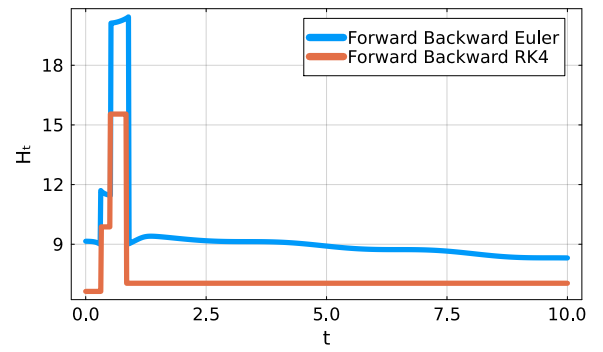
(a) $\Delta t = 0.1$



(b) $\Delta t = 0.05$

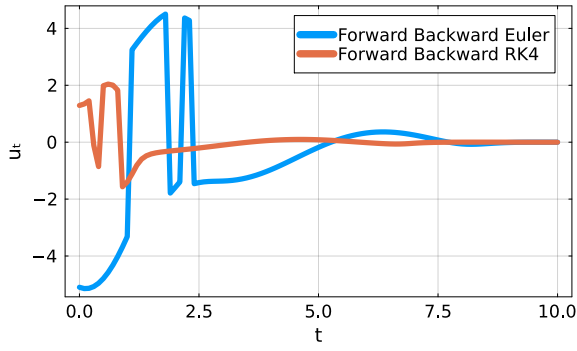


(c) $\Delta t = 0.025$

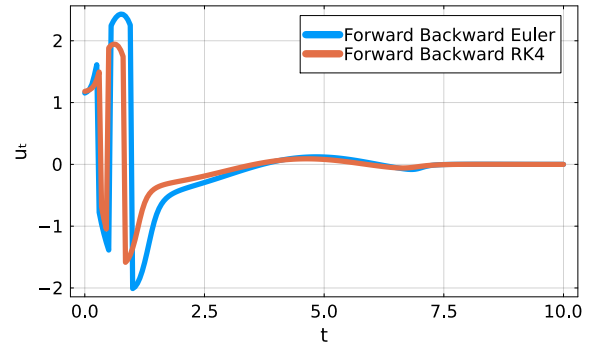


(d) $\Delta t = 0.0125$

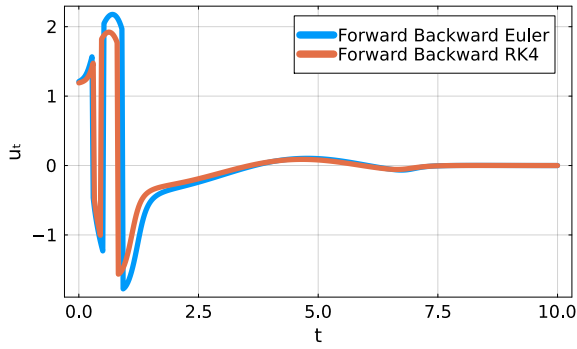
Figure 44: Hamiltonian profile for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 1.0)^T$, as the time step is refined. H_i flattens with smaller Δt , in agreement with its first-integral character along the optimal trajectory.



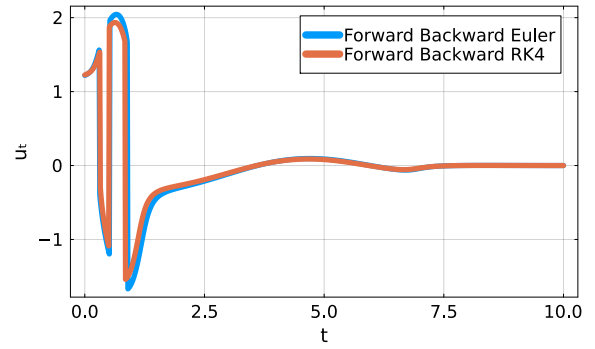
(a) $\Delta t = 0.1$



(b) $\Delta t = 0.05$

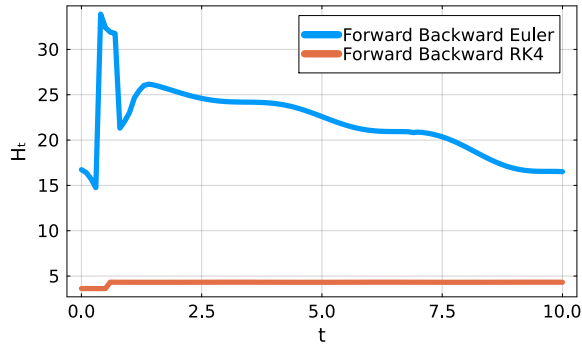


(c) $\Delta t = 0.025$

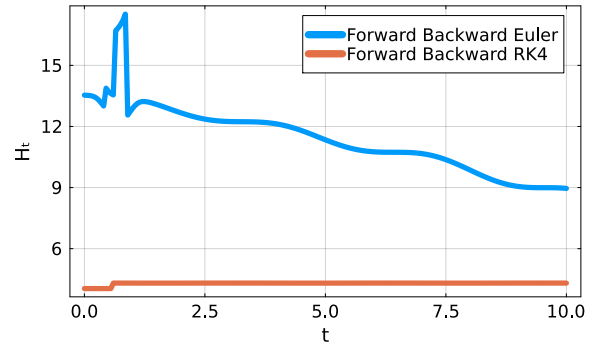


(d) $\Delta t = 0.0125$

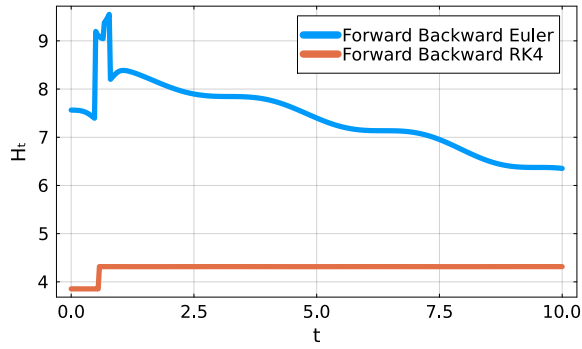
Figure 45: Control u_i for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 1.0)^T$, as the time step is refined. Boundary-layer transients sharpen and the bulk profile stabilises with smaller Δt .



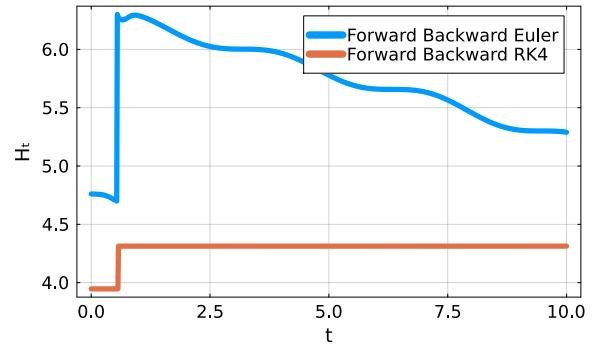
(a) $\Delta t = 0.1$



(b) $\Delta t = 0.05$

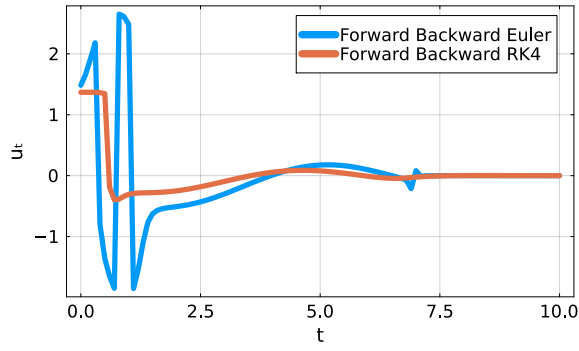


(c) $\Delta t = 0.025$

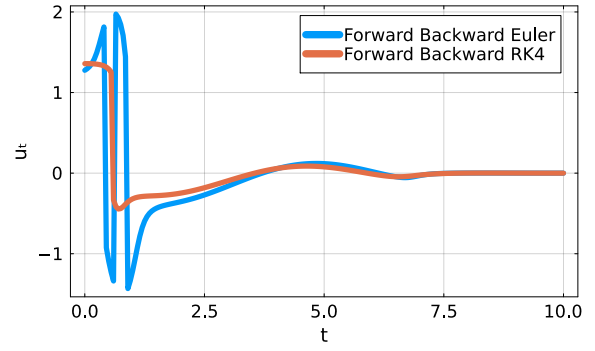


(d) $\Delta t = 0.0125$

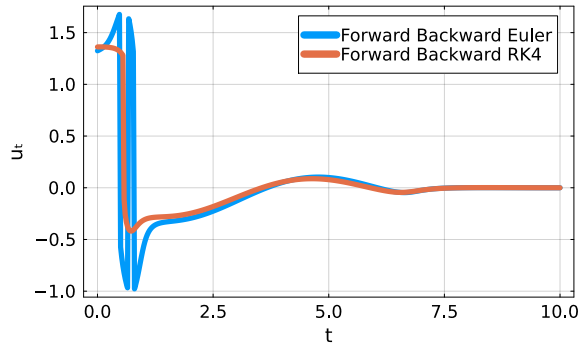
Figure 46: Hamiltonian profile for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 2.0)^T$, as the time step is refined. H_i flattens with smaller Δt , in agreement with its first-integral character along the optimal trajectory.



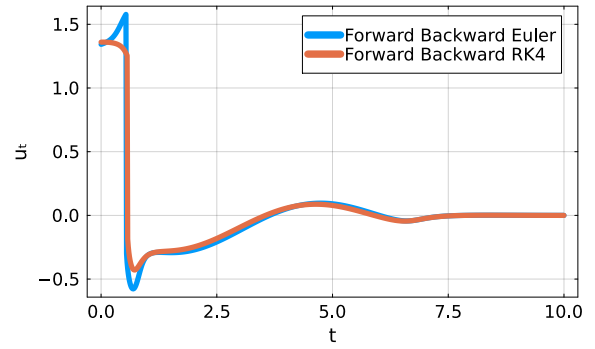
(a) $\Delta t = 0.1$



(b) $\Delta t = 0.05$

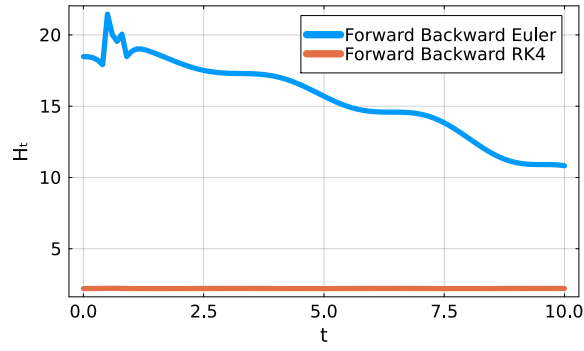


(c) $\Delta t = 0.025$

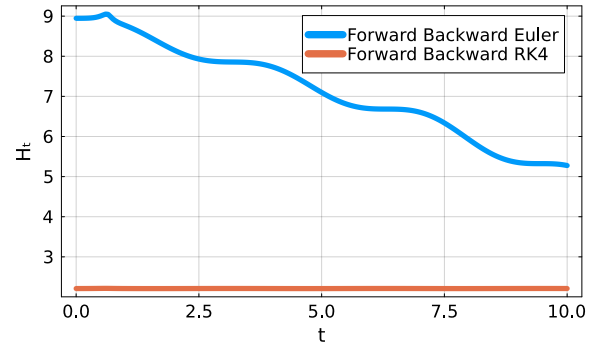


(d) $\Delta t = 0.0125$

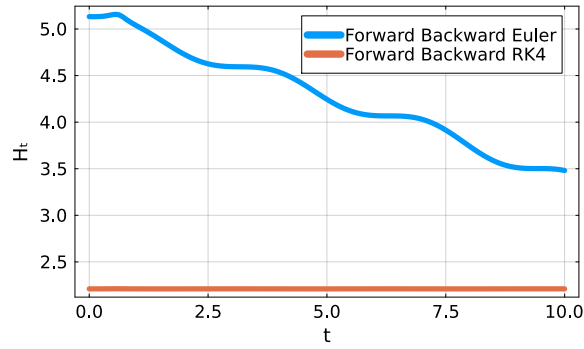
Figure 47: Control u_i for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 2.0)^T$, as the time step is refined. Boundary-layer transients sharpen and the bulk profile stabilises with smaller Δt .



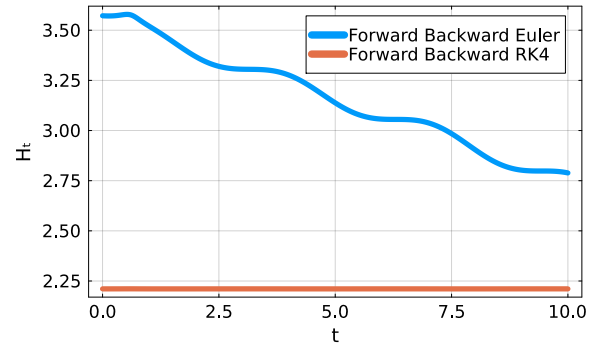
(a) $\Delta t = 0.1$



(b) $\Delta t = 0.05$



(c) $\Delta t = 0.025$



(d) $\Delta t = 0.0125$

Figure 48: Hamiltonian profile for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 3.0)^T$, as the time step is refined. H_i flattens with smaller Δt , in agreement with its first-integral character along the optimal trajectory.

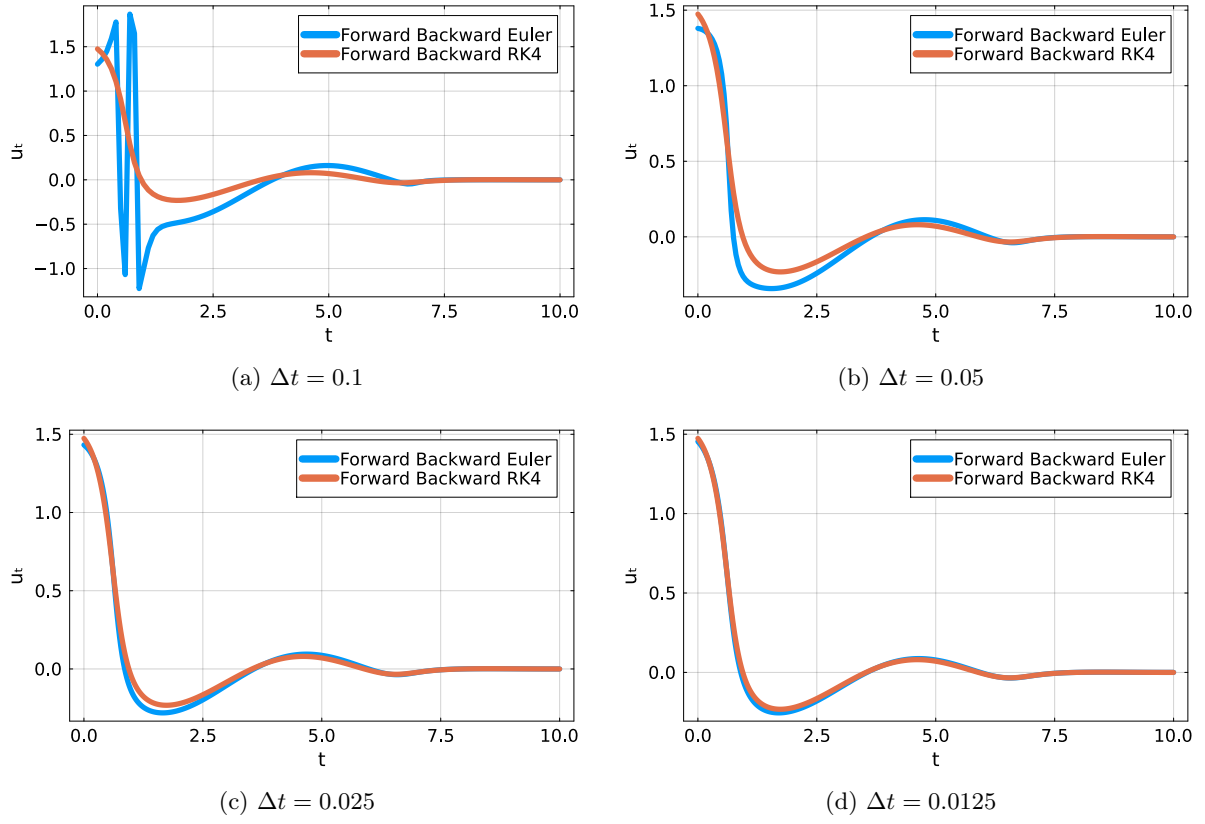


Figure 49: Control u_i for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 3.0)^T$, as the time step is refined. Boundary-layer transients sharpen and the bulk profile stabilises with smaller Δt .

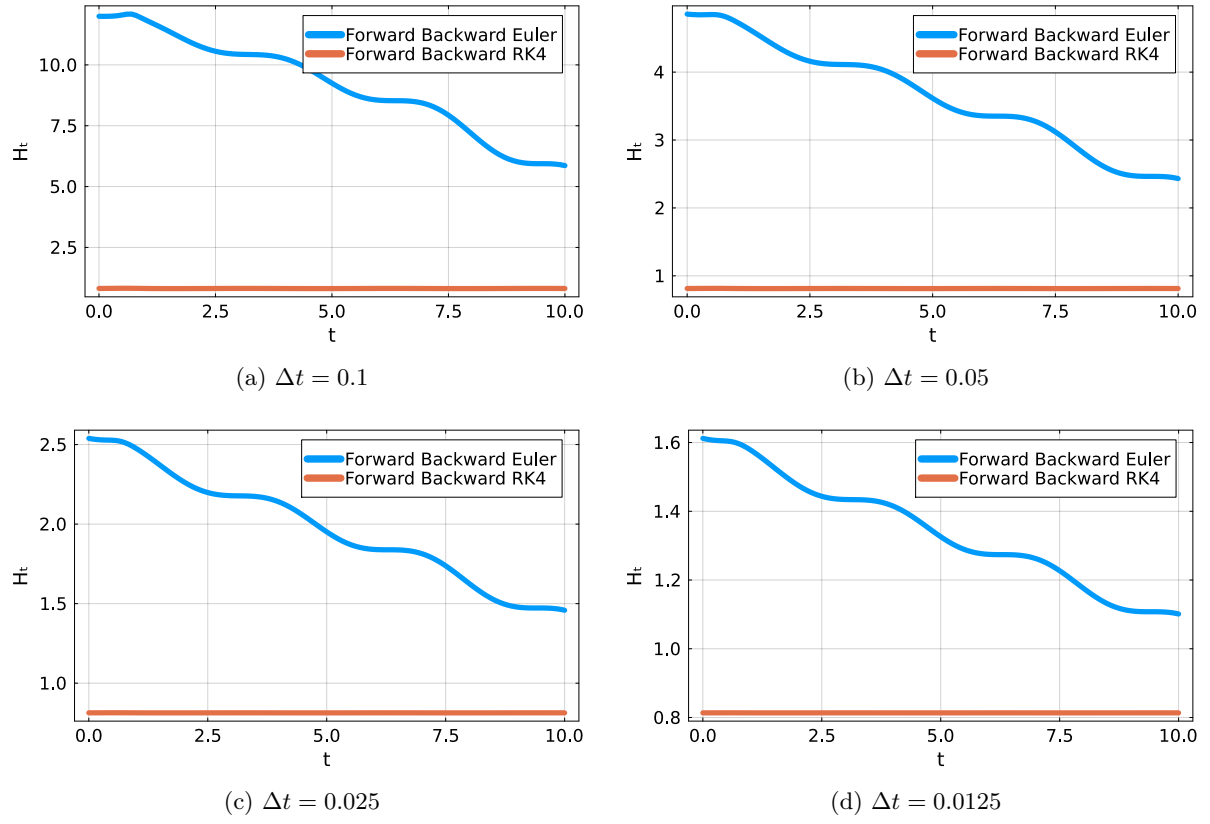


Figure 50: Hamiltonian profile for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 4.0)^T$, as the time step is refined. H_i flattens with smaller Δt , in agreement with its first-integral character along the optimal trajectory.

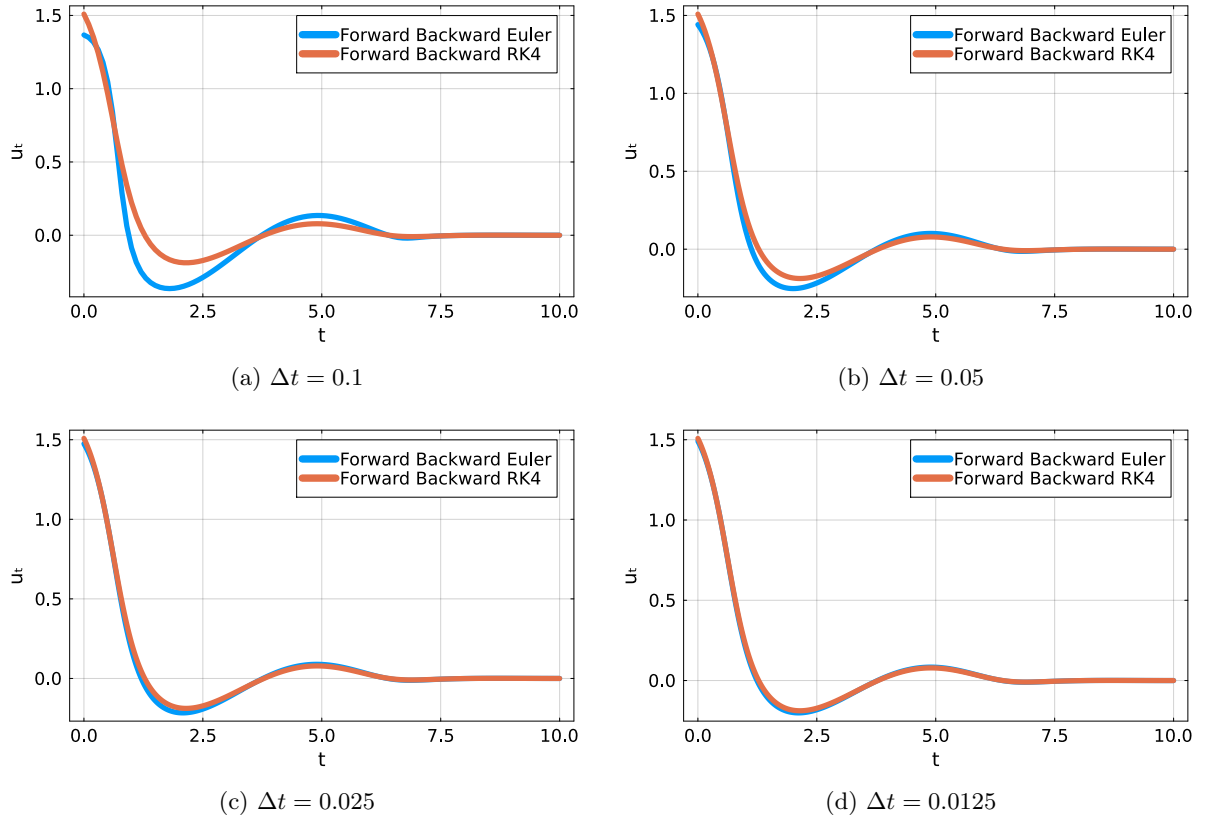


Figure 51: Control u_i for $\alpha = 10.0$, $\mathbf{x}_0 = (1.0, 4.0)^T$, as the time step is refined. Boundary-layer transients sharpen and the bulk profile stabilises with smaller Δt .

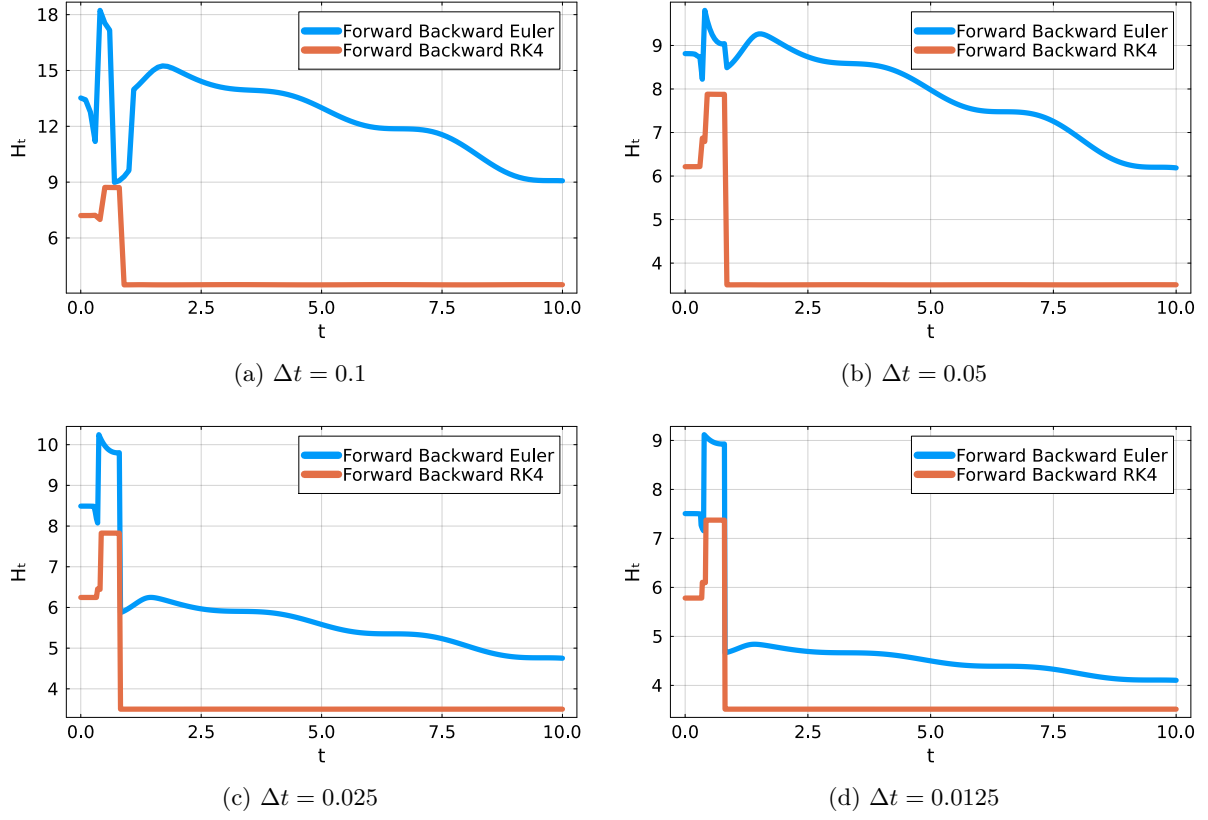
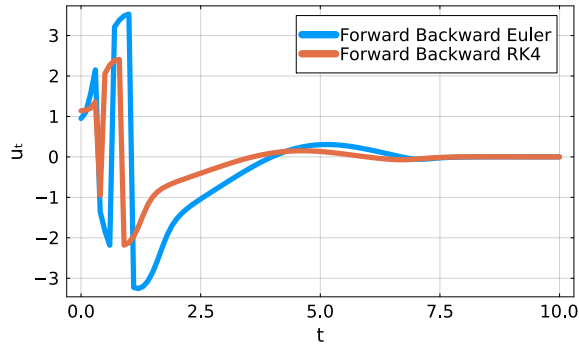
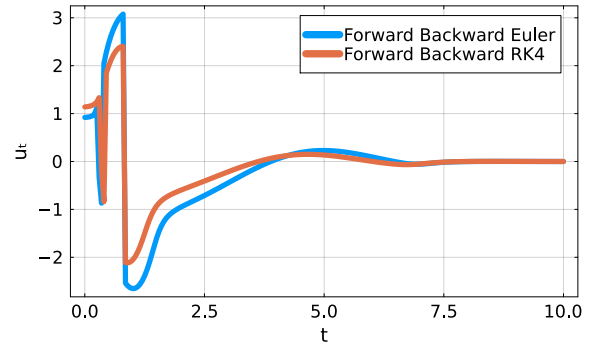


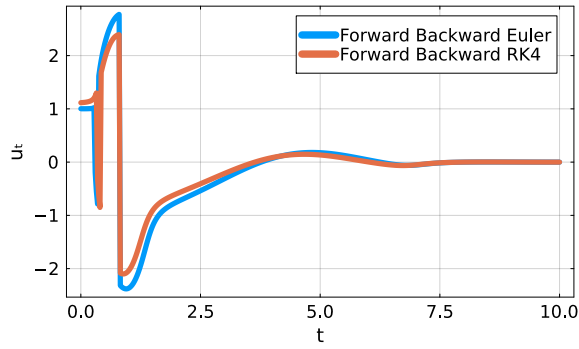
Figure 52: Hamiltonian profile for $\alpha = 5.0$, $\mathbf{x}_0 = (1.0, 2.0)^T$, as the time step is refined. H_t flattens with smaller Δt , in agreement with its first-integral character along the optimal trajectory.



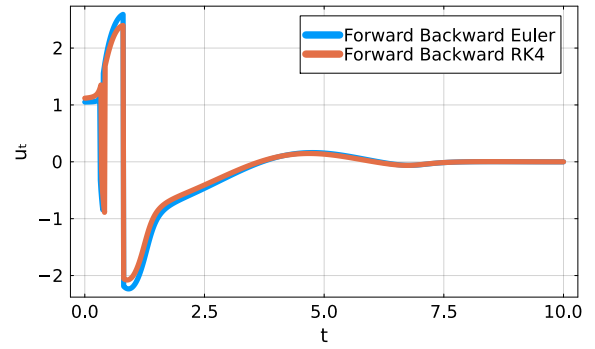
(a) $\Delta t = 0.1$



(b) $\Delta t = 0.05$



(c) $\Delta t = 0.025$



(d) $\Delta t = 0.0125$

Figure 53: Control u_i for $\alpha = 5.0$, $\mathbf{x}_0 = (1.0, 2.0)^T$, as the time step is refined. Boundary-layer transients sharpen and the bulk profile stabilises with smaller Δt .